Democratizing Illiquid Assets: Liquidity Transformation and Performance in Interval Funds*

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ABSTRACT

We study whether regulated semiliquid funds can successfully democratize private assets and improve retail investors' risk-adjusted performance. Using interval funds, which invest in illiquid assets while offering periodic redemptions, we show retail investors face a trade-off between agency frictions and liquidity premia. Managers generate positive risk-adjusted returns in illiquid, information-insensitive markets, especially credit, but underperform in liquid, information-sensitive strategies, consistent with weak flow-based incentives. Outperformance originates from distribution yields rather than NAV appreciation and is amplified by leverage and portfolio illiquidity. Retail investors benefit from co-investing alongside sophisticated investors, whose stable capital supports illiquid exposure and improves liquidity transformation.

KEYWORDS: Democratization of alternative assets, semiliquid funds, interval funds, private credit, private equity, agency frictions, liquidity transformation, retirement portfolio diversification, 401(k) plans.

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1 Introduction

Traditionally, retail investors have been largely excluded from opportunities to invest in private markets and illiquid assets. Recently, policymakers and fund managers have democratized access to private markets through regulated semiliquid funds.[1] With these funds, retail investors gain access to illiquid strategies while retaining regulatory oversight and periodic redemption rights. The Securities and Exchange Commission (SEC) describes these funds as "the optimal way for retail investors to access private market assets" (SEC, 2025) and, among private-market investment vehicles, these funds are growing at the fastest pace, with growth rates of up to 20% per year over the past five years.[2]

Although regulators and asset managers promote semiliquid funds as a gateway to private markets, retail investors face a critical uncertainty: whether these funds deliver private-market diversification and superior performance, or instead expose investors to high fees, agency frictions, and costly liquidity management. To provide periodic redemption rights, fund managers must maintain cash buffers, borrow, or liquidate assets, thus undermining their performance. Conversely, if managers restrict redemption opportunities, investors may suffer from agency costs. Investors and fund managers must navigate these trade-offs across multiple asset classes and heterogeneous clienteles, ranging from retail investors to sophisticated institutional investors. Without understanding the structural interactions between agency frictions and liquidity transformation across asset classes and clienteles, retail investors cannot safely navigate the complexities of alternative investments. To capture the opportunities of a democratized private market while avoiding its risks, retail investors urgently require clear guidance.

To provide this guidance, we examine interval funds, a regulated semiliquid vehicle open to both retail and institutional investors. We show that retail investors face distinct trade-offs depending on the liquidity and information sensitivity of the underlying assets and the composition of the fund's clientele. Interval fund managers generate superior risk-adjusted returns in illiquid asset classes, especially if information-insensitive, such as credit.[3] By allocating 5% to 20% of their retirement savings to credit interval funds, retail investors could improve their risk-return trade-off compared to popular target-date funds.[4] In contrast, managers fail to deliver superior

[1] With the 2025 Executive Order 14330 ("Democratizing Access to Alternative Assets for 401(k) Investors") and the 2026 Proposed Rule 1210-AC38 by the Department of Labor ("Fiduciary Duties in Selecting Designated Investment Alternatives"), the current administration is actively seeking to remove regulatory hurdles to the democratization of private assets.

[2] McKinsey Global Private Equity Report, 2026: https://www.mckinsey.com/industries/private-capital/our-insights/global-private-markets-report/private-equity.

[3] We use the definition of information sensitivity from the theoretical literature on security design, where information sensitivity corresponds to the value of information for pricing a security (Myers and Majluf, 1984; Gorton and Pennacchi, 1990; Holmstrom, 2015; Dang et al., 2013, 2020).

[4] We use the fixed-weight alpha methodology proposed by Brown et al. (2024), Couts and Goncalves (2025), and

performance for retail investors in liquid strategies. Finally, we show that retail investors face a trade-off between fund liquidity and illiquid asset holdings, but institutional investors improve the trade-off by providing more stable funding.

Compared with other semiliquid funds, interval funds offer a particularly suitable empirical setting for evaluating the trade-offs of a democratized private market. Unlike business development companies (BDCs) and real estate investment trusts (REITs), which specialize exclusively in private credit and real estate, respectively, interval funds invest across a wide spectrum of asset classes. And unlike tender-offer funds, which typically serve accredited or qualified investors, interval funds primarily target a retail clientele (see Figure 1). Uniquely among semiliquid vehicles, interval funds commit to providing liquidity at regular intervals, typically quarterly.[5] Moreover, because investors can purchase shares on a daily basis at a price equal to the net asset value (NAV) of the fund's shares, we observe high-frequency, transaction-based NAV and return data that accurately reflect their performance. Finally, using a comprehensive analysis of SEC filings, we document that interval fund managers provide investors with significant exposure to illiquid, hard-to-value assets but impose high structural costs and substantial redemption frictions.

To develop a theoretical framework for our analysis, we hypothesize that interval fund investors gain a comparative advantage in illiquid, information-insensitive strategies, such as credit, but invest at a disadvantage in liquid, information-sensitive strategies, such as publicly traded equities. On the one hand, because of the restricted redemption schedules, interval fund managers mitigate costly fire sales and fragility (Shleifer and Vishny, 1997; Gromb and Vayanos, 2002; Chen et al., 2010; Ma et al., 2022), reduce NAV volatility stemming from forced liquidation (Coval and Stafford, 2007; Greenwood and Thesmar, 2011; Jiang et al., 2022; Dannhauser and Hoseinzade, 2022), and optimize liquidity management (Edelen, 1999; Chernenko and Sunderam, 2016; Choi et al., 2020). Furthermore, interval fund managers exploit longer investment horizons to capture liquidity premiums for their shareholders (Amihud and Mendelson, 1986; Pástor and Stambaugh, 2003; Aragon, 2007; Chen et al., 2007; Sadka, 2010; Ang et al., 2013; Choi et al., 2025). On the other hand, because of the limited redemption pressure, interval fund managers face reduced market-based incentives (Lynch and Musto, 2003; Berk and Green, 2004; Dangl et al., 2008; Wu et al., 2016; Stein, 2005; Berk and DeMarzo, 2026).[6] Consequently, they may exert less effort when investing in information-sensitive securities, whose valuations critically depend on the acquisition and processing of public and private information (Myers and Majluf, 1984; Gorton and Pennacchi,

Goncalves et al. (2025).

[5] At each redemption opportunity, interval funds must offer between 5% and 25% of their total net assets (TNA) for repurchase.

[6] By regulation, interval fund managers can charge performance-based fees only on the income component of their returns, not on capital appreciation. Hence, investors cannot write optimal performance-based contracts to align managerial incentives in information-sensitive strategies in which price appreciation is the primary driver of returns.

1990; Holmstrom, 2015; Dang et al., 2013, 2020). Thus, while fund managers use structured liquidity to generate value in illiquid markets, they operate with weaker incentives when investing in information-sensitive assets.

To evaluate the risk-adjusted performance of interval funds, we build on the literature on hedge fund evaluation (Agarwal and Naik, 2004; Couts et al., 2024; Couts and Goncalves, 2025; Fung and Hsieh, 2004; Fung et al., 2008; Getmansky et al., 2004) and employ an 11-factor model comprising equity, fixed-income, and real-asset factors based on public-market benchmarks. Because retail investors invest primarily in public markets, interval funds' alpha over public-market benchmarks measures their utility gain from accessing illiquid assets through interval funds (Gibbons et al., 1989; Jensen, 1968, 1969). We unsmooth monthly fund returns using both moving-average and autoregressive techniques to address stale pricing and return autocorrelation.

Across a comprehensive set of tests, including fund-level regressions, portfolio sorts, and Fama and MacBeth (1973) regressions, we show that credit funds consistently generate positive and statistically significant risk-adjusted returns after fees, with annualized net alphas exceeding 2% across specifications.[7] Credit funds deliver superior performance to all share classes, although non-retail share classes perform better on average. Private equity and venture capital (PE/VC) funds display positive risk-adjusted performance, although less statistically robust than in credit funds and accruing only to institutional or high-net-worth investors. Finally, funds that engage in liquid strategies deliver severely negative alphas. Overall, the results are consistent with our theoretical framework that predicts a trade-off between illiquidity and agency frictions in more information-sensitive assets.

We implement two sets of tests to examine the tradeoffs between agency frictions and illiquidity in determining the performance of interval funds. In these tests, we employ panel regressions comparing the performance of interval funds to a matched investable (Berk and Van Binsbergen, 2015) sample of index exchange-traded funds (ETFs) with overlapping investment styles. This approach allows us to test the effect of fund clienteles, contractual features, and portfolio allocation on performance, controlling for fund characteristics and specific benchmark-time fixed effects.

In our first set of tests, we find that retail investors in interval funds face substantial agency frictions. First, we document that interval fund managers generate returns primarily through distribution yields (income) rather than through net asset value appreciation. We interpret this behavior as evidence that managers generate value not by selecting undervalued securities, but by harvesting the premiums associated with high-yield, illiquid assets. Second, we find that retail investors in credit funds realize significantly higher returns when they co-invest alongside institutional and

[7] The positive risk-adjusted returns of credit interval funds contrast with those of private-debt drawdown funds, which fail to outperform equity and debt benchmarks after fees (Erel et al., 2024). However, the authors show that these vehicles also hold substantial (information-sensitive) equity alongside debt. Moreover, these funds' managers are not subject to the disciplining effect of quarterly redemptions.

high-net-worth investors. In credit funds, sophisticated investors provide a more stable capital base, allowing managers to preserve illiquid assets and sustain liquidity transformation.[8] Finally, we establish that interval fund managers who charge performance-based fees on income deliver superior returns,[9] whereas managers with less frequent redemption opportunities deliver lower returns.

In our second series of tests, we use portfolio-level data to examine how interval fund managers execute liquidity transformation and balance portfolio illiquidity against investor redemption demands. First, we show that credit fund managers generate significantly higher returns when they take on more leverage and hold illiquid portfolios, indicating that managers earn the illiquidity premium in information-insensitive markets. Second, we find that managers of retail-only funds tend to employ liquid holdings rather than leverage to accommodate more generous redemption policies, highlighting a key trade-off in their liquidity transformation. However, managers relax this trade-off when they attract institutional investors, because institutional clients provide stable capital and utilize redemption opportunities less frequently. We conclude that retail investors capture the full illiquidity premium only when institutional co-investors provide the stable capital necessary for managers to sustain high leverage and large exposures to illiquid assets.

Our paper relates to the growing literature on democratizing access to private and illiquid assets. Clayton and de Fontenay (2026) argue that this democratization should proceed carefully due to high fees, limited understanding by retail investors, and the possibility of diluting the private capital business model with too much capital. Empirical work in structured equity and unlisted REITs finds that retail-focused vehicles offer low returns, limited transparency, and high fees (Célérier and Vallée, 2017; Riddiough, 2022; Henderson et al., 2023). The existing literature on BDCs, many of which are listed on public stock exchanges, shows that they are well-capitalized (Chernenko et al., 2025), beneficial to the firms they lend to (Davydiuk et al., 2024), and that they deliver high risk-adjusted returns (Robinson and Wallskog, 2026). However, Bates (2025) shows that BDCs' reported NAV returns have improbably low volatility and that broadly-sold BDCs perform poorly relative to those reserved for wealthier investors. Moreover, literature on vehicles that cater to both institutions and retail investors finds that institutional investors outperform retail investors (Evans and Fahlenbrach, 2012; Gómez et al., 2024; Begenau and Siriwardane, 2024). Balloch et al. (2025) find that accredited individual investors in drawdown private equity funds perform

[8] Prior work shows that investor sophistication determines investor sensitivity to fund performance, and that these differences shape managerial effort, costs, and fund outcomes (Del Guercio and Reuter, 2014). Funds that attract both retail and institutional capital tend to exhibit improved performance and lower costs, consistent with stronger performance pressure (Evans and Fahlenbrach, 2012). More broadly, costly incentive mechanisms are more likely to be implemented and to be effective when investors are better able to assess managerial skill (Moreno et al., 2018).

[9] By regulation, managers of registered investment companies may not charge performance-based fees on capital appreciation. However, they may charge performance-based fees on income components such as interest or dividends. Robinson and Wallskog (2026) documents that BDCs often employ performance-based fees.

similarly to institutions, although the most affluent individuals outperform the least affluent ones.

Overall, we contribute to this literature by examining an evergreen investment vehicle designed to improve retail access to illiquid assets. In a comprehensive simulation of the return and risk profiles of evergreen funds compared to drawdown funds, Brown and Volckmann (2025) find that the former can offer a better risk-reward trade-off because investor capital is put to work for a greater percentage of the time. Examining such vehicles is increasingly important as these assets become part of more 401(k) retirement plans.

To the best of our knowledge, we are the first to provide empirical evidence on the role of interval funds in democratizing access to private and illiquid assets. In particular, we examine the performance and portfolio holdings of interval funds across clienteles and asset classes. We also document institutional features, fee structure, and liquidity transformation for the entire universe of interval funds. In a more recent, independent paper, Ewens and Faber (2026) study the fee and portfolio difference between private-asset indices and a sample of semiliquid funds specialized in private assets. They find that, after fees, semiliquid funds underperform the private-asset indices. In contrast, we evaluate whether democratizing access to private markets through the interval fund structure improves retail investors' portfolio diversification and risk-adjusted performance. Because retail investors invest primarily in public markets, we measure their utility gain from democratization using interval funds' alphas relative to public benchmarks (Gibbons et al., 1989; Jensen, 1968, 1969). We show that retail investors face trade-offs depending on the liquidity and information sensitivity of the underlying assets and the clientele of the fund.

Lastly, we contribute to the broader literature on governance and agency in delegated portfolio management. One strand of the theoretical literature (Chung et al., 2012; Dangl et al., 2008; Di Tella and Sannikov, 2021; Wu et al., 2016; Pegoraro, 2023; Stein, 2005) shows that fund flows are an effective mechanism for providing managerial incentives. Another strand argues that limited fund liquidity facilitates investment in high-return but illiquid assets (Cherkes et al., 2008; Chordia, 1996; Nanda et al., 2000). In parallel, the empirical literature in asset management also shows that flows influence managerial incentives (Brown et al., 1996; Chevalier and Ellison, 1997; Chung et al., 2012; Del Guercio and Reuter, 2014; Lewellen and Lewellen, 2022; Lim et al., 2016), and that fund families play a key role in shaping internal incentives and strategic interactions across affiliated products (Evans et al., 2020). An additional body of research, primarily focused on hedge funds, finds a positive relationship between liquidity restrictions, such as lock-up periods, and performance (Agarwal et al., 2009; Aragon, 2007; Liang, 1999). However, hedge funds also charge incentive fees based on their total returns. We contribute to this literature by showing that, in the absence of performance fees based on capital gains, the limited liquidity of interval funds can enhance performance in illiquid, information-insensitive assets such as private credit but leads to underperformance in liquid, information-sensitive assets.

2 Institutional Background

Interval funds are increasingly popular with investors, who have driven their aggregate TNA to $158 billion in 2025, a twofold increase from 2021.[10] Moreover, fund managers continue to launch new interval funds at a sustained pace: 168 new interval funds were launched between 2016 and March 2026, and another 67 are currently awaiting approval from the Securities and Exchange Commission (Figure 2). For reference, using SEC filings on EDGAR, we identify a total of 212 interval funds active between 2010 and March 2026, as we describe in this section.

Because this is the first paper to comprehensively examine interval funds, we document the institutional features of interval funds and their liquidity provision.Specifically, we discuss their regulatory foundations, fee structure, redemption opportunities, and portfolio holdings and present descriptive statistics from the complete universe of regulatory filings.

Data for the empirical patterns described in this section are sourced from EDGAR and represent the complete universe of interval funds. We analyze forms filed between 2010 and March 2026. From EDGAR, we identify 212 interval funds based on the submission of Form N-23c-3, which is filed solely by interval funds to notify investors of a periodic repurchase offer (see the discussion of repurchase offers below).

The Investment Company Act of 1940 governs the structure and operations of registered investment companies. Interval funds were introduced in the early 1990s following the adoption of Rule 23c-3 under this Act, establishing a new class of closed-end funds offering limited, periodic liquidity to investors.

While open-end mutual funds are generally restricted from investing more than 15 percent of their assets in illiquid securities under Rule 22e-4, interval funds are not subject to this limitation. The Act imposes requirements such as board oversight, diversification standards, and mandated disclosures, similar to mutual funds. For an interval fund, the Board oversees adherence to a fundamental policy requiring periodic repurchases, typically quarterly.

2.1 Liquidity Provision and Redemption Mechanism

Before each repurchase offer under Rule 23c-3, interval funds must file a notice on Form N-23c-3 with the SEC. We use these filings to identify interval funds in EDGAR and to extract data on their repurchase activity.

[10] See https://xainvestments.com/document/monthly-non-listed-cef-market-update/.

2.1.1 Repurchase Offers

Interval fund repurchase offers specify three key dates: the *offer start date*, the *repurchase request deadline*, and the *repurchase pricing date*. Investors may submit redemption requests during the window between the start of the offer and its deadline. For standard interval funds (quarterly, semiannual, or annual), shareholders must receive notification between 21 and 42 days prior to the repurchase request deadline. However, funds offering monthly repurchases typically operate under exemptive relief that permits a shorter notification period, often as brief as seven days, to accommodate the higher frequency of offers.

Shares are repurchased at the share class's net asset value (NAV) per share determined on the pricing date. In an interval fund, the repurchase pricing date must occur no later than 14 days after the repurchase request deadline. The funds are then required to pay redeeming investors within seven days of the pricing date.

Interval fund repurchase offers must specify the percentage of total net assets (TNA) the fund will offer to repurchase (between 5% and 25% of TNA). To ensure sufficient liquidity for the periodic repurchases, interval funds are required to maintain assets that the board reasonably believes can be converted to cash within seven days, in an amount no less than the value offered for repurchase.

From the EDGAR database, we obtain data on redemption offers for 212 interval funds covering the period from 2010 to March 2026. Panel A of Table 1 presents summary statistics for these redemption offers. The median time between consecutive redemption opportunities is 91.00 days, reflecting the quarterly nature of most interval fund repurchases. While the majority of interval funds offer quarterly redemptions, a cluster offers monthly liquidity, and very few utilize semiannual or annual schedules.

Interval funds typically allow approximately one month between the offer start date and the repurchase request deadline (median of 29.00 days). Although some funds do not specify the pricing date in their redemption offers, most interval funds price their shares on or very close to the repurchase request deadline (median of 0.00 days from deadline to pricing).

Finally, most interval funds offer the statutory minimum of 5.00% of TNA for redemption. However, interval funds typically retain the discretion to increase the repurchase amount by up to 2.00% of outstanding shares if requests exceed the initial offer.

Over our sample period, interval funds have steadily offered an average of close to four redemption opportunities each year. However, the total amount of liquidity they provide to investors has declined. We measure total liquidity provision as the percentage of TNA offered for redemption in a given calendar year. Figure IA.1(a) shows that, up to 2013, interval funds offered over 50% of their TNA for redemption each year. Since then, average TNA liquidity has been trending down and is currently around 20% of the TNA each year.

2.1.2 Oversubscription and Prorated Redemption

In their annual shareholder reports (Form N-CSR), interval funds summarize the outcomes of their redemption offers. We define a redemption as *oversubscribed* if investors tendered more shares than the amount offered by the fund. A redemption is *prorated* if it is oversubscribed and the fund does not fulfill all redemption requests. A fund can increase the offered amount at its discretion to accommodate excess requests. Finally, for prorated offers, we collect information on the *proration*; that is, the fraction of tendered shares repurchased by the fund.

Summary statistics appear in Panels B and C of Table 1. Panel B presents fund-level statistics. Over our sample, 15.7% of the redemption offers were oversubscribed, but only 9.00 % resulted in proration (these two are indicator variables). Among prorated offers, the mean proration was 53.21%. In panel C, we present share-class-level data. On average, investors in individual share classes tender approximately 4.20 % of the shares outstanding at each redemption event. The median, however, is only 2.20%, indicating the existence of a right tail of large redemption requests.

Figure IA.1(b) shows the time series of oversubscribed and prorated offers each year. Overall, the frequency of oversubscriptions and prorations shows a stable but modest upward trend starting from 2014, with a peak in 2020, when 25% of the redemption offers were oversubscribed, and 15% were prorated.

2.2 Fee Structure

Interval funds are relatively expensive investment products. We collect expense ratios from the annual N-CSR forms filed by interval funds between 2010 and March 2026. Panel A of Table 2 reports summary statistics for annual expense ratios for interval funds. Over this period, the average net expense ratio across all interval fund share classes was 3.02%, with a median of 2.74%. The reported expense ratios include management fees, borrowing costs, other annual expenses,[11] and incentive fees.

Like other investment products available to retail investors, interval funds charge annual management fees. Panel A of Table 2 shows that the average management fee for interval funds is 1.39% of the fund's TNA, with a median of 1.30%. Although management fees are disclosed as a percentage of TNA, they are often calculated based on the fund's assets under management (AUM). As a result, higher leverage tends to increase management fees as a share of TNA. Using

[11]Other annual expenses include, among others, distribution fees, servicing fees, and administrative fees. Some interval funds may also apply fee reductions or caps to limit investor costs.

those observations for which the statutory fee on the AUM is disclosed, we calculate that, on average, investors' management fees are increased by 36 basis points because of funds' leverage (see "Fee on leverage" row in Panel A).

Furthermore, if an interval fund uses leverage, it may incur interest expenses and other borrowing-related costs. Like other registered investment companies, interval funds are subject to a leverage limit of one-third of their AUM. However, unlike mutual funds, which rarely employ leverage, 70% of our share-class-year observations report positive borrowing costs. Among these, borrowing costs average 0.84% of annual TNA, with a median of 0.58%, as shown in Panel A of Table 2.

For those interval funds that invest in other funds, the average acquired fund fees are 0.45%, although the median is only 0.06%, reflecting a distinction between funds of funds that invest primarily in other funds and accumulate large acquired fund fees, and funds that invest only a portion of their portfolio in other funds as part of a broader allocation strategy.

Depending on the distribution channel of a given share class, interval funds may also charge front-end sales loads, which can range from 1% to 8%. Transaction fees may apply to redeeming investors as well. Some of these fees take the form of a contingent deferred sales charge (CDSC), which is triggered only if shares are redeemed within a specified period after purchase. Panel A of Table 2 shows that, among interval funds that charge redemption fees, the average (median) redemption fee is 1.67% (2.00%). When fees apply only to early redemptions, the average (median) period during which the fee applies is 1.19 years (1.00 years) from the purchase date (see "Early withdrawal period" row in Panel A).

Interestingly, as shown in Panel B of Table 2, 17.5% of the interval funds charge incentive fees. These fees are based on Net Investment Income, defined as interest income, dividend income, and other income, less operating expenses such as management fees and fund costs. Although Section 205 of the Investment Advisers Act of 1940 prohibits registered investment advisers from charging fees "on the basis of a share of capital gains upon or capital appreciation of" a fund, performance fees based on income are permitted.

Among interval funds that charge performance-based fees, the median incentive fee rate is 15.00% (see Panel A of Table 2). Of these funds, 81.6% apply a hurdle rate, typically set at 6.00%. Among those with a hurdle, almost all of them include a catch-up provision.[12] In addition, 14.9% of interval funds with incentive fees include a high-water mark provision.

[12] A catch-up provision allows the adviser to receive a greater share of the fund's dividend and interest income (often 100% of such income) once the hurdle is met. This catch-up phase typically continues until the stated incentive fee rate is met. Beyond that point, the standard incentive fee rate applies to any additional dividend and interest income.

2.3 Portfolio Composition

We analyze the portfolio composition of interval funds using Form N-PORT, which registered investment companies are required to file with the SEC on a quarterly basis. Form N-PORT provides detailed holdings-level information, including asset classifications, fair value hierarchy levels, and security-level characteristics, and is available from the fourth quarter of 2019. We match N-PORT filings to our sample using CIK identifiers.

For each filing, we compute the portfolio share allocated to different asset classes: equity (which includes equity investments in real estate and infrastructure), fixed income, asset-backed securities (ABS), and cash and cash equivalents. We also identify investments in private funds (unregistered investment vehicles) and registered funds (such as mutual funds and ETFs) based on issuer-type classifications in the filings. To measure portfolio liquidity, we extract the share of assets classified under each level of the fair value hierarchy (Levels 1, 2, and 3) and the share of restricted securities. All these portfolio exposures are measured as a percentage of AUM. We also obtain data on leverage, which we measure as total liabilities as a percentage of AUM. Figure 3 presents the distribution of portfolio characteristics for interval funds.

2.3.1 Asset Allocation

The asset allocation of interval funds reflects their distinct investment mandate to invest in illiquid assets. As Panel 3(a) shows, interval funds are predominantly fixed-income vehicles: the median interval fund allocates 31% of its AUM to debt securities and loans, with a mean allocation of 40%. Asset-backed securities represent another significant component, with a mean allocation of 17%. Equity holdings are more modest, with a median of only 0.4% and a mean of 13%, reflecting the concentration of equity exposure in a subset of funds. Interval funds have a median allocation of 0% to private funds, though the mean of 11% indicates that some interval funds pursue a fund-of-fund strategy.

2.3.2 Liquidity and Leverage

The fair value hierarchy and restricted securities status provide complementary insights into portfolio liquidity. The fair value hierarchy, established under ASC 820, classifies assets based on the observability of inputs used in their valuation. Level 1 assets are valued using quoted prices in active markets for identical assets, such as exchange-traded equities, and are typically the most liquid. Level 2 assets are valued using observable inputs other than quoted prices—for example, dealer quotes, matrix pricing, or prices of similar assets—and include most corporate bonds, structured products, and over-the-counter derivatives. Level 3 assets are valued using significant unobservable inputs, such as internal models or management estimates, and typically include private assets, illiquid structured products, and other hard-to-value securities.

Restricted securities, by contrast, are classified based on legal constraints on resale rather than valuation methodology. These include Rule 144A securities, which may only be sold to qualified institutional buyers, and private placements that have not been registered with the SEC. A security can be both Level 2 (if valued using observable inputs) and restricted (if subject to resale limitations), or Level 3 and freely tradeable.

As Panel 3(b) shows, interval funds hold a substantial share of Level 2 assets (mean 36%, median 20%). Level 3 assets represent a mean of 31% and a median of 16%. Level 1 assets account for a mean of only 12%. Moreover, interval funds have a mean restricted securities allocation of 31% and a median of 10%. Finally, the median leverage ratio for interval funds is 8%, with a mean of 13%.

3 Data

3.1 Data Sources and Sample Construction

Our analysis draws on two complementary data sources: regulatory filings collected from the SEC's EDGAR system and Morningstar, which provides returns and fund characteristics. From EDGAR, we use five forms. Form N-23c-3, filed before every repurchase offer, identifies the universe of interval funds and provides the size and frequency of their redemption windows. Registration statements and prospectus amendments (Forms N-2, 486BPOS, and 486APOS) are the source for investor eligibility and clientele information, including the explicit accredited-investor requirements used in our alternative clientele classification. Form N-CSR, the annual shareholder report, provides expense ratio data. Form N-CEN, the annual census filing of registered investment companies, reports average annual net assets. Form N-PORT, which registered investment companies have filed quarterly since the fourth quarter of 2019, provides security-level holdings with fair-value hierarchy classifications, restricted-security flags, and balance-sheet items used to measure portfolio liquidity and leverage.

Morningstar provides the monthly share-class panel used in our regressions. From it, we obtain monthly total returns, net asset value (NAV), and total net assets (TNA), together with time-invariant share-class and fund-level characteristics, including inception date, minimum investment, share class type, fund-of-funds status, Morningstar category, and benchmark index. The initial Morningstar extract covers 443 interval fund share classes (issued by 184 funds). As comparison groups, we obtain data on 252 tender offer share classes (147 funds), 71 BDC share classes (25 funds), 157 nontraded REIT share classes (44 funds), 3,685 active ETFs, and 3,838 index ETFs.[13] To ensure comparability across fund types, we restrict the sample to Morningstar investment categories and benchmark indices that contain at least one interval fund. This filter leaves the interval, BDC, and REIT samples unchanged and narrows the active and index ETF samples to 1,546 and 1,192 funds, respectively. After requiring non-missing values for the dependent variable and lagged controls in our baseline specification, the final estimation sample covers January 2010 through March 2026 and includes 356 interval fund share classes (150 funds), 1,248 active ETFs, 1,014 index ETFs, 104 tender offer share classes (51 funds), 54 BDC share classes (22 funds), and 93 nontraded REIT share classes (23 funds).

For the regression analysis, we consolidate the Morningstar category groups into five broader investment categories that group funds pursuing similar strategies: *Credit* (Morningstar Private Debt, Fixed Income, and Municipal Fixed Income groups, capturing private credit, traditional fixed income, and municipal bond strategies); *Real Asset* (Direct Real Estate and Direct Infrastructure groups, together with Equity funds specialized in real estate or infrastructure); *PE/VC* (Private Equity and Venture Capital groups); *Multi-Asset* (Allocation and Private Allocation groups, capturing diversified portfolios that combine equities, fixed income, and alternative exposures); and *Liquid Strategies* (Liquid Alternatives and the remaining Equity funds, which implement hedge-fund-like or listed equity strategies). Among interval funds in the estimation sample, Credit is the largest category with 244 share classes (102 funds), followed by Real Asset (63 share classes, 20 funds), PE/VC (20 share classes, 11 funds), Liquid Strategies (15 share classes, 8 funds), and Multi-Asset (14 share classes, 9 funds).

We further classify interval fund share classes by investor clientele based on Morningstar share class labels and minimum investment requirements. Share classes carrying retail-oriented labels (e.g., Class A, B, C, D, Investor, No Load, Retirement) with a minimum investment at or below \$25,000 are classified as *Retail*. Share classes carrying an institutional label, or any non-ETF share class with a minimum investment above \$25,000, are classified as *Inst/HNWI*, a single category that pools institutional and high-net-worth investors. Because ETFs trade freely on exchanges and are accessible to any investor, we classify all ETF share classes as Retail. In the estimation sample, 200 of the 356 interval fund share classes are Retail, and 156 are Inst/HNWI. As a robustness check, because retail labels and low minimum investments do not guarantee that a share class is open to unrestricted investors, we construct an alternative classification that adds a third category, *Accredited*, identified from explicit accreditation requirements disclosed in EDGAR prospectus

[13] Like interval funds, the tender offer funds, BDCs, and REITs in our sample are nontraded and transact at NAV. However, unlike interval funds, they offer discretionary and irregular repurchase programs subject to board approval. Investor eligibility requirements are generally more restrictive for these funds than for interval funds, reflecting their orientation toward high-net-worth and institutional investors. Table IA.1 summarizes the differences between investment vehicles.

filings.

3.2 Summary Statistics

Figure 4 shows the time-series evolution of the number of interval fund products and their aggregate TNA in the Morningstar sample. Panels 4(a) and 4(b) indicate that interval funds have grown rapidly in recent years, both in the number of products and in assets under management. Figure 5 plots the distribution of interval fund share classes across the five consolidated investment categories and by clientele. Credit funds account for about two-thirds of share classes in the estimation sample.[14] Within each asset class, the sample is split between retail and Inst/HNWI share classes in a balanced way, although retail share classes outnumber Inst/HNWI share classes by a modest margin.[15]

Table 3 reports share-class-level summary statistics for interval funds January 2010 through March 2026. Interval funds manage an average of $264 million in TNA per share class and belong to relatively small fund families, with a mean family TNA of $1.22 billion. They are relatively young, with a mean age of 4.99 years, and only a small fraction operate as fund-of-funds structures (8%). Interval funds receive positive average monthly net flows of 3.25%, although flows are dispersed across the distribution. Mean annualized net returns are 5.56%, and are largely driven by distributions: average annualized distribution returns are 7.22%, while NAV returns are negative on average at -1.66%. Similar summary statistics for active ETFs, index ETFs, BDCs, REITs, and tender-offer funds are in Table IA.2 of the Internet Appendix.

Most of our specifications are estimated at the share-class level, and a few variables require construction steps beyond what Morningstar reports directly. Total net assets are primarily sourced from Morningstar, with Form N-CEN used to fill in missing values. Monthly fund flows are computed as the percentage change in TNA net of price appreciation, and annualized NAV and distribution returns are constructed from month-over-month price changes and from the residual component of total returns, respectively. We winsorize monthly net returns at the 1st and 99th percentiles within fund type and month, and monthly fund flows at the 1st and 99th percentiles. Slow-moving share-class characteristics such as TNA and NAV are forward-filled within the share class to handle irregular reporting.

Some tests require fund-level observations. For these, we construct a portfolio-level version of the panel by aggregating share classes within each fund-month, taking TNA-weighted averages of share-class returns and summing TNA across share classes. While the summary statistics

[14] The distribution of active ETFs, index ETFs, BDCs, REITs, and tender-offer funds across asset classes is in Figure IA.2 of the Internet Appendix.

[15] The pattern closely matches the distribution inferred from the language of EDGAR prospectus filings (Figure 1(a)).

focus on net returns, the fund-level analysis relies on gross returns to abstract from fee dispersion across share classes. Expense ratios are obtained primarily from Morningstar, with Form N-CSR used to fill missing values,[16] and when gross returns are not directly available, we impute them as net returns plus one-twelfth of the contemporaneous annual net expense ratio. We use this portfolio-level panel for tests whose variables of interest are defined at the fund level (e.g., portfolio holdings from N-PORT, redemption policy, and incentive-fee structure). Appendix Table A.1 lists all variables and their construction.

3.3 Factors

To evaluate the risk-adjusted performance of interval funds, we construct a set of 11 factors spanning equity and fixed-income indices. Interval funds engage in investment strategies that partially overlap with those of registered funds, such as mutual funds, and with those of unregistered funds, such as hedge funds and private equity funds. Following the factor-model literature for alternative investment vehicles (Agarwal and Naik, 2004; Fung and Hsieh, 2004; Fung et al., 2008; Couts et al., 2024; Couts and Goncalves, 2025), we construct a set of eleven factors (seven equity and four fixed income) designed to capture exposures across asset classes, market capitalizations, investment styles, and credit quality.[17]

The equity factors include the excess return on the S&P 500 index (Datastream mnemonic SPXT$), a large-cap value-minus-growth factor constructed as the difference between the S&P 500 Value and S&P 500 Growth indexes (SPXV$ minus SPXG$), the excess return on the Russell 2000 index (RU20INTR$), a small-cap value-minus-growth factor constructed as the difference between the Russell 2000 Value and Russell 2000 Growth indexes (R2VLINTR$ minus R2GLINTR$), the excess return on REITs using the FTSE NAREIT All Equity REITs index (FNAREVG$),[18] the excess return on global equities using the FTSE Global Ex US index (FAWXUS$), and the excess return on emerging market equities using the FTSE Emerging index (AWALEG$).

The fixed income factors are constructed from changes in yield spreads sourced from FRED and Bloomberg. First, we include the change in the risk-free rate.[19] Then, a term spread factor captures the monthly change in the spread between the 10-year Treasury yield (FRED series DGS10) and the risk-free rate. The credit spread factor captures the monthly change in the spread between Moody's Baa corporate bond yield (FRED series DBAA) and the 10-year Treasury yield.

[16]Figure IA.3 in the Internet Appendix shows average expense ratios by investment category and by clientele.

[17]Compared to Agarwal and Naik (2004), Fung and Hsieh (2004), and Fung et al. (2008), we do not include option-based factors because interval funds do not typically engage in the type of dynamic trading strategies that generate option-like returns.

[18]Couts et al. (2024) use the same REIT index in their factor model to evaluate the performance of hedge funds and commercial real estate funds.

[19]The risk-free rate is obtained from Kenneth French's data library, available at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Finally, the high-yield spread factor captures the monthly change in the spread between the ICE BofA CCC & Lower US High Yield Index Effective Yield (FRED series BAMLH0A3HYCEY) and Moody's Baa yield.

4 Theoretical Framework

We develop a theoretical framework to derive predictions about the fund-level excess payoff of interval funds relative to a passive benchmark with zero expected excess return. The framework highlights that, because of their structured liquidity, the performance of interval funds depends critically on the information sensitivity and maturity structure of the underlying asset.

4.1 Set-Up

We consider an interval fund and assess its performance against a passive benchmark whose expected excess return is normalized to zero. We assume that the fund is subject to real-world regulatory requirements and, hence, cannot charge performance-based fees on capital appreciation.

There are three periods indexed by $t \in \{0,1,2\}$. Without loss of generality, we normalize the fund's net asset value (NAV) at $t = 0$ to one and set initial assets under management (AUM) to $k_0 = 1$. At time $t = 1$, the interval fund can limit the maximum amount of redemption requests it satisfies. We denote the upper limit on redeemed shares by $\lambda \in (0,1)$.

The fund can invest in two assets. Short investments mature every period and earn a deterministic premium δ_S per period. Long investments mature at $t = 2$ and earn a deterministic premium δ_L, with $\delta_L > 2\delta_S \geq 0$. Let $\ell \in [0,1]$ denote the share of the portfolio allocated to long investments, so that $1-\ell$ is allocated to short investments. At $t = 1$, the NAV remains equal to one, and the income from the short-investment portion is paid as distributions.

To meet redemption requests, the fund uses maturing short investments first, but it can also liquidate long investments at $t = 1$ at their NAV of one. For any realized redemption measure $m \in [0,\lambda]$, the amount of long investment liquidated at $t = 1$ is

$$q(\ell,m) := \max\{0, m-(1-\ell)\}.$$

This implies that no period-1 long-investment liquidation occurs when $\ell \leq 1-\lambda$.

Long assets are less liquid than short assets, so the fund incurs a liquidation cost of γ per unit of long investment liquidated at $t = 1$. We assume that, after redemptions, the fund rebalances its portfolio to its target allocation ℓ of illiquid assets.[20] The fund then incurs a liquidation cost of γ per unit of long investment liquidated at $t = 2$ to rebalance the portfolio after redemptions.

[20] With this assumption, we avoid counter-intuitive outcomes whereby redemptions improve performance by mechanically increasing the share of long investments in the portfolio. The assumption can be microfounded in

Given redemption risk, the manager chooses the allocation between short and long investments to maximize expected long-term returns. The optimal allocation ℓ^* is characterized below after the distribution of redemptions is specified.

For a given allocation ℓ and realized capped redemption measure m, define the fund-level cumulative excess return, net of fees, as

$$R_2(\ell,m) = \underbrace{(1-\ell)\delta_S - q(\ell,m)\gamma - f}_{\text{first-period return}} + \underbrace{\ell(\iota v+\delta_L)+(1-\ell)(\iota v+\delta_S) - \gamma\frac{\max\{0,\ell m - q(\ell,m)\}}{1-m} - f}_{\text{second-period return}}, \tag{1}$$

where $\iota \in \{0,1\}$ indicates whether the investment is information-sensitive; v represents the stochastic capital appreciation of the investment with cumulative distribution function (c.d.f.) $F(\cdot;e)$ that depends on managerial effort $e \in \{0,1\}$; $\max\{0,\ell m - q(\ell,m)\}$ is the amount of long investment liquidated to rebalance the portfolio after redemptions and dividing by $1-m$ converts this to the share of post-redemption AUM,; and f is the fund's expense ratio.

Let $v_0 := \mathrm{E}[v \mid e=0]$ and $v_1 := \mathrm{E}[v \mid e=1]$, with $v_1 > 0 > v_0$. We assume the following:

$$v_0 + \delta_L < f, \quad \delta_S \geq f + \gamma.$$

The first condition implies that, absent effort, net continuation payoff on the information-sensitive investment is negative. The second condition is a low-liquidation-cost restriction: short-illiquidity premium per period covers fees and a unit liquidation-cost buffer. It rules out self-fulfilling "runs" by redeeming investors and multiple equilibria. Together with $v_1 > 0$ and $\delta_L > 2\delta_S$, these conditions imply $v_1 + \delta_L > f$.

The manager earns fees based on AUM and incurs a private cost c if she chooses to exert effort. The cost of effort is private information and, ex ante, is drawn from a cumulative distribution function $G(\cdot)$ with support on $[0,\bar{c}]$, where $\bar{c} > 0$.

For any effort level $e \in \{0,1\}$ and allocation ℓ, the manager's expected payoff is given by

$$f + f\mathrm{E}[1+\Delta k(\ell,e,z) \mid e] - c\,e,$$

where Δk denotes the change in AUM due to redemptions at $t=1$. The manager chooses to exert effort if and only if

$$c \leq f\left(\mathrm{E}[\Delta k(\ell,1,z)] - \mathrm{E}[\Delta k(\ell,0,z)]\right). \tag{2}$$

Fund capital is divided into two types: informed and uninformed. Informed capital comprises

an infinite-horizon model where the fund faces a sequence of redemption shocks and rebalances after each shock. To improve tractability and intuition, we focus on a two-period model with a single redemption shock at $t=1$ and rebalancing at $t=2$.

a fraction η of AUM and uninformed capital comprises the remaining fraction $(1-\eta)$, with $\eta + (1-\eta) = 1$. Informed investors observe the manager's effort e. Given our assumptions, when the investment is information-sensitive, informed investors redeem after observing $e = 0$ and remain invested after observing $e = 1$. When the investment is information-insensitive, effort has no effect on expected capital appreciation, so informed capital remains unchanged.

Uninformed investors do not observe effort. Instead, a random measure z of uninformed investors is hit by a liquidity shock and submits redemption requests at $t = 1$. We assume z has cumulative distribution function $H_\eta(\cdot)$ with support on $[0,(1-\eta))$. We assume that a higher share of informed capital reduces exposure to uninformed liquidity shocks; that is, if $\eta' > \eta$, then $H_{\eta'}(x) \geq H_\eta(x)$ for all x, so the distribution of z shifts weakly to the left.

Redemption requests in period $t = 1$ are therefore given by $T(e,z) = z + \iota(1-e)\eta$. Let $m_e(z) := \min\{\lambda, T(e,z)\}$ denote realized capped redemptions. Accordingly, flows at time 1 are given by $\Delta k(\ell,e,z) = -m_e(z)$. The fund uses short investments to meet redemptions and liquidates long investments as needed.

4.2 Predictions

Next, we derive predictions for the performance of the interval fund, beginning with managerial incentives to exert effort. Using equation (2) and noting that $\Delta k(\ell,e,z) = -m_e(z)$, we find that the manager exerts effort if and only if

$$c \leq \iota f \Psi(\eta,\lambda),$$

where

$$\Psi(\eta,\lambda) := \mathrm{E}_z\big[\min\{\lambda, z+\eta\} - \min\{\lambda, z\}\big] = \int_0^\lambda \min\{\eta, \lambda - z\}\, dH_\eta(z).$$

The term $\Psi(\eta,\lambda)$ is the expected additional redemption outflow generated by informed investors when the manager shirks, after accounting for the fact that liquidity shocks may already use part of the redemption cap. Define the probability of exerting effort as

$$\pi(\eta) := P(e = 1 \mid \eta) = G\big(\iota f \Psi(\eta,\lambda)\big).$$

Because the redemption cap λ limits the outflow penalty from shirking, $\pi(\eta)$ is depressed compared to an open-end fund with no redemption cap: the manager faces a smaller penalty from investor outflows when choosing not to exert effort. Liquidity shocks further reduce the marginal penalty from shirking in states where they consume part of the redemption cap. When $\iota = 0$, effort does not affect investor payoffs, so the effort channel drops out of expected returns.

The manager chooses the allocation ℓ to maximize expected fund-level excess payoff, taking into account both the higher payoff on long investments and the possibility that long investments may be liquidated after large redemptions. Thus,

$$\ell^* \in \arg\max_{\ell \in [0,1]} \left\{ \pi(\eta)\, \mathrm{E}_z\big[R_2(\ell, m_1(z)) \mid e=1\big] + (1-\pi(\eta))\, \mathrm{E}_z\big[R_2(\ell, m_0(z)) \mid e=0\big] \right\}.$$

We focus on interior solutions and assume $\ell^* \in (0,1)$.

Evaluating expected fund-level excess payoff at the optimal allocation yields

$$\mathcal{V}^* := \pi(\eta)\, \mathrm{E}_z\big[R_2(\ell^*, m_1(z)) \mid e=1\big] + (1-\pi(\eta))\, \mathrm{E}_z\big[R_2(\ell^*, m_0(z)) \mid e=0\big]. \tag{3}$$

These expressions yield predictions for the interval fund's expected excess return based on asset characteristics. Setting $\iota = 0$, informed redemptions are inactive, so $m_0(z) = m_1(z) = \min\{\lambda, z\}$.

PREDICTION 1. *If the investment is information-insensitive ($\iota = 0$), the interval fund's expected excess return satisfies*

$$\mathcal{V}^* \geq \gamma(2-\lambda) > 0.$$

Expected excess return is strictly increasing in both δ_S and δ_L.

All proofs and derivations are in the Internet Appendix. When the investment is information-insensitive ($\iota = 0$), managerial effort does not affect performance. A higher δ_L raises continuation payoff on the long sleeve through $\ell^*(\delta_L - 2\delta_S)$. A higher δ_S raises both short-maturity contributions across periods. A larger liquidation-cost parameter γ lowers performance through higher expected liquidation costs, which capture both period-1 liquidation and period-2 rebalancing liquidation. Under $\delta_S \geq f + \gamma$, the model delivers a strictly positive level prediction even after liquidation costs.

According to the literature (Myers and Majluf, 1984; Gorton and Pennacchi, 1990; Holmstrom, 2015; Dang et al., 2013, 2020), non-distressed debt is a canonical example of an information-insensitive security. Debt resembles a short put option on the value of the firm. As long as this put option remains sufficiently out of the money, marginal changes in information about the firm's fundamentals have little impact on the value of the firm's debt.

The information-sensitive case introduces an additional channel through which effort matters, yielding the next prediction.

PREDICTION 2. *If the investment is information-sensitive ($\iota = 1$), the interval fund's expected excess return depends on the illiquidity premium of the underlying investments. In a limiting case where illiquidity premia, liquidation costs, and fees are zero ($\delta_L \to \delta_S = \gamma = f = 0$), the interval fund underperforms the benchmark iff*

$$\pi(\eta) < \frac{-v_0}{v_1 - v_0}.$$

In that case, positive expected excess return requires illiquidity premia, especially the long-investment premium δ_L, to be sufficiently large.

When the investment is information-sensitive ($\iota = 1$), managerial effort shapes capital appreciation, making shirking more costly to investors. However, because the redemption cap weakens the manager's effort incentives, the fund cannot rely on capital appreciation alone. Outside this limiting case, expected liquidation burdens also depress returns. The effect of the redemption cap λ is non-monotone: a tighter cap weakens effort incentives by limiting $\Psi(\eta,\lambda)$, but it also changes expected liquidation burdens and therefore alters the optimal long allocation.

Equity is a canonical example of an information-sensitive security. As the residual claimant on the value of the firm, equity is directly affected by changes in the valuation of the firm's assets. Therefore, marginal changes in information about the firm's fundamentals have a direct, immediate effect on the value of its equity.

Beyond asset characteristics, the composition of the fund's capital base also shapes performance. We collect all comparative statics with respect to η in the final prediction. A useful decomposition is

$$\frac{d\mathcal{V}^*}{d\eta} = \underbrace{\pi(\eta)\,\frac{\partial A_1(\ell;\eta)}{\partial\eta}\bigg|_{\ell=\ell^*} + (1-\pi(\eta))\,\frac{\partial A_0(\ell;\eta)}{\partial\eta}\bigg|_{\ell=\ell^*}}_{\text{liquidity transformation channel}} + \underbrace{(A_1^* - A_0^*)\frac{d\pi(\eta)}{d\eta}}_{\text{incentive channel}},$$

where $A_e(\ell;\eta) \equiv \mathrm{E}_z[R_2(\ell, m_e(z)) \mid e]$ and $A_e^* \equiv A_e(\ell^*;\eta)$. The first term captures how informed capital changes expected liquidation burdens across effort states; the second captures how informed capital affects performance by changing effort incentives.

PREDICTION 3. *If the investment is information-insensitive ($\iota = 0$), the incentive channel is inactive and*

$$\frac{\partial A_1(\ell;\eta)}{\partial\eta}\bigg|_{\ell=\ell^*} = \frac{\partial A_0(\ell;\eta)}{\partial\eta}\bigg|_{\ell=\ell^*} = \frac{d\mathcal{V}^*}{d\eta} \geq 0,$$

so higher informed capital weakly raises expected excess returns through the liquidity transformation channel.

If the investment is information-sensitive ($\iota = 1$), both the liquidity transformation and incentive channels are active. The incentive channel is weakly positive and $\frac{\partial A_1(\ell;\eta)}{\partial\eta}\Big|_{\ell=\ell^*} \geq 0$. However, the sign of $\frac{\partial A_0(\ell;\eta)}{\partial\eta}\Big|_{\ell=\ell^*}$ is ambiguous, so the overall sign of $d\mathcal{V}^*/d\eta$ is ambiguous in general. The effect is weakly positive when equation (IA.2)*, specified in the proof, holds.*

Prediction 3 highlights a distinctive feature of the interval fund structure under this rebalancing specification: investor sophistication affects performance not only through the incentive channel, but also through the liquidity transformation channel. More patient capital can improve fund performance, as also shown by Maurin et al. (2023). Empirically, the prediction maps to a stronger positive association between investor sophistication and risk-adjusted performance when the incentive channel complements the favorable liquidity transformation channel.

5 Risk-Adjusted Performance

Interval funds present a trade-off for investors, particularly retail ones. Their structured liquidity enables exposure to high-yield, illiquid assets: Choi et al. (2025) show that actively managed bond funds outperform benchmarks by taking on greater exposure to illiquid securities. At the same time, investors in interval funds may face greater agency frictions due to weaker managerial incentives, and concerns have emerged in the media that retail participants may be harmed by adverse selection and high fees.[21]

In this section, we estimate interval funds' alphas using multi-factor models that allow for heterogeneous risk exposures across funds and asset classes. For retail investors with broad exposure to public markets, these alphas represent their marginal utility gain from increasing their allocation to interval funds (Gibbons et al., 1989; Jensen, 1968, 1969) in a mean-variance framework. They therefore provide a measure of the welfare gain associated with the democratization of private assets.

5.1 Unsmoothing Returns

Interval funds frequently invest in illiquid assets whose valuations are based on appraisals or infrequent market transactions rather than continuous market prices. This practice may induce serial correlation in reported returns, a phenomenon known as return smoothing (Getmansky et al., 2004).[22] Ercan et al. (2025) find that buyout and venture capital funds with the most smoothed NAVs perform worse in the future. Thus, we employ two standard unsmoothing procedures (Couts et al., 2024; Geltner, 1991, 1993; Getmansky et al., 2004): the moving average (MA) approach and the autoregressive (AR) approach.[23] We report summary statistics on raw and unsmoothed returns

[21] Wall Street Journal: https://www.wsj.com/finance/investing/the-fees-on-these-funds-will-leave-you-high-and-dry-4b556475.

[22] Figure IA.4 in the Internet Appendix shows the one-month autocorrelation of returns for individual funds across various asset classes. Interestingly, raw returns of Credit, PE/VC, and Real-Asset interval funds display a surprisingly low level of autocorrelation compared to other investment vehicles, such as hedge funds and private commercial real estate funds (Couts et al., 2024; Couts and Goncalves, 2025).

[23] Couts et al. (2024) advocate for using a three-step unsmoothing procedure, which requires unsmoothing returns for the average performance of each investment category and then for individual deviations from the average. In the paper, we include results using standard (one-step) unsmoothing procedures. Because interval funds are a growing investment vehicle, our panel is highly unbalanced, with periods in which some of the smallest categories include a single fund. In this context, deviations from the category average would not provide an economically meaningful measure of fund-specific return smoothness.

in Table IA.3 of the Internet Appendix.

In the MA approach, the reported returns are modeled as a weighted average of current and past true returns. Specifically, the observed return R_t^o is expressed as:

$$R_t^o = \theta_0 R_t^* + \theta_1 R_{t-1}^* + \theta_2 R_{t-2}^* + \cdots + \theta_N R_{t-N}^*$$

where R_t^* represents the true (unobserved) return at time t, and the weights $\{\theta_j\}$ satisfy $\sum_{j=0}^{N} \theta_j = 1$ and $\theta_j \in [0,1]$ for all j. As in Couts et al. (2024) and Getmansky et al. (2004), we use an MA(3) model with $N = 3$ lags and follow the methodology in Couts et al. (2024) to compute unsmoothed returns.

The AR approach provides an alternative perspective on return smoothing by modeling the reported return as a function of its own past values. Under this framework, the observed return follows an autoregressive process:

$$R_t^o - \mu = \phi_1 (R_{t-1}^o - \mu) + \phi_2 (R_{t-2}^o - \mu) + \cdots + \phi_N (R_{t-N}^o - \mu) + \varepsilon_t$$

where μ is the mean return for the fund, $\{\phi_j\}$ are the AR coefficients, and ε_t represents the true innovation (unsmoothed shock) at time t. We follow Couts et al. (2024) in using an AR(2) process with $N = 2$ lags, and we implement their estimation methodology.

5.2 Alphas and Factor Exposures

We estimate share-class level factor regressions of the form

$$R_{it} - r_t = \alpha_i + \beta_i' F_{it} + u_{it}, \tag{4}$$

where R_{it} are annualized returns, r_t is the risk-free rate, and F_{it} is a vector of the factors described above. We consider three return specifications: raw returns, MA-unsmoothed returns, and AR-unsmoothed returns. Because we estimate a total of 12 parameters (alpha and 11 factor loadings), we require at least 24 months of data for each fund.

Figure 6 presents average alpha estimates by asset class.The results reveal substantial heterogeneity in risk-adjusted performance across interval fund categories. The degree of cross-category dispersion stands in sharp contrast to the mutual fund literature, where average alphas are close to zero across nearly all investment styles (Fama and French, 2010), and to the hedge fund literature, where the cross-sectional distribution of alphas is narrower once returns are properly unsmoothed (Couts et al., 2024). Qualitatively, the findings are insensitive to return unsmoothing, suggesting that potential return smoothing does not materially affect our performance conclusions.

Credit funds emerge as strong and consistent performers, with average annualized alphas exceeding 3% across the three return specifications. These magnitudes are economically large relative to the broader private debt landscape. Erel et al. (2024) find that traditional private debt drawdown funds do not outperform equity and debt benchmarks after fees, while Robinson and Wallskog (2026) document that BDCs—the closest institutional comparator to private credit interval funds—generate meaningful risk-adjusted returns in middle-market lending. Importantly, the performance is consistent across funds. As shown in Figure IA.5 in the Internet Appendix, over 60% of private credit funds exhibit statistically significant alphas with the same sign as the category mean using raw returns. This rate far exceeds 5%, which is the upper bound on the probability of a statistically significant estimate with the same sign as the average estimate under the null hypothesis of zero abnormal performance.

In Figure 6, PE/VC funds also show positive average alphas exceeding 3.50% across specifications, with 27–35% exhibiting statistically significant performance consistent with the category mean (Figure IA.5). Real-asset funds exhibit positive, although more modest performance on average, never exceeding 2% average alpha across specifications. Multi-asset funds display even more modest alphas, although they remain positive.

In stark contrast, liquid-strategy funds display negative average alphas across all specifications of approximately -6%. Over 70% of the liquid-strategy funds exhibit negative and statistically significant performance. The magnitude of this underperformance exceeds that typically documented for actively managed equity mutual funds, where average net-of-fee alphas range from -0.60% to -1.60% per year (Fama and French, 2010; Barras et al., 2010). The worst performance of liquid-strategy interval funds likely reflects the additional costs of the interval fund structure—including higher expense ratios and weaker incentive alignment—without the offsetting illiquidity premium that benefits more illiquid strategies, such as credit, PE/VC, and real assets (Aragon, 2007).

Figure 7 summarizes the factor exposures of interval funds across investment categories using index-based factors. Each panel displays mean factor loadings from fund-level regressions, with a color gradient indicating the share of funds whose statistically significant factor loadings have the same sign as the category mean, a measure of the consistency of factor exposures within each category.

The heatmaps reveal economically intuitive patterns. Credit funds display strong negative loadings on the risk-free rate, term-spread, credit-spread, and high-yield spread factor, reflecting their credit-intensive strategies (because fixed-income factors are defined as changes in yields or spreads, a negative loading on those factors represents a long position in fixed-income assets). Real-asset funds demonstrate the largest positive loading on the real assets factor, confirming their orientation toward real estate and infrastructure. Liquid-strategies funds show exposures to a variety of factors, including long positions in small caps and small-cap value stocks and a tilt toward large growth stocks. They also reveal some fixed-income exposures. Multi-asset and PE/VC funds generally exhibit modest and mostly insignificant factor exposures, reflecting their diversified strategies.

5.3 Performance in a Retirement Portfolio

To complement the baseline factor-regression evidence, we run a retirement-allocation exercise. The goal is to assess whether interval funds can improve the risk-return trade-off of a realistic retirement portfolio built around a target-date fund. In particular, for each interval fund i and target-date benchmark fund b from the Vanguard Target Retirement series, we ask whether allocating a modest share of one's retirement portfolio to the interval fund improves risk-adjusted performance.

Because policymakers and asset managers are currently proposing to democratize alternative assets by including them in retirement plans, target-date funds represent a natural benchmark to evaluate the risk-return trade-off of interval funds. Moreover, by benchmarking interval funds against target-date funds, we evaluate their performance against an investable and diversified portfolio, as advised by Berk and Van Binsbergen (2015).

We evaluate practical allocation sizes $w \in \{5\%, 10\%, 20\%\}$ to the interval funds and construct blended portfolio with returns

$$R_{ibt}^{\text{blend}}(w) = w R_{it}^{\text{IF}} + (1-w) R_{bt}^{\text{BM}},$$

where R_{it}^{IF} is the return of interval fund i in month t and R_{bt}^{BM} is the returns of the target-date fund b in month t.

Using each fund-benchmark pair, we compute the Sharpe ratio of the blended portfolio, $SR_{ib}^{\text{blend}}(w)$, and the Sharpe ratio of the target-date fund, SR_{b}^{BM}. We then convert the improvement in Sharpe ratio of the blended portfolio into a *fixed-weight alpha* following the methodology in Brown et al. (2024), Couts and Goncalves (2025), and Goncalves et al. (2025):

$$\alpha_{ib}^{\text{FW}}(w) = \text{sign}\Big(SR_{ib}^{\text{blend}}(w) - SR_{b}^{\text{BM}}\Big)\, \sigma(\hat{\varepsilon}_{ib}) \sqrt{\left| \big(SR_{ib}^{\text{blend}}(w)\big)^2 - \big(SR_{b}^{\text{BM}}\big)^2 \right|}, \tag{5}$$

where $\sigma(\hat{\varepsilon}_{ib})$ is the standard deviation of the residuals in a regression of the excess return of interval fund i on the excess return of the target-date fund b. Equation (5) converts the Sharpe-ratio improvement into annualized return units (alpha) by scaling with residual volatility, so the effect is directly comparable to the alpha estimates reported elsewhere in the paper. Moreover, unlike standard alpha measures, fixed-weight alphas are less susceptible to short-sample bias (Brown et al., 2024; Couts and Goncalves, 2025; Goncalves et al., 2025)

Figure 8 reports raw-return results for Credit, PE/VC, and Real Asset funds.[24] The exercise indicates that adding interval fund allocations can improve retirement-portfolio risk-adjusted performance, but the magnitude differs sharply by category. In raw returns, average fixed-weight alpha rises with the interval fund weight from 5% to 20% in both credit (up to 0.8% annualized) and PE/VC (up to 1.9% annualized), with PE/VC delivering the largest gains. Real Asset improvements are much smaller on average and are not uniformly positive across target-date vintages. Internet Appendix Figures IA.6 and IA.7 show the same broad ranking and weight-sensitivity under MA-unsmoothed and AR-unsmoothed returns: PE/VC remains strongest, credit remains positive, and real asset remains modest.

5.4 Alphas of Asset-Class Level Portfolios

We next aggregate fund returns at the asset-class level by forming equal-weighted portfolios for each asset class, then estimating regression (4) for each portfolio. Table 4 reports the results. We use Newey and West (1987) standard errors with the Newey and West (1994) automatic lag-selection rule.

Credit funds demonstrate the most robust aggregate performance. Equal-weighted portfolios deliver annualized alphas of 4.472% (raw returns), 4.366% (MA-unsmoothed), and 4.347% (AR-unsmoothed), all statistically significant at the 1% level. The high R-squared values indicate that the factor model explains substantial return variation, making the detected alphas particularly noteworthy. These aggregate alphas substantially exceed those documented for institutional asset managers across all asset classes, such as the 31 basis points net of fees as documented by Gerakos et al. (2021). However, they are consistent with the presence of an illiquidity premium (Aragon, 2007; Ang et al., 2013) and the specialized lending capabilities emphasized by Robinson and Wallskog (2026).

PE/VC funds display positive alphas across all three specifications: 4.842% (raw), 4.255% (MA-unsmoothed), and 4.290% (AR-unsmoothed). The raw return estimate is statistically significant at the 1% level, while the MA-unsmoothed and AR-unsmoothed estimates are significant at the 5% level. The lower R-squared values for private equity indicate substantial idiosyncratic risk in these strategies.

Real-asset funds deliver positive estimated alphas, but they are statistically significant only for raw returns, and marginally so. Equal-weighted portfolios show annualized alphas of 2.086% (raw), 1.917% (MA-unsmoothed), and 2.103% (AR-unsmoothed).

Liquid strategies funds, which include listed equity, warrant particular attention. Equal-

[24] Because of their lack of risk-adjusted performance, we omit results using liquid strategies and multi-asset funds in retirement portfolios. These results are available upon request. In Figures IA.6 and IA.7, we report results for Credit, PE/VC, and Real Asset funds using MA-unsmoothed and AR-unsmoothed returns.

weighted portfolios exhibit negative alphas of -6.109% (raw), -6.546% (MA-unsmoothed), and -6.417% (AR-unsmoothed), all statistically significant at the 1% level. This pattern suggests that the typical liquid-strategy interval fund severely underperforms, raising questions about the suitability of interval fund structures for liquid strategies.

5.5 Cross-Sectional Evidence: Fama–MacBeth Results

In a final set of tests, we estimate abnormal returns while accounting for heterogeneity in factor exposures across funds. We use Fama and MacBeth (1973) regressions with asset class-specific intercepts. We use the full-sample loadings obtained by estimating equation (4). We winsorize loadings on each individual factor at the 1% and 99% level to mitigate the influence of outliers, as standard in the literature (Ang et al., 2006; Knez and Ready, 1997; Levi and Welch, 2020).

Columns (1)–(3) of Table 5 implement Fama and MacBeth (1973) regressions, estimating cross-sectional regressions of fund returns on their factor loadings for each time period and then averaging the coefficients over time. We use Newey and West (1987) standard errors with the Newey and West (1994) automatic lag-selection rule. The reported asset class-specific intercepts can be interpreted as the excess return of a zero-beta fund (that is, a fund with no systematic risk exposures) in each asset class, capturing the component of performance that cannot be attributed to systematic factor exposures.

The Fama-MacBeth results strongly corroborate our previous findings. Credit funds exhibit annualized abnormal returns of 4.024% (raw returns), 3.768% (MA-unsmoothed), and 4.118% (AR-unsmoothed), all statistically significant at the 1% level. These estimates represent the performance of a hypothetical zero-beta private credit fund and thus isolate the value added beyond any systematic risk exposures. In contrast, liquid strategies exhibit deeply negative abnormal returns, although not statistically significant. Multi-asset, PE/VC, and real-asset funds continue to display positive estimates of abnormal returns, but without consistent statistical significance.

Columns (4)–(6) present results separately for retail share classes, and columns (7)–(9) present share classes for institutions and high-net-worth investors. Results reveal that the full-sample finding of outperformance for credit funds exists for both types of share classes, but is stronger in magnitude for institutional share classes. For credit funds, annualized abnormal returns of 2.070% (raw), 2.631% (MA-unsmoothed), and 2.097% (AR-unsmoothed) for retail share classes, and 4.505% (raw), 4.367% (MA-unsmoothed), and 4.235% (AR-unsmoothed) for institutional share classes.

For the PE/VC category, unsmoothed outperformance is not significant for retail share classes. Instead, performance in this asset class is entirely attributable to institutional share classes, which display abnormal returns of 10.280% (raw), 9.900% (MA-unsmoothed), and 9.423% (AR-unsmoothed).

Overall, the pattern is consistent with the theoretical prediction that semiliquid structures generate value primarily in information-insensitive, illiquid asset classes, where fund managers can exploit the longer investment horizons enabled by restricted redemptions (Aragon, 2007). Conversely, asset managers struggle to add value in liquid, efficiently priced markets.

6 Agency Frictions

The performance analysis of Section 5 showed that interval funds earn large positive risk-adjusted returns in credit and consistently negative ones in Liquid Strategies. We now ask whether these patterns are consistent with the agency frictions embedded in the interval fund structure. The tests in this section first benchmark interval funds against matched investable vehicles and then examine three channels through which the interval fund structure can affect performance: the composition of returns between NAV appreciation and distributions, the investor clientele, and contractual incentives.

6.1 Investable Benchmarks

Our main specification is a share-class-by-month panel regression of the form

$$R_{i,t} = \alpha + \beta \,\text{Interval}_i + \gamma' X_{i,t-1} + \phi_{b(i),t} + \varepsilon_{i,t}, \tag{6}$$

where $R_{i,t}$ is the annualized net return of share class i in month t. The variable Interval_i is an indicator equal to one for interval fund share classes and zero for index ETFs, which serve as the omitted comparison group. $X_{i,t-1}$ is a vector of lagged share-class characteristics that includes the log of share-class TNA, the log of aggregate fund family TNA within the same fund type, the log of one plus share-class age in years, an indicator for fund-of-funds status, and one-month lagged net flows. $\phi_{b(i),t}$ denotes benchmark-by-month fixed effects, where $b(i)$ is the Morningstar Category Index associated with share class i, and absorbs time-varying style exposures common to all funds that track the same index in the same month. Standard errors are clustered at the month-year level. As in Section 5, we estimate equation (6) separately for each of the five consolidated investment categories (Credit, Liquid Strategies, Multi-Asset, PE/VC, and Real Asset). In each regression, we restrict the interval fund sample to the named category but keep the full comparison group, since the benchmark-by-month fixed effects already identify the Interval coefficient from ETFs that track the same Morningstar Category Index as those interval funds.

The coefficient of interest is β, which measures the average net-return differential between interval fund share classes and comparable index ETFs that track the same benchmark index in the same month, after partialling out the controls. Under the null that the interval fund structure is neutral for investor returns, $\beta = 0$ in every category. The theoretical framework in Section 4 predicts a conditional sign pattern. In illiquid, information-insensitive markets, the interval fund structure allows managers to earn liquidity premia, so we expect $\beta > 0$. In information-sensitive markets, the sign depends on whether liquidity premia are large enough to offset the weaker incentives created by limited redemption pressure. Thus, the model predicts $\beta < 0$ when liquidity premia are insufficiently large, as is likely for liquid strategies.

Table 6 reports the results. In credit, interval fund share classes outperform benchmark-matched index ETFs by 3.457% per year ($t = 3.61$), significant at the 1% level. In Liquid Strategies, which capture listed equity and liquid alternatives, they underperform by 4.550% per year ($t = -1.93$), significant at the 10% level. The Multi-Asset, PE/VC, and Real Asset estimates are statistically indistinguishable from zero. The Credit and Liquid Strategies results match the sign pattern predicted by the framework. Appendix Table IA.4 shows that the credit premium is robust at the portfolio level using annualized gross returns ($\beta = 3.612\%$, $t = 4.96$), and Appendix Table IA.5 shows that the pattern survives when active ETFs replace index ETFs as the comparison group. These baseline estimates anchor the remaining tests.

6.2 Sources of Returns: NAV versus Distributions

A positive interval fund premium in credit can reflect two distinct channels. Managers can generate value by selecting undervalued securities, which would show up as higher NAV appreciation; or they can harvest an illiquidity or credit premium, which would accrue primarily as higher income on illiquid, information-insensitive holdings (note that income also includes PIK income and amortization of any original issue discount). Our framework predicts the second channel: interval fund managers, facing weak market-based incentives, tilt toward assets with high distribution yields rather than toward security selection.

To retain Regulated Investment Company (RIC) status,[25] interval funds must pay out 90% of their income to investors, thus the majority of income translates into distributions. We decompose the annualized net return into an NAV return, defined as the annualized percentage change in price per share, and a distribution return, defined as the residual (annualized net return minus NAV return). We then re-estimate equation (6) with each component as the dependent variable. Panel A of Table 7 reports the NAV-return regressions and Panel B the distribution-return regressions. In credit, the NAV-return coefficient is -0.221 and statistically indistinguishable from zero, while the distribution-return coefficient is 3.678%, significant at the 1% level. Interval fund managers in credit do not generate value through asset appreciation; the entire net premium accrues

[25] See 26 U.S.C. § 852(a)(1), which conditions a RIC's pass-through treatment on the dividends-paid deduction equaling or exceeding 90% of its investment company taxable income for the taxable year. Available at https://www.law.cornell.edu/uscode/text/26/852.

through higher distribution yields on illiquid, income-generating holdings. Liquid Strategies tells a different story: interval fund managers still collect a positive distribution premium (4.779%, significant at 1%), but NAV returns are sharply negative (−9.329%, significant at 1%) and more than offset it, producing the net underperformance.

These patterns are consistent with the view that interval fund managers harvest illiquidity premia rather than selecting undervalued securities, and with the broader literature showing that the illiquidity premium in credit markets accrues primarily as yield (Amihud and Mendelson, 1986; Aragon, 2007; Ang et al., 2013). They also suggest that the underperformance of interval funds in Liquid Strategies is a failure of active management in information-sensitive markets, not a failure to collect the yield.

6.3 Clientele Effects

A central prediction of our framework is that a more sophisticated investor base helps sustain managerial incentives when redemption pressure is limited. We test this prediction by interacting the interval fund indicator with clientele type in a specification that extends equation (6):

$$R_{i,t} = \alpha + \beta_I \left(\text{Interval}_i \times \text{Inst/HNWI}_i\right) + \beta_R \left(\text{Interval}_i \times \text{Retail}_i\right) + \gamma' X_{i,t-1} + \phi_{b(i),t} + \varepsilon_{i,t}, \tag{7}$$

where the dependent variable $R_{i,t}$ and the controls $X_{i,t-1}$ and fixed effects $\phi_{b(i),t}$ are defined as in equation (6). The indicator Inst/HNWI_i equals one for share classes that carry an institutional label or require a minimum investment above \$25,000, and zero otherwise. The indicator Retail_i equals one for share classes with retail-oriented labels and a minimum investment at or below \$25,000, and zero otherwise. Index ETFs serve as the omitted comparison group.

Our coefficients of interest are β_I and β_R, which capture the interval fund premium separately for institutional/HNWI share classes and for retail share classes. Panel A of Table 8 reports the results. In credit, institutional share classes of interval funds outperform benchmark index ETFs by 4.188% per year and retail share classes by 3.010% per year, both significant at the 1% level. The institutional premium exceeds the retail premium by 1.2 percentage points, consistent with sophisticated investors helping sustain the liquidity-transformation strategy in credit funds.

To sharpen the test, we exploit portfolio-level variation in the investor base. Interval fund managers often structure a single portfolio with share classes targeting different clienteles, so retail investors in a multi-clientele fund co-invest alongside institutional and high-net-worth investors within the same portfolio. Panels B and C of Table 8 estimate equation (7) separately for single-clientele interval funds (Panel B) and for multi-clientele interval funds (Panel C). In single-clientele credit funds, retail share classes earn a point estimate of only 0.801% relative to index ETFs, statistically indistinguishable from zero, while institutional share classes outperform by 3.072% (significant at 1%). In multi-clientele credit funds, both clienteles outperform by similar magnitudes: 4.809% for retail share classes and 5.452% for institutional share classes, both significant at the 1% level. Retail investors, therefore, capture the illiquidity premium only when they co-invest alongside sophisticated investors within the same portfolio. In credit funds, this result is best interpreted through the liquidity-transformation channel: sophisticated co-investors provide a more stable funding base, allowing the same portfolio to maintain illiquid exposure and leverage rather than shifting toward liquid assets to accommodate redemptions.

The Morningstar classification relies on share class labels and minimum investment thresholds, which are proxies for investor sophistication: some share classes with retail-oriented labels and low minimums are, in fact, restricted to accredited investors through prospectus language, while some high-minimum share classes simply target wealthy but otherwise unrestricted investors. To further refine our proxies for sophistication, Appendix Table IA.6 re-estimates equation (7) under an alternative classification that uses the explicit investor eligibility restrictions disclosed in EDGAR prospectus filings. This classification separates three groups: true retail share classes with no accreditation requirement, accredited-investor share classes, and institutional/HNWI share classes. The pattern from Table 8 holds under this sharper classification. In single-clientele credit funds, accredited and institutional share classes earn premia of 2.969% and 2.342% relative to index ETFs (both significant at 1%), while true retail share classes earn an insignificant 1.147%. In multi-clientele credit funds, all three share-class types capture the premium (4.817% for retail, 5.441% for institutional, and 5.103% for accredited, all significant at 1%). The EDGAR test confirms that the credit premium accrues only to investors in share classes with documented sophistication, or to retail investors who co-invest alongside them in a multi-clientele portfolio.

As a placebo, we compare the performance of interval funds with other nontraded semiliquid funds, which tend to have more accreditation requirements. Overall, we find no evidence that interval funds outperform them. If anything, they tend to underperform tender-offer funds (in Multi-Asset and PE/VC categories) and REITs (in Real Assets).

6.4 Contractual Features

If interval fund performance depends on the trade-off between liquidity transformation and weak redemption-based discipline, contractual features can affect performance by shaping both managerial incentives and the fund's ability to sustain exposure to high-yield illiquid assets. We test whether interval funds that charge an incentive fee[26] earn a higher gross return by augmenting

[26] A second contractual feature that should affect performance is redemption frequency. Appendix Table IA.8 reports the relation between performance and the length of the redemption window. Because most interval funds adopt the statutory default of quarterly repurchase offers, variation in redemption frequency is limited. However, the sign in credit funds is consistent with longer redemption windows, weakening the liquidity-transformation channel and lowering performance.

equation (6) with an incentive-fee interaction and estimating the regression at the portfolio level using annualized gross returns: [27]

$$R_{p,t}^{\text{gross}} = \alpha + \beta_1 \,\text{Interval}_p + \beta_2 \,(\text{Interval}_p \times \text{IncentiveFee}_p) + \gamma' X_{p,t-1} + \phi_{b(p),t} + \varepsilon_{p,t},$$

where $R_{p,t}^{\text{gross}}$ is the annualized gross return of portfolio p in month t. The indicator IncentiveFee_p equals one for interval funds that charge an incentive fee and zero otherwise. We set IncentiveFee_p to zero for index ETFs, so the comparison group is the pooled sample of index ETFs. Controls $X_{p,t-1}$ and benchmark-by-month fixed effects $\phi_{b(p),t}$ are defined as in equation (6), aggregated to the portfolio level. Standard errors are clustered at the month-year level.

Our coefficients of interest are β_1 and β_2. The first measures the premium of an interval fund portfolio without an incentive fee relative to index ETFs tracking the same benchmark, and the second measures the additional premium earned by interval fund portfolios that charge an incentive fee. Under the null that contractual incentives are irrelevant, $\beta_2 = 0$. Under the alternative that incentive fees improve alignment and support the fund's income-generating credit strategy, $\beta_2 > 0$.

Table 9 reports the results. In credit, $\beta_1 = 3.213\%$ per year ($t = 4.42$) and $\beta_2 = 2.248\%$ per year ($t = 4.21$), both significant at the 1% level. Credit interval funds that charge an incentive fee outperform comparable index ETFs by $\beta_1 + \beta_2 = 5.461\%$ per year, roughly 70% higher than the premium earned by credit interval funds without such a fee. Income-based performance compensation provides a meaningful incremental source of alignment in credit funds, complementing the clientele evidence and supporting the fund's ability to earn income from illiquid credit assets.

7 Portfolio Illiquidity and Liquidity Transformation

By limiting redemptions to periodic windows, interval funds can avoid the fire sales and fragility that afflict open-end funds holding illiquid assets (Shleifer and Vishny, 2011; Coval and Stafford, 2007; Chen et al., 2010; Ma et al., 2022), but they also limit investors' ability to discipline managers through redemptions (Stein, 2005; Dangl et al., 2008). Sections 5 and 6.3 showed that the net effect of this trade-off varies across asset classes and clienteles. This section uses portfolio-level data to examine the underlying portfolio mechanics. We first document the cross-section of leverage, liquidity, and asset allocation at the interval fund portfolio level. We then test whether leverage and portfolio illiquidity are associated with higher returns in credit funds, where the illiquidity premium is largest. Finally, we examine how interval fund managers balance liquidity offered to investors against portfolio illiquidity, and how this trade-off varies by clientele.

[27] We use gross returns because, although explicit contracts should improve gross performance, they may not necessarily allocate the increased performance to investors.

7.1 Portfolio Holdings and Illiquidity Exposure

The N-PORT filings deliver quarterly portfolio composition, leverage, and fair-value hierarchy shares for registered investment companies. The reporting requirement became effective for interval funds in 2019, so the holdings panel runs from October 2019 to March 2026. We observe total assets under management, total liabilities, and the share of holdings in each level of the ASC 820 fair-value hierarchy. From these data, we construct two measures. *Leverage* is total liabilities as a percentage of AUM. *Liquidity* is a portfolio liquidity index defined as Level 1 plus one-half of Level 2, where Level 1 assets are valued using quoted prices in active markets and Level 2 assets use other observable inputs; Level 3 assets, which require unobservable inputs and typically represent private or hard-to-value holdings, carry zero weight. Related papers (Chen et al., 2010; Goldstein et al., 2017; Giannetti and Kahraman, 2018; Gómez et al., 2024) use market-based measures such as the Amihud illiquidity measure or bid-ask spreads, which are not available for the private and structured holdings that dominate interval fund portfolios.

Table IA.9 in the Internet Appendix reports summary statistics for the interval fund portfolios in the holdings sample, split by clientele. Panel A pools all interval fund portfolios, and Panels B through D restrict to retail-only, institutional-only, and multi-clientele funds, respectively. Portfolio composition differs meaningfully across clienteles. Retail-only interval funds hold an average Level 3 share of 45.11%, far higher than the 25.92% average for institutional-only funds, yet their liquidity indices are comparable (27.70 versus 27.18). The difference is absorbed by Level 2 holdings, which are larger for institutional-only funds (31.30% versus 18.50%). Retail-oriented interval funds, therefore, tilt more heavily toward private or hard-to-value assets. Multi-clientele funds carry both the highest leverage (mean 16.34%) and the highest Level 2 share (42.52%), a pattern consistent with a tilt toward leveraged credit strategies. Figure IA.8 in the Internet Appendix plots average portfolio liquidity and leverage across asset categories. Credit interval funds carry the highest leverage (mean 18%) and the largest Level 2 exposure (58%), together with a substantial Level 3 exposure of 32%.

7.2 Performance, Leverage, and Portfolio Illiquidity

We next ask whether the credit premium documented in Section 5 is stronger in interval funds that take more leverage and hold more illiquid portfolios. Our framework predicts that, in information-insensitive markets, the premium should scale with the fund's ability to harvest the illiquidity premium. That ability is higher when leverage is larger and when portfolio liquidity is lower.

Our specification interacts the interval fund indicator with a portfolio characteristic of interest:

$$R^{\text{gross}}_{p,t} = \alpha + \beta_1 \, \text{Interval}_p + \beta_2 Z_{p,t} + \beta_3 \left(\text{Interval}_p \times Z_{p,t}\right) + \gamma' X_{p,t-1} + \phi_{b(p),t} + \varepsilon_{p,t}, \tag{8}$$

where $R^{\text{gross}}_{p,t}$ is the annualized gross return of portfolio p in month t. The portfolio characteristic $Z_{p,t}$ takes two forms. In Panel A of Table 10, $Z_{p,t} = \text{Leverage}_{p,t}$, the ratio of total liabilities to assets under management expressed as a percentage. In Panel B, $Z_{p,t} = \text{Liquidity}_{p,t}$, the portfolio liquidity index defined above. The indicator Interval_p equals one for interval fund portfolios and zero for index ETFs, which serve as the omitted comparison group. $X_{p,t-1}$ contains the portfolio-level controls (log TNA, log family TNA, log of one plus age, a fund-of-funds indicator, and one-month lagged fund flows), and $\phi_{b(p),t}$ denotes benchmark-by-month fixed effects. Standard errors are clustered at the month-year level. Because N-PORT holdings coverage begins in October 2019, all regressions in this subsection use the subsample from October 2019 through March 2026.

Our coefficient of interest is β_3, which measures how the interval fund premium varies with portfolio leverage in Panel A and with portfolio liquidity in Panel B. Under the null that the interval fund premium does not depend on the fund's exposure to illiquidity, $\beta_3 = 0$. Under the alternative that the premium is driven by harvesting the illiquidity premium, we expect $\beta_3 > 0$ in Panel A (leverage amplifies the premium) and $\beta_3 < 0$ in Panel B (lower liquidity is associated with higher returns).

Panel A reports the leverage interaction. In credit, interval fund portfolios generate significantly higher gross returns than index ETFs unconditionally ($\beta_1 = 4.119\%$, $t = 3.26$, significant at 1%), and the leverage interaction is positive and significant ($\beta_3 = 0.081\%$, $t = 1.88$, significant at 10%). Economically, a one standard deviation increase in the leverage ratio (13.26) is associated with roughly a 1.1% increase in annualized gross returns. Panel B reports the complementary liquidity interaction. In credit, $\beta_1 = 8.323\%$ ($t = 5.38$, significant at 1%) and $\beta_3 = -0.063\%$ ($t = -2.90$, significant at 1%). The negative sign indicates that credit portfolios with lower liquidity (more Level 3 assets) deliver higher gross returns. In economic terms, a one standard deviation decrease in the liquidity index (20.74) implies an increase of approximately 1.3% in annual returns relative to the comparison group. In the other four categories, the leverage and liquidity interactions are not statistically significant, except in the case of multi-asset funds, where the coefficient is negative and weakly significant.

The credit results provide direct evidence that the interval fund premium in Section 5 comes from harvesting the illiquidity premium through leverage and Level 3 exposure, not from category-wide fund effects. The absence of comparable patterns in the other categories is consistent with the framework: in information-sensitive markets, manager effort, not illiquidity exposure, drives returns, and there the interval fund structure weakens incentives rather than helping.

7.3 Liquidity Transformation

We now ask how interval fund managers balance the liquidity they offer to investors against the illiquidity of the portfolio they hold, and whether this trade-off varies across clienteles. Fund managers who promise more frequent or larger redemption windows must either hold more cash, reduce leverage, or tilt toward more liquid assets, any of which erodes the illiquidity premium. A more stable investor base relaxes this constraint.

We measure the liquidity a fund offers by $\text{OfferedAmount}_{p,t}$, defined as the percentage of AUM that the fund makes available for repurchase at each redemption window. Because offered amounts are persistent within a fund, this variable captures the fund's standing redemption policy. We estimate the following specification separately for retail-only, institutional-only, and multi-clientele interval fund portfolios:

$$Y_{p,t} = \alpha + \lambda \, \text{OfferedAmount}_{p,t} + \gamma' X_{p,t-1} + \phi_{b(p),t} + \varepsilon_{p,t},$$

where the dependent variable $Y_{p,t}$ takes three forms. In Panel A of Table 11, $Y_{p,t}$ is the fund's leverage. In Panel B, $Y_{p,t}$ is the fund's portfolio liquidity index. In Panel C, $Y_{p,t}$ is the fund's monthly percentage fund flow. $X_{p,t-1}$ is the same vector of portfolio-level controls used in equation (8), and $\phi_{b(p),t}$ denotes benchmark-by-month fixed effects. Standard errors are two-way clustered at the month-year and portfolio levels. Panels A and B use the October 2019 to March 2026 holdings subsample, while Panel C uses the full January 2010 to March 2026 portfolio panel in which the offered amount is observed but the holdings are not.

Our coefficient of interest is λ, which measures how the fund's leverage, portfolio liquidity, or fund flows covary with the liquidity the fund promises to its investors. Under the null that the liquidity trade-off does not depend on the investor base, λ is identical in the three clientele columns. Under the alternative that a stable investor base relaxes the trade-off, λ differs in sign or magnitude across clienteles.

The results in Table 11 show a sharp split by clientele. In retail-only interval funds, a one-percentage-point increase in Offered Amount is associated with a 2.401-percentage-point decrease in leverage (Panel A, $t = -2.94$, significant at 1%) and a 3.538-percentage-point increase in the liquidity index (Panel B, significant at 5%). Retail-only managers facing more demanding redemption schedules respond by deleveraging and shifting into more liquid holdings. The response is much more muted in institutional-only interval funds: a one-percentage-point increase in Offered Amount is associated with a 0.509-percentage-point *increase* in leverage (Panel A, significant at 1%) and a 0.859-percentage-point increase in the liquidity index (Panel B, significant at 5%). The leverage coefficient flips sign relative to the retail response, which is consistent with institutional managers meeting higher promised liquidity through credit facilities rather than by deleveraging the portfolio. Multi-clientele funds display attenuated and statistically insignificant effects of Offered Amount on both leverage and the liquidity index, consistent with the funding stability that the co-investment of institutional and retail investors provides to the portfolio.

Panel C documents the corresponding effect on fund flows. In retail-only funds, Offered Amount is positively and significantly associated with fund flows ($\lambda = 0.785$, significant at 5%), indicating that retail investors respond to more generous redemption policies by contributing more capital. Institutional and multi-clientele flows show no significant relationship to Offered Amount. Combined with the summary statistics in Table IA.10 in the Internet Appendix, which show that retail share classes tender with greater frequency and greater volatility than institutional share classes, the results paint a consistent picture. Retail investors demand and use liquidity; institutional investors subsidize it.

In sum, interval funds transform liquidity along two margins: a portfolio margin (leverage and asset liquidity) and a funding margin (the composition of the investor base). When the funding base is retail, managers absorb the promised liquidity on the portfolio margin by holding more liquid assets and less leverage, and the illiquidity premium we documented in credit is partly foregone. When the funding base includes institutional or high-net-worth investors, managers can meet the promised liquidity through credit lines and marginal cash without cutting illiquid exposure, and the premium remains intact.

8 Recent Developments

One potential concern is that credit interval funds deliver apparent overperformance because they systematically overvalue their holdings and because they do not mark down their value when they are distressed. Starting in the fourth quarter of 2025, high redemption requests among non-traded BDCs have cast doubts on the quality of their holdings and on whether their values are correctly reported.[28] Current events primarily involved BDCs, but similar concerns may also extend to interval funds. While we look forward to future research specifically investigating whether asset valuations in semiliquid funds are overvalued, we believe that overvalued and stale NAVs are unlikely to drive our results. First, our results are robust to unsmoothing returns, with interval fund returns being only weakly autocorrelated. Smooth returns are an indicator of manipulated NAVs (Couts et al., 2024; Couts, 2025, 2024). Second, our sample includes the Covid period, which was another stressful period in private credit. Figure IA.9 shows that, as one would expect, credit funds delivered negative abnormal returns, from which they later recovered, suggesting that NAVs were revised down during the Covid shock. Abnormal returns were positive in the fourth quarter of 2025 and near zero in the first quarter of 2026. Third, the credit funds in our sample typically originate their own loans, and are less reliant on purchases of secondaries at a discount that are then marked

[28] See, for example: https://www.wsj.com/finance/investing/the-private-credit-party-turns-ugly-for-individual-investors-287356f9, https://www.wsj.com/finance/investing/private-credit-investors-are-cashing-out-in-droves-bc1e8cbc, and https://www.wsj.com/finance/investing/private-credit-warning-signs-flash-after-blue-owl-unloads-1-4-billion-in-assets-02494fab.

up to the sponsor's stated NAV. As shown in section 6.2, credit funds' outperformance originates from distributions, not NAV returns. Fourth, loans are not showing signs of distress as of the end of 2025. Using N-PORT data, Figure IA.10(a) shows the share of loans experiencing default or exercising any option to defer interest payments (including via pay-in-kind, PIK, toggles). This share has always remained well below 4.5% of the interval funds' debt portfolio, and it has declined in 2025. Finally, Figure IA.10(b) shows that even in the last quarter of 2025, net flows to interval funds remained positive. Morningstar data on TNA, shown in Figure 4(b), also illustrate that net flows have also been positive in recent months. This continued inflow is consistent with the idea that interval funds have benefited from a stable funding base, which can support liquidity transformation by reducing the need to liquidate illiquid assets to meet redemptions.

9 Conclusions

We show that interval funds can deliver value to retail investors, but only under specific conditions. Consistent with our theoretical framework, retail investors face a trade-off between agency frictions and liquidity premia. Interval funds create value when they invest in illiquid, information-insensitive assets, especially credit, but destroy value when they invest in liquid, information-sensitive strategies. Consistent with this trade-off, credit interval funds generate risk-adjusted returns exceeding 2% per year across specifications, risk-factor models, and investable benchmarks. Moreover, allocating 5% to 20% of retirement savings to credit interval funds meaningfully improves the risk-return trade-off of retirement portfolios.

Several findings point to the role of agency frictions. First, investors gain primarily through high distribution yields rather than NAV appreciation, suggesting that managers create value mainly by harvesting income from illiquid assets rather than by selecting underpriced securities. Second, retail-oriented interval funds perform as well as accredited-investor-oriented semiliquid funds in credit strategies, but underperform them in more information-sensitive strategies, such as private equity and real assets. Third, retail share classes underperform share classes catering to sophisticated investors. At the same time, retail investors benefit when they co-invest in funds that also serve institutional or high-net-worth investors. These sophisticated investors generate positive externalities for retail investors, especially in credit strategies, by providing a more stable capital base that helps funds sustain liquidity transformation. Fourth, interval funds with incentive fees and more frequent redemption opportunities outperform their peers, further supporting the view that contractual features matter for performance.

We also find evidence consistent with an illiquidity premium. Credit interval funds perform better when they allocate more capital to illiquid assets and use more leverage. However, this benefit comes with a liquidity-transformation trade-off. Retail investors value periodic redemption opportunities, but generous redemption terms can force managers to hold more liquid assets or reduce leverage, thereby weakening their ability to earn the illiquidity premium. Institutional and high-net-worth investors help relax this constraint because they provide more stable capital and make less intensive use of redemption opportunities. As a result, retail investors face a more favorable trade-off when they co-invest alongside sophisticated investors.

These findings inform the ongoing policy debate over retail access to alternative assets and, in particular, the inclusion of semiliquid vehicles in defined-contribution retirement plans, such as 401(k) and 403(b) plans. Policymakers, regulators, and industry participants are actively exploring how to expand access to private and illiquid assets for regular households. Our evidence shows that retail investors can obtain material benefits from this expansion, but access alone is not sufficient. Retail investors gain when fund structures allow them to capture liquidity premia in information-insensitive assets and when sophisticated co-investors strengthen incentives and provide stable capital. They lose value when weak managerial incentives matter most for performance, as in liquid or information-sensitive strategies. Therefore, to successfully democratize private assets, policymakers and industry professionals must do more than simply expand the menu of eligible investments. They must design fund structure, investment mandates, and compensation contracts to strengthen managerial incentives, improve liquidity transformation, and direct portfolio allocation toward asset classes where semiliquid structures are most likely to add value.

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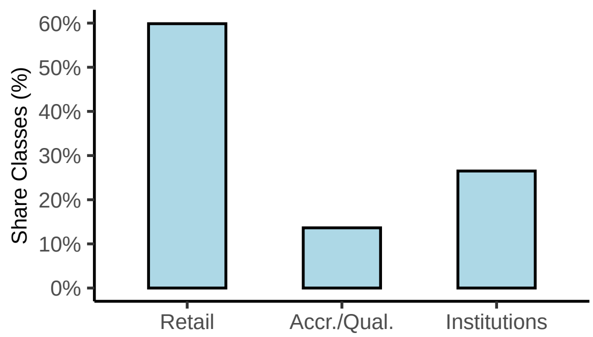
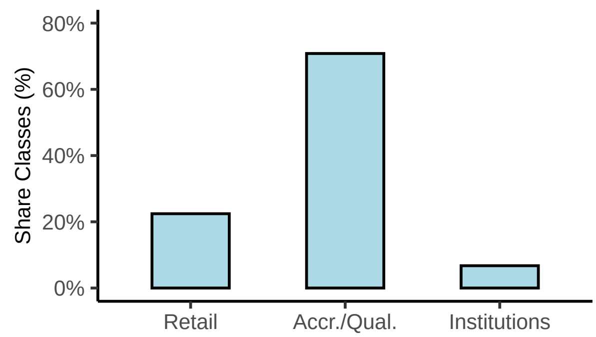
Figures and Tables



(a) Clientele - interval funds



(b) Clientele - tender-offer funds

Figure 1. Distribution of clienteles in interval funds and tender-offer funds. Based on the description in the funds' prospectuses and amendments, share classes are classified as open for retail investors, qualified or accredited investors, or institutional investors. Data are from forms 486BPOS, 486APOS, and N-2 available on EDGAR from 2010 to March, 2026. Based on the language of the forms, we classify the clientele of each share class as *retail*, accredited or qualified (abbreviated as *HNW*, for high net worth), or *institutional*.

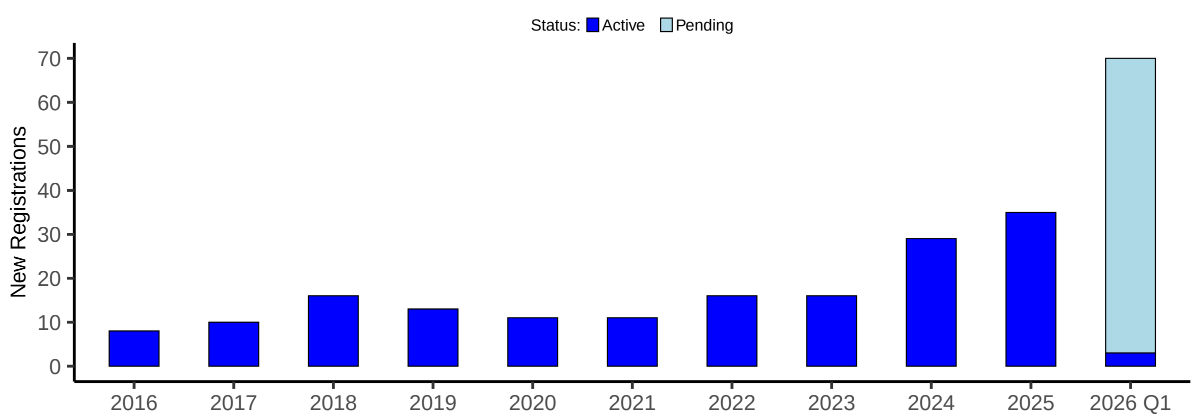
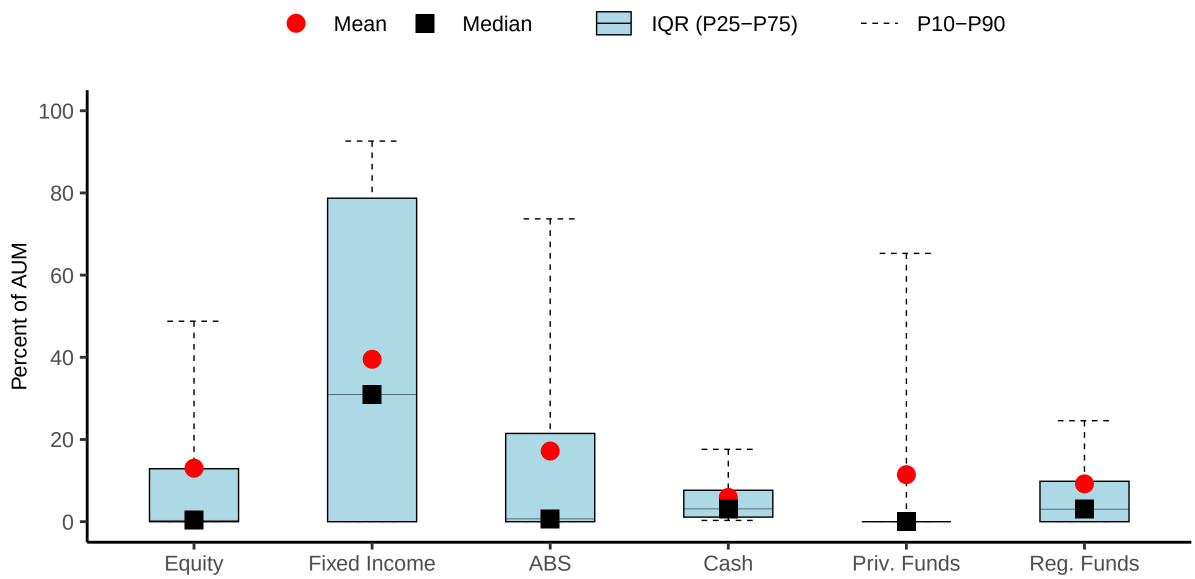
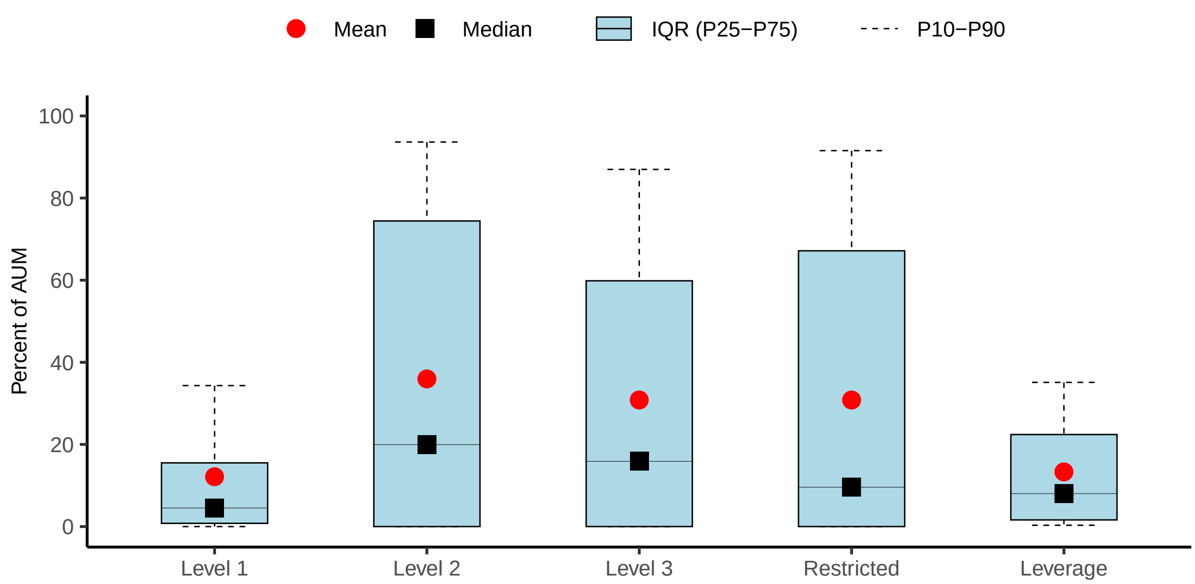


Figure 2. Active interval funds and pending interval fund registrations. The plot shows the number of new interval fund registrations for every year from 2016 to 2025 and for the first quarter of 2026. The plot also shows pending interval fund registrations as of March 31, 2026. Data are from publicly filed initial N-2 forms and are available at https://intervalfundtracker.com/.

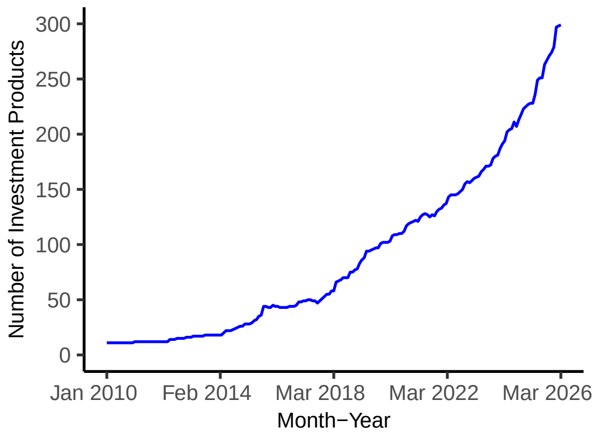


(a) Asset Allocation

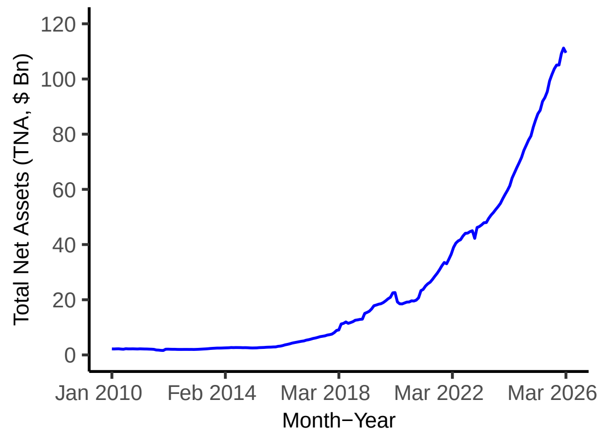


(b) Liquidity, Restrictions, and Leverage

Figure 3. Portfolio composition of interval funds. Panel (a) shows the distribution of asset allocation across the following asset classes: equity, fixed income, asset-backed securities (ABS), cash and cash equivalents, private funds (unregistered investment vehicles), and registered funds. Panel (b) shows holdings distributions across fair value hierarchy levels and restricted securities, and the distribution of leverage. Level 1 assets are valued using quoted prices in active markets and are typically the most liquid; Level 2 assets are valued using observable inputs other than quoted prices, such as dealer quotes or matrix pricing, and are generally less liquid; Level 3 assets are valued using unobservable inputs, such as internal models or management estimates, and are typically the least liquid. Restricted securities are securities with resale restrictions, such as Rule 144A securities or private placements. Leverage is total liabilities as a share of total assets under management (AUM). All variables are expressed as percentages of AUM. The red dot marks the mean, the black square marks the median, the light blue box shows the interquartile range, and whiskers extend to the 10th and 90th percentiles. Data are from N-PORT filings available on EDGAR from October 2019 to March 2026.



(a) Number of Products



(b) Total Net Assets

Figure 4. The figures show the number of investment products and their total net assets for Interval Funds from 2010 to March 2026 in the Morningstar sample. Panel (a) shows the number of interval fund share classes, and Panel (b) shows the total net assets.

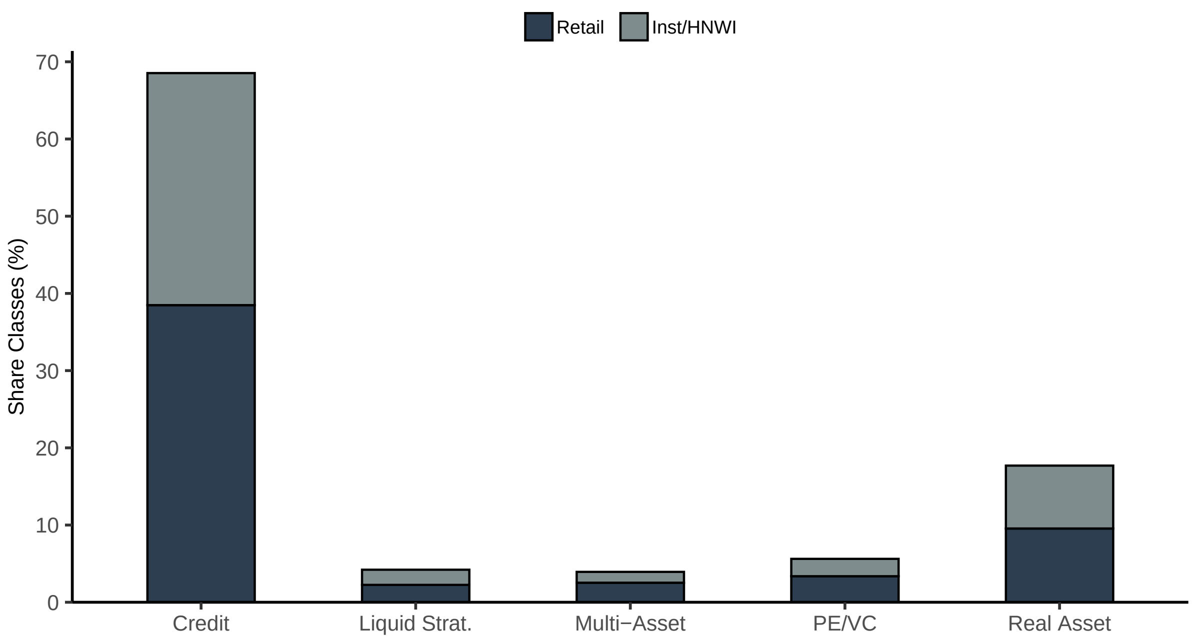


Figure 5. The figure displays the frequency distribution of investment categories and clienteles among the sample of Morningstar interval funds from 2010 to March 2026.

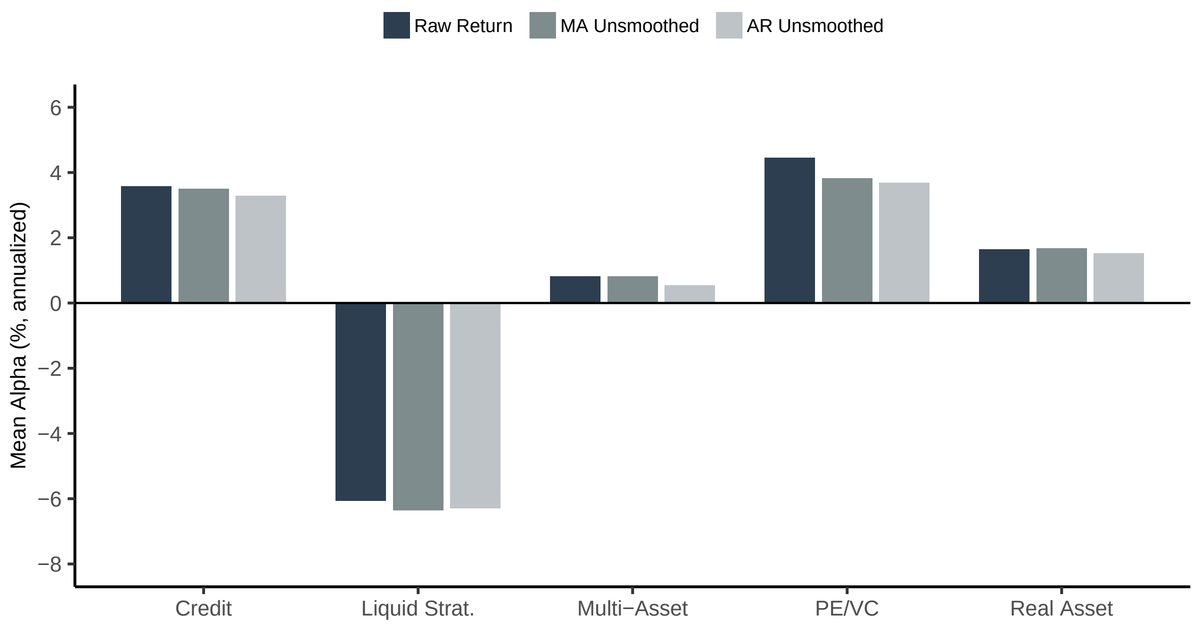
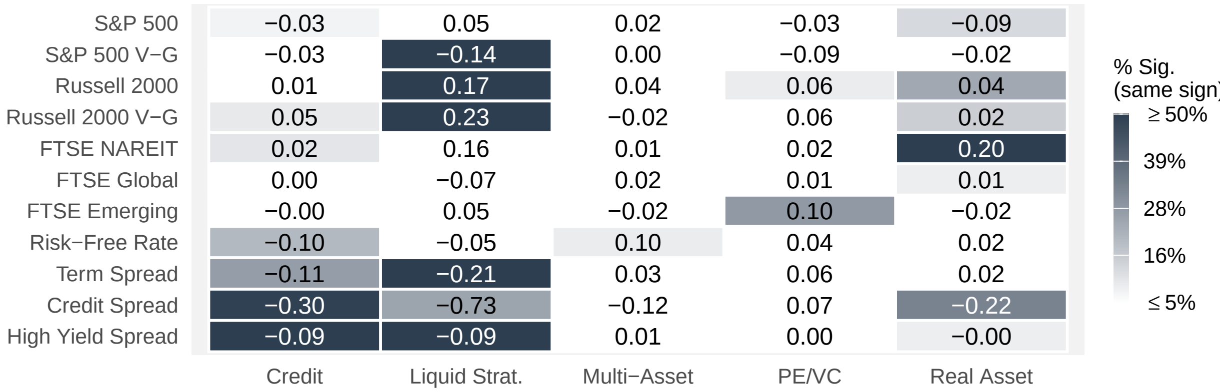
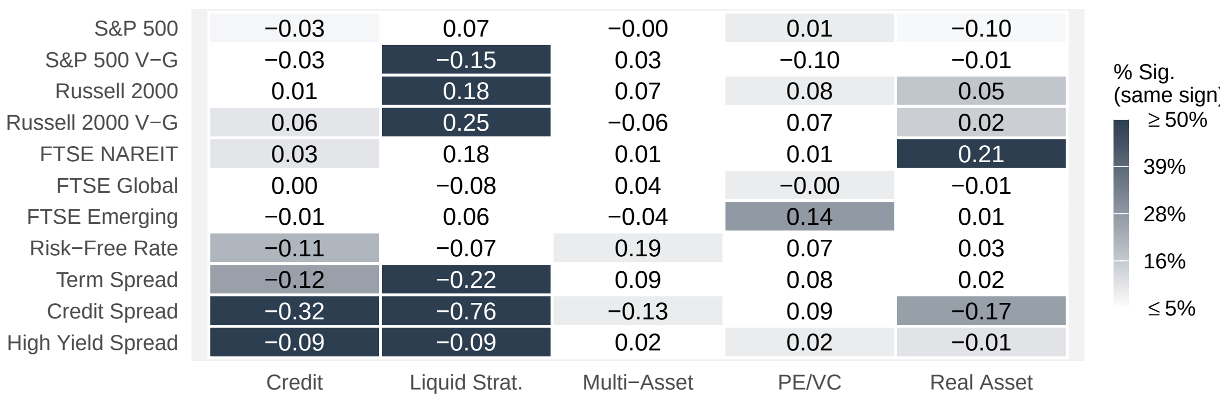


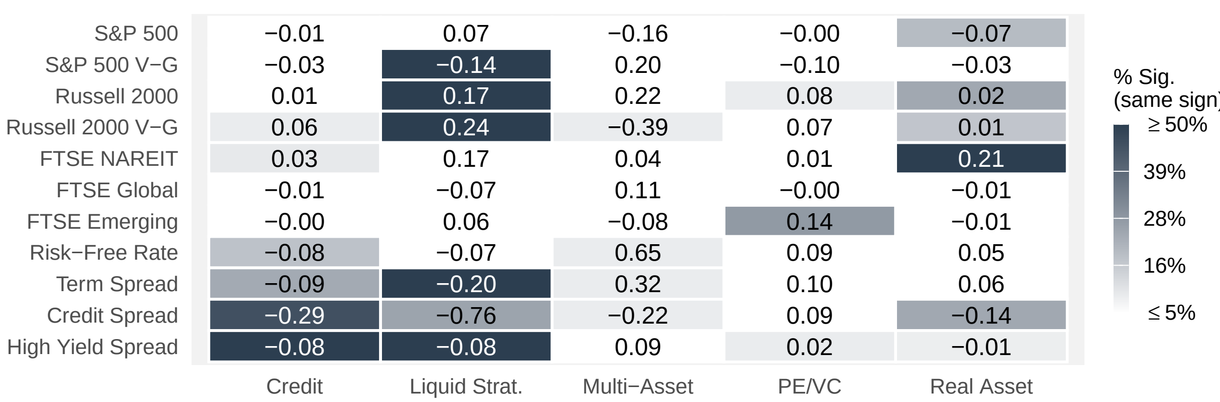
Figure 6. The figure plots average risk-adjusted performance (alpha) for interval funds across investment categories using multi-factor regressions with index-based factors. For each interval-fund share class, we estimate alpha from a time-series regression of excess returns on equity factors derived from equity indexes (S&P 500, SP500 Value-minus-Growth, Russell 2000, Russell 2000 Value-minus-Growth, FTSE NAREIT, FTSE Global, and FTSE Emerging) and fixed income factors derived from monthly changes in rates (Risk-Free Rate, Term Spread, Credit Spread, and High Yield Spread). Results are displayed for three return specifications: raw returns (dark grey), MA-unsmoothed returns (medium grey), and AR-unsmoothed returns (light grey).



(a) Raw Returns

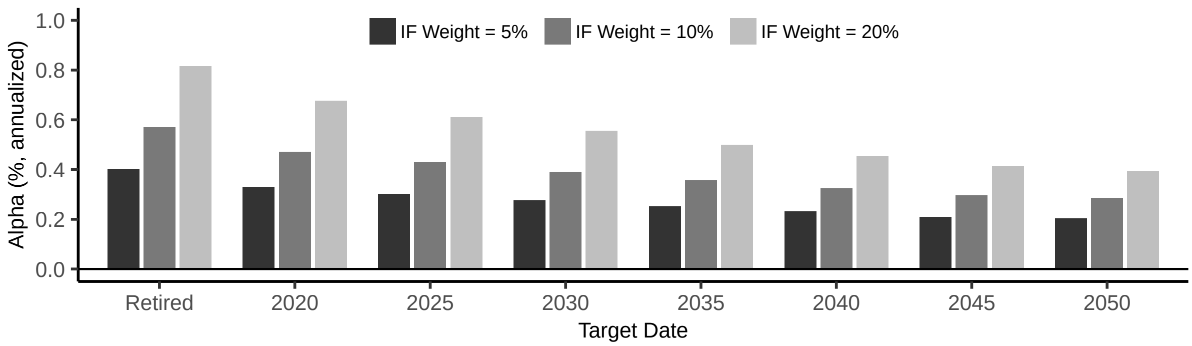


(b) MA-Unsmoothed Returns

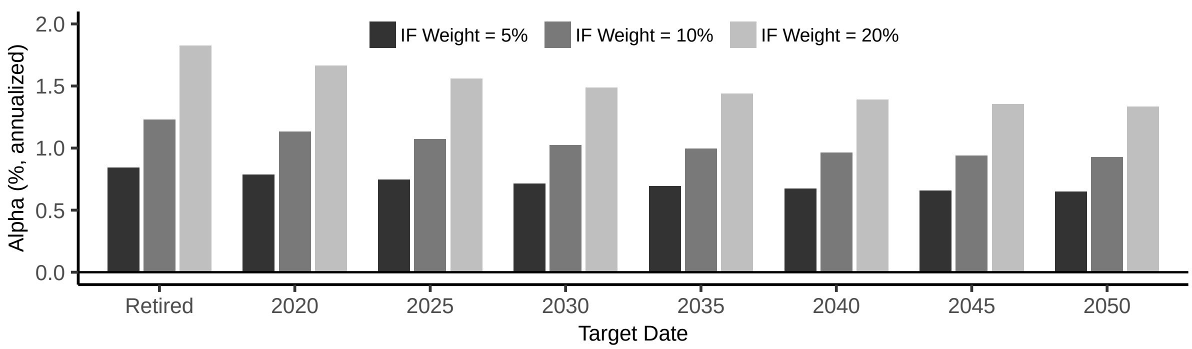


(c) AR-Unsmoothed Returns

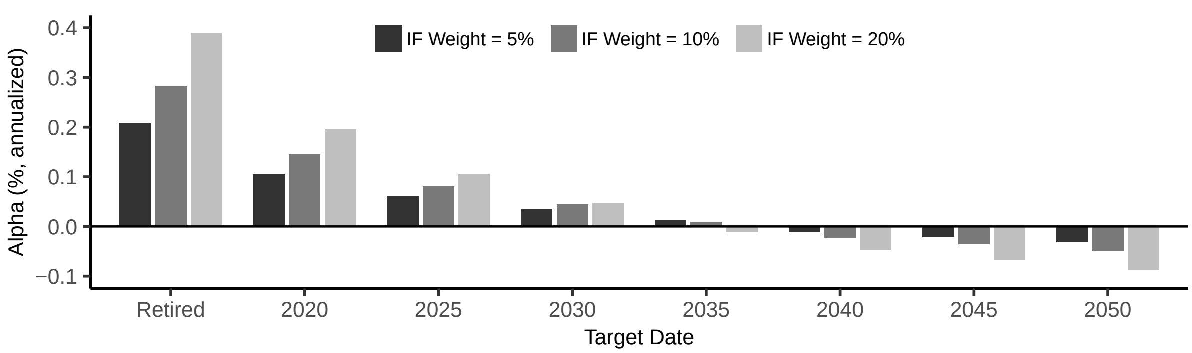
Figure 7. Factor Loading Heatmaps (Index-Based Factors). Each cell shows the mean factor loading from share class-level regressions of excess returns on equity factors derived from equity indexes (S&P 500, SP500 Value-minus-Growth, Russell 2000, Russell 2000 Value-minus-Growth, FTSE NAREIT, FTSE Global, and FTSE Emerging) and fixed income factors derived from monthly changes in rates (Risk-Free Rate, Term Spread, Credit Spread, and High Yield Spread). Cell shading indicates the share of funds with significant loadings ($|t| > 1.96$) whose sign agrees with the category mean, ranging from 5% (white) to 50%+ (dark red). Panel (a) uses raw returns, Panel (b) uses MA(3)-unsmoothed returns, Panel (c) uses AR(2)-unsmoothed returns.



(a) Fixed-Weight Alpha of Credit Interval Funds



(b) Fixed-Weight Alpha of PE/VC Interval Funds



(c) Fixed-Weight Alpha of Real Asset Interval Funds

Figure 8. Risk-adjusted performance of interval-fund allocations in retirement portfolios (raw returns). Each panel reports average fixed-weight alpha for interval-fund shares in a given asset class (Credit, PE/VC, and Real Asset) against target-date funds from the Vanguard Target Retirement series, where the x-axis reports target retirement dates (with "Retired" for the income fund). For each fund share class and target-date pair, we first compute the Sharpe ratio of the benchmark target-date fund and the Sharpe ratio of a blended portfolio that combines the interval fund and the target-date fund. We consider three blended portfolios in which the interval fund represents 5% (dark grey), 10% (medium grey), and 20% (light grey) of the total asset allocation, with the remaining share invested in the target-date fund. We then convert the Sharpe-ratio gap into fixed-weight alpha using equation (5). Bars show averages across individual funds within each category.

Table 1. Redemption Characteristics of Interval Funds

This table reports summary statistics on the redemption terms and outcomes of interval funds. Panel A shows summary statistics for redemption offers. "Days between redemption offers" measures the number of days between the start dates of two consecutive redemption offers, "Days from start to deadline" measures the length of the period during which investors may submit redemption requests, and "Days from deadline to pricing" measures the number of days between the redemption-request deadline and the date on which the NAV is used to price redemptions if a pricing date is specified in the redemption offer. "Offered amount" is the percentage of TNA made available for redemption in each offer. "Potential increase" is the percentage by which the offered amount may be increased at the fund's discretion. Panel B shows summary statistics for redemption outcomes at the fund level. "Oversubscribed" is the percentage of redemption offers in which investor redemption requests exceeded the offered amount. "Prorated" is the percentage of offers in which redemptions were prorated. "Proration" is the percentage of requested shares that were redeemed, conditional on proration occurring. Panel C shows summary statistics for investor transactions at the share-class level. "Percent tendered" is the percentage of outstanding shares submitted for redemption by investors. "Percent repurchased" is the percentage of outstanding shares actually redeemed. Data are from forms N-23c-3 and N-CSR available on Edgar from 2010 to March 2026.

Panel A: Fund-Level Redemption Offers

	Mean	SD	P25	P50	P75	N. Obs
Days between redemption offers	82.18	34.32	86.00	91.00	91.00	3125
Days from start to deadline	25.84	9.76	21.00	29.00	32.00	3282
Days from deadline to pricing	2.30	5.03	0.00	0.00	0.00	2848
Offered amount (% TNA)	7.47	5.75	5.00	5.00	6.00	3282
Potential increase (% TNA)	1.97	0.22	2.00	2.00	2.00	3282

Panel B: Fund-level Redemption Outcomes

	Mean	SD	P25	P50	P75	N. Obs
Oversubscribed (%)	15.7					2429
Prorated (%)	9.0					2429
Proration (%, conditional on prorated)	53.21	34.12	21.59	45.73	94.98	216

Panel C: Share Class-Level Redemption Outcomes

	Mean	SD	P25	P50	P75	N. Obs
Percent tendered (% outstanding)	4.20	7.74	0.55	2.20	5.00	2874
Percent repurchased (% outstanding)	3.05	3.52	0.54	2.14	4.93	2864

Table 2. Fee Structure of Interval Funds

This table reports summary statistics for key fee and compensation features of interval funds. Panel A shows the distribution of expense items. Net expense ratios include management fees, borrowing costs, and other fund expenses. Management fees refer to the base advisory fee. The fee on leverage measures the management fee charged on leveraged assets. Borrowing costs include interest and fees on borrowed funds. Acquired fund fees are expenses from investing in other funds. Redemption fees include deferred sales loads, back-end loads, and early redemption fees. Incentive fees apply to a subset of funds and are often subject to hurdle rates and catch-up provisions. Catch-up thresholds are the level of return above the hurdle rate up to which increased incentive fees apply. Catch-up rate denotes the share of returns allocated to the advisor during this range. Panel B shows the prevalence of features associated with performance-based fee structures among interval funds. The first row shows the share of all funds that charge an incentive fee. Subsequent rows report the frequency of additional contractual features (high-water marks, hurdle rates, and catch-up provisions) within the subset of funds that charge incentive fees. The final row shows the frequency of catch-up provisions among funds that apply a hurdle rate. Data are annual observations at the share-class level from forms N-CSR, N-2, 486BPOS, and 486APOS available on Edgar from 2010 to March 2026.

Panel A: All expenses

Feature	Mean	SD	P25	P50	P75	N. Obs
Net expense ratio (%)	3.02	1.61	2.02	2.74	3.60	2326
Management fee (%)	1.39	0.44	1.10	1.30	1.61	2326
Fee on leverage (%)	0.36	0.27	0.20	0.30	0.43	673
Borrowing cost (%)	0.84	0.85	0.27	0.58	1.25	1634
Acquired fund fees (%)	0.45	0.85	0.01	0.06	0.60	1021
Front load (%)	3.88	1.41	3.00	3.50	5.75	759
Redemption fees (%)	1.67	0.74	1.00	2.00	2.00	875
Early withdrawal period (years)	1.19	0.75	1.00	1.00	1.00	831
Incentive fee (%)	16.32	3.73	15.00	15.00	20.00	408
Hurdle rate (%)	6.20	1.96	6.00	6.00	8.00	333
Catch-up threshold (%)	7.48	2.54	7.04	7.27	9.41	325
Catch-up rate (%)	99.24	6.14	100.00	100.00	100.00	393
Catch-up threshold minus hurdle rate (%)	1.28	0.62	1.04	1.27	1.50	325

Panel B: Incentive-fee features

Feature	Frequency	Sample
Has incentive fee	17.5%	All funds (2326 observations)
Has high-water mark	14.9%	Funds with incentive fee (408 observations)
Has hurdle rate	81.6%	Funds with incentive fee (408 observations)
Has both high-water mark and hurdle rate	0.2%	Funds with incentive fee (408 observations)
Has catch-up provision	97.6%	Funds with hurdle rate (333 observations)

Table 3. Summary Statistics

This table presents share-class-level summary statistics for interval funds, including the mean, standard deviation, and the 25th, 50th, and 75th percentiles. The sample covers monthly data from January 2010 through March 2026. Variables are defined in Appendix Table A.1. Variables not applicable to a given fund structure are omitted. The final column reports the total number of monthly share-class observations.

	Mean	SD	P25	P50	P75	N.Obs
TNA (millions)	264.03	1215.03	6.37	47.13	167.39	17525
Family TNA (billions)	1.22	2.88	0.11	0.36	1.17	17525
Age (years)	4.99	4.93	1.64	3.55	6.56	17525
Fund of Funds (indicator)	0.08	0.27	0.00	0.00	0.00	17525
Flows (monthly %)	3.25	14.45	-0.78	0.04	2.95	17525
Net Return (annualized %)	5.56	25.36	-0.95	6.86	13.82	17525
NAV Return (annualized %)	-1.66	36.07	-8.66	0.46	8.98	17525
Distribution Return (annualized %)	7.22	25.20	0.00	5.83	8.98	17525

Table 4. Interval Fund Alphas: Category Portfolios

This table reports alphas equal-weighted portfolios of interval funds against index factors. Panel A uses annualized raw returns, Panel B uses annualized MA(3)-unsmoothed returns, and Panel C uses annualized AR(2)-unsmoothed returns. Alphas are estimated from time-series regressions on 11 index factors. As factors, we use excess returns of equity indexes (S&P 500, SP500 Value-minus-Growth, Russell 2000, Russell 2000 Value-minus-Growth, FTSE NAREIT, FTSE Global, and FTSE Emerging) and fixed income factors derived from monthly changes in rates (Risk-Free Rate, Term Spread, Credit Spread, and High Yield Spread). t-statistics are reported in parentheses. t-statistics use Newey-West adjustment. * denotes significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Panel A: Raw Returns

	Credit	Liquid Strat.	Multi-Asset	PE/VC	Real Asset
	(1)	(2)	(3)	(4)	(5)
Alpha	4.472***	−6.109***	1.287	4.842***	2.086*
	(7.20)	(−2.78)	(0.78)	(2.86)	(1.71)
Observations	195	183	102	141	168
Adjusted R^2	0.782	0.659	0.522	0.162	0.681

Panel B: MA-Unsmoothed Returns

	Credit	Liquid Strat.	Multi-Asset	PE/VC	Real Asset
	(1)	(2)	(3)	(4)	(5)
Alpha	4.366***	−6.546***	0.746	4.255**	1.917
	(6.89)	(−2.92)	(0.40)	(2.26)	(1.50)
Observations	195	183	102	141	168
Adjusted R^2	0.777	0.660	0.504	0.140	0.685

Panel C: AR-Unsmoothed Returns

	Credit	Liquid Strat.	Multi-Asset	PE/VC	Real Asset
	(1)	(2)	(3)	(4)	(5)
Alpha	4.347***	−6.417***	−1.046	4.290**	2.103
	(6.73)	(−2.88)	(−0.24)	(2.19)	(1.49)
Observations	195	183	102	141	168
Adjusted R^2	0.763	0.658	0.177	0.138	0.590

Table 5. Interval Fund Alphas: Abnormal Return Estimates

This table reports abnormal return estimates by asset class for interval funds against index factors using Fama and MacBeth (1973) regressions. Columns 1–3 show results for the full sample, columns 4–6 show results for retail share classes, and columns 7-9 show results for institutional or high-net-worth investors' share classes. Three return specifications are used: annualized raw returns (columns 1, 4, and 7), annualized MA(3)-unsmoothed returns (columns 2, 5, and 8), and annualized AR(2)-unsmoothed returns (columns 3, 6, and 9). The reported estimates represent asset class-specific intercepts from monthly cross-sectional regressions and can be interpreted as the excess return of a zero-beta fund in each asset class. As factors, we use excess returns of equity indexes (S&P 500, SP500 Value-minus-Growth, Russell 2000, Russell 2000 Value-minus-Growth, FTSE NAREIT, FTSE Global, and FTSE Emerging) and fixed income factors derived from monthly changes in rates (Risk-Free Rate, Term Spread, Credit Spread, and High Yield Spread). *t*-statistics are reported in parentheses. *t*-statistics use Newey-West adjustment. * denotes significance at the 10% level, ** at the 5% level, and *** at the 1% level.

	Full Sample			Retail			Institutional/HNWI		
	Raw Ret.	MA-Unsm.	AR-Unsm.	Raw Ret.	MA-Unsm.	AR-Unsm.	Raw Ret.	MA-Unsm.	AR-Unsm.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Credit	4.024***	3.768***	4.118***	2.070***	2.631***	2.097***	4.505***	4.367***	4.235***
	(6.51)	(6.74)	(6.71)	(3.32)	(3.86)	(3.85)	(15.95)	(13.54)	(12.18)
Liquid Strat.	−5.200	−4.694	−2.947	−3.577	0.380	−1.631	−1.037	−2.447	−5.034
	(−1.30)	(−1.18)	(−0.76)	(−0.68)	(0.08)	(−0.28)	(−0.22)	(−0.50)	(−1.02)
Multi-Asset	1.451	1.423	0.995	0.649	1.238	0.857	1.927*	1.522	0.855
	(1.49)	(1.59)	(0.71)	(0.67)	(1.33)	(0.69)	(1.66)	(1.36)	(0.69)
PE/VC	3.899**	2.327	1.746	3.312*	1.608	−0.992	10.280***	9.900***	9.423***
	(2.07)	(1.20)	(0.95)	(1.76)	(0.67)	(−0.37)	(5.52)	(5.48)	(4.39)
Real Asset	1.733	1.144	1.730	1.699	1.528	1.760	0.905	1.160	1.981
	(1.56)	(1.01)	(1.43)	(0.96)	(1.05)	(1.10)	(0.58)	(0.81)	(1.35)
Avg. Adj. R^2	0.764	0.733	0.721	0.790	0.757	0.744	0.748	0.718	0.694
N months	128	128	128	103	103	103	82	82	82

Table 6. Performance across Investment Categories

This table presents results on the performance of interval funds relative to index ETFs, estimated using OLS regressions at the share-class level. The dependent variable is annualized net returns, and regressions are estimated separately by investment category. The sample includes interval funds and index ETFs from January 2010 to March 2026. The key independent variable is *Interval*, an indicator equal to one for interval fund share classes, with index ETFs as the omitted comparison group. All regressions include benchmark-by-time fixed effects, where benchmarks correspond to Morningstar Category Indices and time is measured monthly. Variables are defined in Appendix Table A.1. t-statistics, based on standard errors clustered at the month-year level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Interval	3.457***	-4.550*	-1.028	2.371	-2.179
	(3.61)	(-1.93)	(-0.27)	(0.39)	(-0.52)
TNA	-0.004	-0.048	-0.048	-0.056	-0.045
	(-0.05)	(-0.54)	(-0.54)	(-0.60)	(-0.52)
Family TNA	0.188***	0.171***	0.174***	0.185***	0.175***
	(3.85)	(3.18)	(3.31)	(3.40)	(3.33)
Age	-0.201	0.086	0.099	0.033	0.043
	(-0.51)	(0.17)	(0.19)	(0.06)	(0.09)
FoF	0.535	-0.070	0.141	0.137	-0.023
	(1.12)	(-0.15)	(0.29)	(0.28)	(-0.05)
Flows	0.002	0.004	0.006	0.001	0.007
	(0.26)	(0.43)	(0.61)	(0.13)	(0.71)
Benchmark $\times$ Time FE	Yes	Yes	Yes	Yes	Yes
Observations	90356	79443	79475	79663	82738
Adjusted R^2	0.737	0.737	0.735	0.725	0.716

Table 7. NAV and Distribution Returns across Investment Categories

This table decomposes total returns into NAV returns (Panel A) and distribution returns (Panel B) and re-estimates the interval fund premium of Table 6 with each component as the dependent variable. The dependent variables are annualized at the share-class level, and regressions are estimated separately by investment category. The sample includes interval funds and index ETFs from January 2010 to March 2026. Variables are defined in Appendix Table A.1. All regressions include benchmark-by-time fixed effects, where benchmarks correspond to Morningstar Category Indices and time is measured monthly. *t*-statistics, based on standard errors clustered at the month-year level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: NAV Returns

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Interval	-0.221	-9.329***	-2.524	4.989	-4.079
	(-0.19)	(-3.81)	(-0.66)	(0.80)	(-0.98)
TNA	-0.024	-0.071	-0.075	-0.081	-0.056
	(-0.31)	(-0.77)	(-0.82)	(-0.86)	(-0.63)
Family TNA	0.173***	0.147**	0.147***	0.155***	0.160***
	(3.15)	(2.54)	(2.61)	(2.66)	(2.85)
Age	-0.071	0.346	0.393	0.326	0.297
	(-0.15)	(0.65)	(0.74)	(0.59)	(0.56)
FoF	0.509	-0.256	0.042	-0.065	0.297
	(0.94)	(-0.49)	(0.08)	(-0.12)	(0.57)
Flows	0.007	0.010	0.011	0.007	0.014
	(0.73)	(0.92)	(1.04)	(0.64)	(1.30)
Benchmark × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	90356	79443	79475	79663	82738
Adjusted R^2	0.701	0.722	0.719	0.709	0.695

Panel B: Distribution Returns

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Interval	3.678***	4.779***	1.496***	-2.618***	1.900***
	(4.73)	(5.07)	(4.11)	(-3.64)	(3.53)
TNA	0.020	0.023	0.027*	0.025	0.011
	(0.73)	(1.54)	(1.81)	(1.65)	(0.57)
Family TNA	0.014	0.024*	0.027*	0.031**	0.015
	(0.89)	(1.68)	(1.92)	(2.22)	(1.02)
Age	-0.131	-0.260***	-0.295***	-0.294***	-0.254***
	(-0.90)	(-3.42)	(-4.26)	(-4.10)	(-3.48)
FoF	0.025	0.186	0.099	0.203	-0.321
	(0.13)	(1.02)	(0.63)	(1.19)	(-1.32)
Flows	-0.005**	-0.005**	-0.005**	-0.006**	-0.007***
	(-2.21)	(-2.31)	(-2.13)	(-2.40)	(-2.66)
Benchmark × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	90356	79443	79475	79663	82738
Adjusted R^2	0.126	0.248	0.249	0.234	0.171

Table 8. Performance by Investor Clientele

This table presents results on the performance of interval funds relative to index ETFs for different investor clientele. The dependent variable is annualized net returns at the share-class level. The sample includes interval funds and index ETFs from January 2010 to March 2026. The key independent variables are interactions between the interval fund indicator and clientele type (*Inst/HNWI* or *Retail*), with index ETFs as the omitted comparison group. Controls include TNA, Family TNA, Age, a Fund-of-Funds indicator, and Flows. All variables are defined in Appendix Table A.1. All regressions include benchmark-by-time fixed effects, where benchmarks correspond to Morningstar Category Indices and time is measured monthly. *t*-statistics, based on standard errors clustered at the month-year level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: All Interval Funds

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Inst/HNWI × Interval	4.188***	-6.197*	-1.090	5.412	-2.340
	(4.01)	(-1.95)	(-0.29)	(0.73)	(-0.54)
Retail × Interval	3.010***	-3.315	-1.004	0.656	-2.067
	(3.25)	(-1.51)	(-0.25)	(0.12)	(-0.50)
Controls	Yes	Yes	Yes	Yes	Yes
Benchmark × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	90356	79443	79475	79663	82738
Adjusted R^2	0.737	0.737	0.735	0.725	0.716

Panel B: Single-Clientele Interval Funds

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Inst/HNWI × Interval	3.072***	-6.372	0.072	5.694	-3.092
	(3.20)	(-1.41)	(0.02)	(0.68)	(-0.64)
Retail × Interval	0.801	-1.862	1.000	-2.049	0.083
	(0.95)	(-0.70)	(0.25)	(-0.40)	(0.02)
Controls	Yes	Yes	Yes	Yes	Yes
Benchmark × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	82417	79122	79207	79345	79843
Adjusted R^2	0.735	0.737	0.736	0.730	0.729

Panel C: Multi-Clientele Interval Funds

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Inst/HNWI × Interval	5.452***	-6.056	-2.749	5.061	-1.890
	(4.12)	(-1.48)	(-0.56)	(0.72)	(-0.43)
Retail × Interval	4.809***	-4.101	-3.532	4.728	-2.378
	(3.85)	(-1.13)	(-0.73)	(0.67)	(-0.58)
Controls	Yes	Yes	Yes	Yes	Yes
Benchmark × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	86807	79139	79086	79136	81713
Adjusted R^2	0.740	0.738	0.737	0.732	0.721

Table 9. Performance and Incentive Fees

This table presents results on the relationship between fund incentive fees and interval fund performance, estimated at the portfolio level using annualized gross returns as the dependent variable. Regressions are estimated separately by investment category. The sample includes interval fund portfolios and index ETFs from January 2010 to March 2026. The key independent variables are the interval fund indicator and its interaction with *Incentive Fee*, an indicator equal to one for funds that charge incentive fees. Variables are defined in Appendix Table A.1. Control variables are included but omitted from the table for brevity. Control variables include the one-month lag of the logarithm of portfolio TNA (*TNA*), the logarithm of aggregate fund family TNA (*Family TNA*), the logarithm of one plus portfolio age in years (*Age*), an indicator for fund-of-funds structures (*FoF*), and one-month lagged net percentage flows (*Flows*). All regressions include benchmark-by-time fixed effects. *t*-statistics, based on standard errors clustered at the month-year level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Interval	3.213***	-3.020	3.004	3.378	-0.687
	(4.42)	(-0.78)	(0.72)	(0.45)	(-0.16)
Interval × Incentive Fee	2.248***	2.503	-4.166		-4.908
	(4.21)	(0.55)	(-1.35)		(-0.45)
Controls	Yes	Yes	Yes	Yes	Yes
Benchmark × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	80427	76026	76033	76165	77063
Adjusted R^2	0.764	0.766	0.764	0.756	0.754

Table 10. Performance and Portfolio Characteristics: Leverage and Liquidity

This table presents results on the relationship between interval fund performance and portfolio characteristics, estimated at the portfolio level using annualized gross returns as the dependent variable. Regressions are estimated separately by investment category. The sample includes interval fund portfolios and index ETFs over October 2019 to March 2026, corresponding to the period for which N-PORT holdings data are available. Panel A examines the interaction between the interval fund indicator and *Leverage*. Panel B examines the interaction between the interval fund indicator and the *Liquidity* index. Variables are defined in Appendix Table A.1. Control variables are included but omitted from the table for brevity. All regressions include benchmark-by-time fixed effects. *t*-statistics, based on standard errors clustered at the month-year level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Leverage

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Interval	4.119***	-10.223	2.991	-0.941	-1.362
	(3.26)	(-1.51)	(0.72)	(-0.09)	(-0.22)
Leverage	-0.002	-0.000	-0.002	-0.002	-0.001
	(-0.06)	(-0.00)	(-0.05)	(-0.05)	(-0.02)
Interval × Leverage	0.081*	2.320	-0.216*	0.315	-0.155
	(1.88)	(1.29)	(-1.75)	(0.27)	(-0.55)
Controls	Yes	Yes	Yes	Yes	Yes
Benchmark × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	42929	39438	39623	39683	40252
Adjusted R^2	0.753	0.755	0.753	0.739	0.740

Panel B: Portfolio Liquidity

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Interval	8.323***	1.842	2.429	-2.584	-2.668
	(5.38)	(0.27)	(0.50)	(-0.24)	(-0.39)
Liquidity	0.016	0.016	0.016	0.015	0.016
	(0.89)	(0.92)	(0.92)	(0.82)	(0.91)
Interval × Liquidity	-0.063***	-0.100	0.009	0.199	0.066
	(-2.90)	(-0.81)	(0.12)	(1.65)	(0.99)
Controls	Yes	Yes	Yes	Yes	Yes
Benchmark × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	42929	39438	39623	39683	40252
Adjusted R^2	0.753	0.755	0.753	0.739	0.740

Table 11. Fund Liquidity Offering and Portfolio Holdings Decisions

This table presents results on the relationship between liquidity offered at redemption windows and portfolio holdings decisions for interval funds. Regressions are estimated separately for retail-only, institutional-only, and multi-clientele funds. Panel A reports results with *Leverage* as the dependent variable, and Panel B with the portfolio *Liquidity* index; both panels use the sample from October 2019 to March 2026. Panel C reports results with fund flows as the dependent variable, using the full portfolio-level sample from January 2010 to March 2026. The key independent variable is *Offered Amount*, defined as the percentage of TNA offered for repurchase at each redemption window. Variables are defined in Appendix Table A.1. Control variables are included but omitted from the table for brevity. All regressions include benchmark-by-time fixed effects. *t*-statistics, based on standard errors two-way clustered at the month-year and portfolio levels, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Leverage

	Retail	Inst/HNWI	Multi-class
Offered Amount	-2.401***	0.509***	0.423
	(-2.94)	(2.71)	(1.12)
Controls	Yes	Yes	Yes
Benchmark × Time FE	Yes	Yes	Yes
Observations	874	1679	1926
Adjusted R^2	0.286	0.098	0.360

Panel B: Portfolio Liquidity

	Retail	Inst/HNWI	Multi-class
Offered Amount	3.538**	0.859**	0.550
	(2.40)	(2.49)	(1.15)
Controls	Yes	Yes	Yes
Benchmark × Time FE	Yes	Yes	Yes
Observations	874	1679	1926
Adjusted R^2	0.308	0.317	0.275

Panel C: Fund Flows

	Retail	Inst/HNWI	Multi-class
Offered Amount	0.785**	0.288	-0.233
	(2.62)	(1.50)	(-1.50)
Controls	Yes	Yes	Yes
Benchmark × Time FE	Yes	Yes	Yes
Observations	1122	2014	2456
Adjusted R^2	0.148	0.285	0.412

Appendix

Table A.1. Variable Definitions

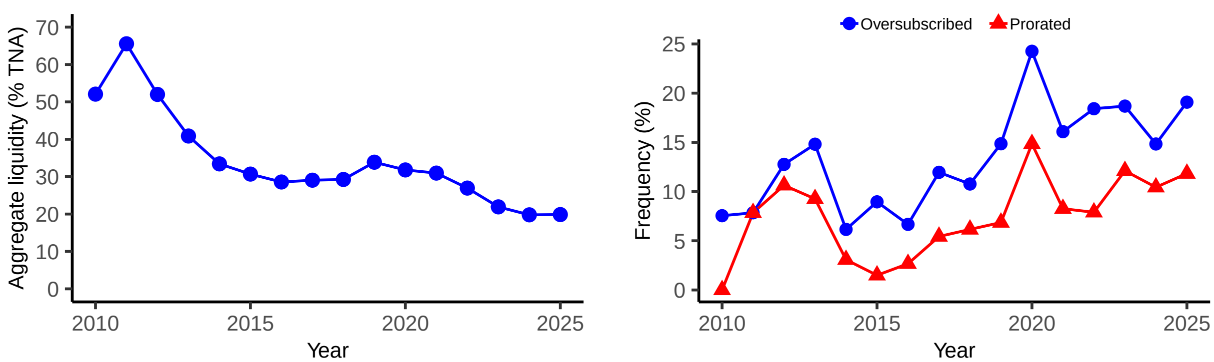
This table defines the variables used in the empirical analysis. Sources are Morningstar (monthly share-class panel), EDGAR (SEC filings N-23c-3, N-CSR, N-2, 486BPOS, 486APOS, TO-I), N-CEN (annual report of registered investment companies), and N-PORT (quarterly portfolio holdings of registered investment companies).

Variable	Description	Source
Returns		
Net Return	Monthly total net return, annualized by multiplying by 12. Winsorized at the 1st and 99th percentiles within fund type and month.	Morningstar
Gross Return	Monthly total gross return, annualized by multiplying by 12. When missing in Morningstar, imputed as the net return plus one-twelfth of the annual net expense ratio. Winsorized at the 1st and 99th percentiles within fund type and month.	Morningstar, EDGAR
NAV Return	Annualized percent change in price per share from the prior month.	Morningstar
Distribution Return	Annualized total net return minus annualized NAV return. Captures the income component of total returns.	Morningstar
Fund and share-class characteristics		
TNA	Share-class total net assets in millions of dollars. In regressions, entered as the one-month lag of the log of TNA.	Morningstar, N-CEN
Family TNA	Aggregate TNA in billions of dollars across all share classes managed by the same firm within the same fund type. In regressions, entered as the one-month lag of the log of Family TNA.	Morningstar
Age	Years since inception date. In regressions, entered as the log of one plus age.	Morningstar
Fund of Funds	Indicator equal to one if the share class is classified as a fund of funds.	Morningstar
Flows	Monthly net percentage flow, computed as $100 \times (\mathrm{TNA}_t - \mathrm{TNA}_{t-1}(1+r_t))/\mathrm{TNA}_{t-1}$, where r_t is the monthly net return. Winsorized at the 1st and 99th percentiles. In regressions, entered as the one-month lag.	Morningstar
Expense Ratio	Annual net expense ratio, including management fees, borrowing costs, other annual expenses, and incentive fees.	Morningstar, EDGAR
Fund type and classification		
Interval	Indicator equal to one for interval fund share classes, zero otherwise.	Morningstar
Fund Type	Categorical variable identifying interval funds, active ETFs, index ETFs, tender offer funds, BDCs, and nontraded REITs.	Morningstar
Investment Category	Consolidated category assigned to each share class: Credit, Liquid Strategies, Multi-Asset, PE/VC, or Real Asset. Constructed from Morningstar category group and category.	Morningstar
Benchmark	Morningstar Category Index associated with each share class. Used to construct benchmark-by-time fixed effects.	Morningstar
Clientele		
Retail	Indicator equal to one for share classes with retail-oriented labels and minimum investment at or below $25,000, and for all ETF share classes.	Morningstar
Inst/HNWI	Indicator equal to one for share classes carrying an institutional share class label, or with minimum investment above $25,000.	Morningstar
Accredited	Indicator equal to one for share classes that require investors to meet accredited investor standards, as disclosed in prospectus filings. Used in the robustness classification.	EDGAR
Multi-class	Indicator equal to one for funds that offer both Retail and Inst/HNWI share classes within the same portfolio.	Morningstar
Portfolio holdings (N-PORT)		
Level 1	Percentage of portfolio holdings valued using quoted prices in active markets for identical assets.	N-PORT
Level 2	Percentage of portfolio holdings valued using observable inputs other than quoted prices.	N-PORT
Level 3	Percentage of portfolio holdings valued using unobservable inputs.	N-PORT
Liquidity	Portfolio liquidity index is defined as Level 1 plus one-half of Level 2.	N-PORT
Leverage	Total liabilities as a percentage of assets under management.	N-PORT
Redemption features		
Offered Amount	Percentage of TNA offered for repurchase at each redemption window.	EDGAR, N-23c-3
Long Interval	Indicator equal to one for funds that offer annual or semiannual redemption windows, zero for funds that offer quarterly or monthly redemptions.	EDGAR, N-23c-3
Incentive Fee	Indicator equal to one for interval funds that charge a performance-based fee on net investment income. Set to zero for index ETFs.	EDGAR

Internet Appendix for
"Democratizing Illiquid Assets: Liquidity Transformation and Performance in Interval Funds"

This Internet Appendix provides additional tables, figures, and proofs that supplement the analysis in the main paper.

A Additional Figures and Tables



(a) Aggregate liquidity provision per fund

(b) Oversubscribed Redemptions

Figure IA.1. Liquidity provision over time. Panel IA.1(a) shows the aggregate amount of TNA made available for redemption per fund per year. Aggregate liquidity is computed as the sum of all offered amounts (as a percentage of TNA) in a given year, divided by the number of funds filing N-CSR reports in that year. Panel IA.1(b) shows the fraction of redemption offers that were oversubscribed and prorated each year. A redemption offer is oversubscribed if redemption requests exceed the offered amount. A redemption offer is prorated if it is oversubscribed and the fund does not increase the amount offered to accommodate the excess redemption requests. Data are from forms N-23c-3 and N-CSR available on EDGAR.

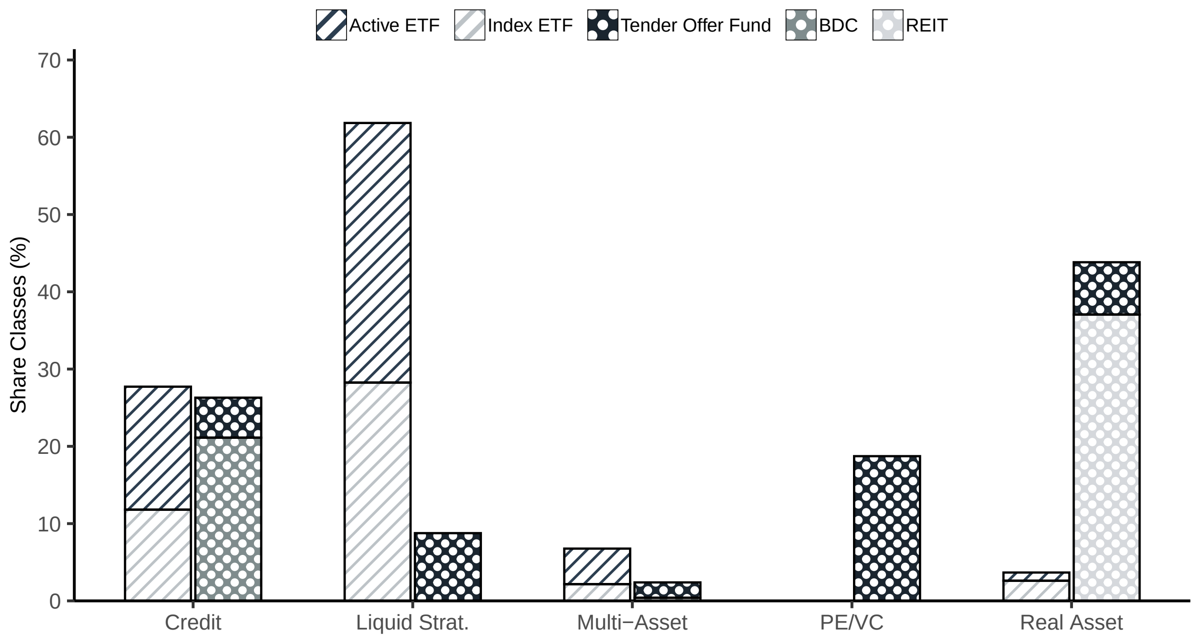
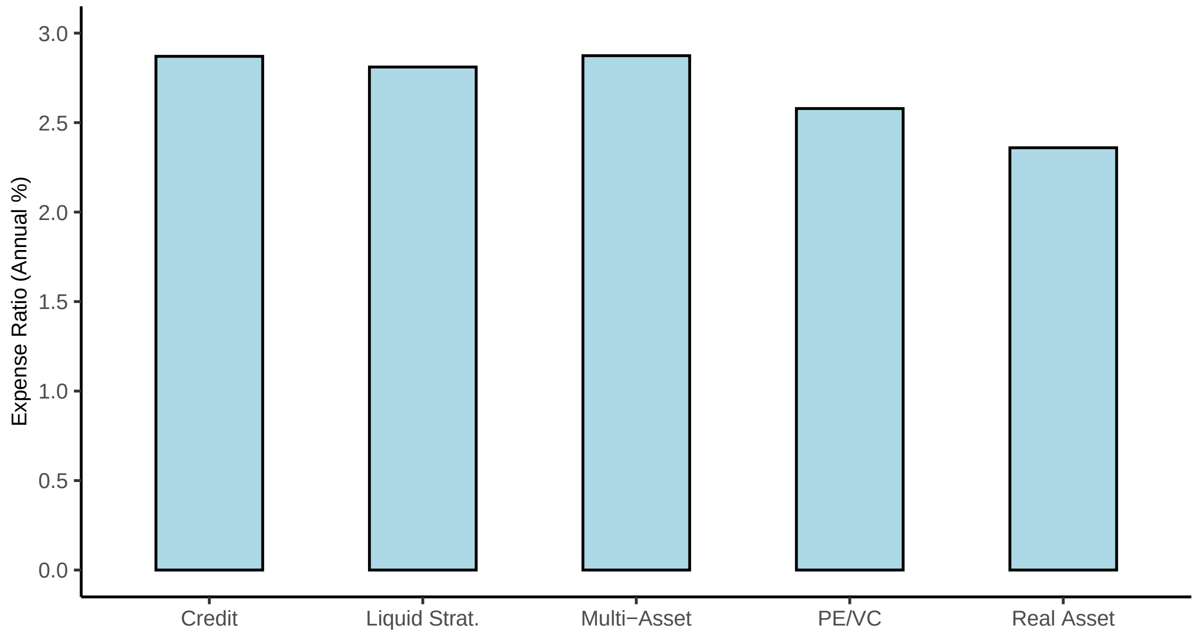
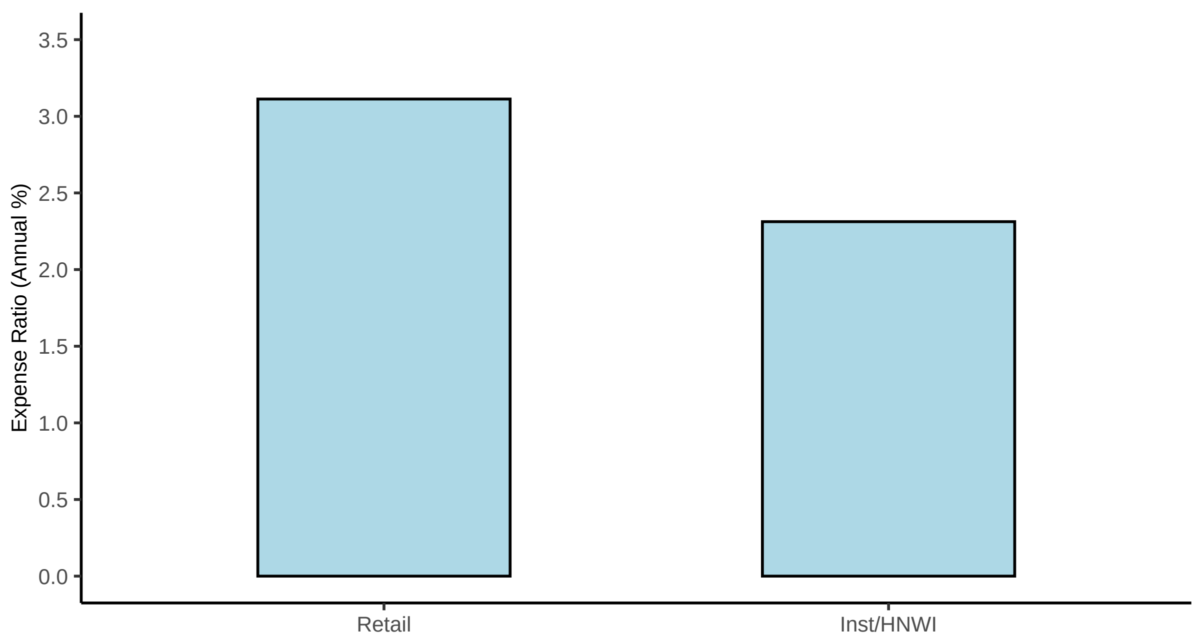


Figure IA.2. The figure displays the frequency distribution of investment categories among the sample of ETFs and semiliquid funds in Morningstar from 2010 to March 2026.



(a) Average Expense Ratios by Investment Category



(b) Average Expense Ratios by Clientele

Figure IA.3. Average annual expense ratio by investment category and clientele for interval funds in the Morningstar sample from 2010 to March 2026. Missing expense ratios from Morningstar are obtained from forms N-CSR available on EDGAR.

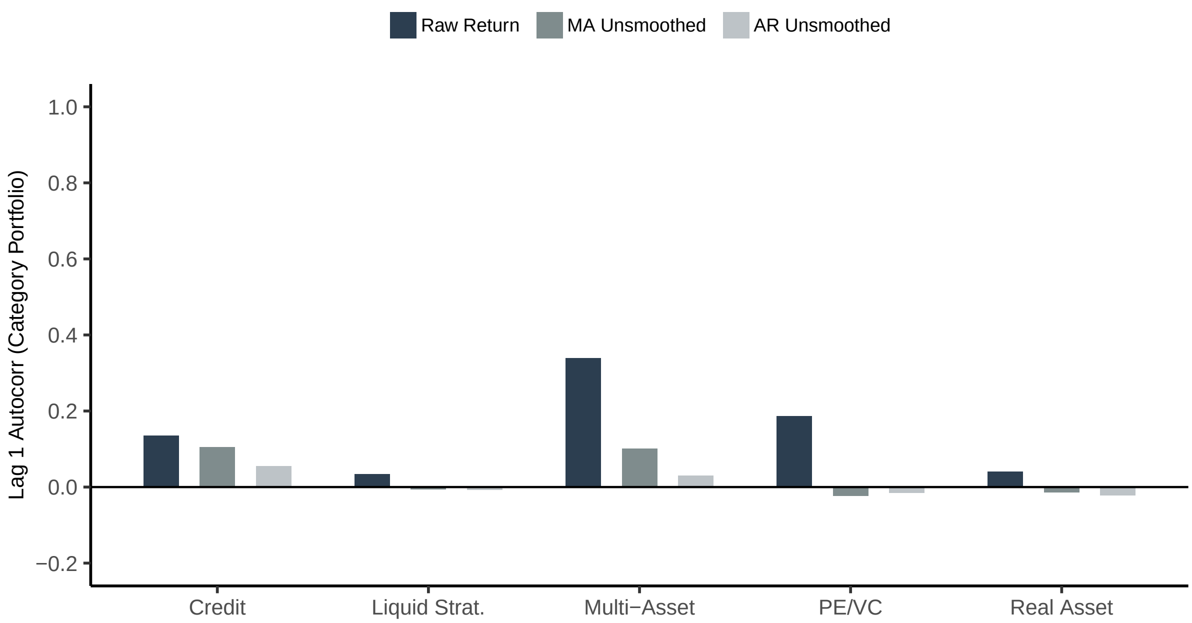


Figure IA.4. The figure plots average one-month autocorrelations of monthly returns for interval funds across investment categories. For each interval fund share class, we calculate the autocorrelation between returns and their one-month lags for each fund, and then compute the mean autocorrelation within each category. Results are displayed for three return specifications: raw returns (dark grey), MA-unsmoothed returns (medium grey), and AR-unsmoothed returns (light grey).

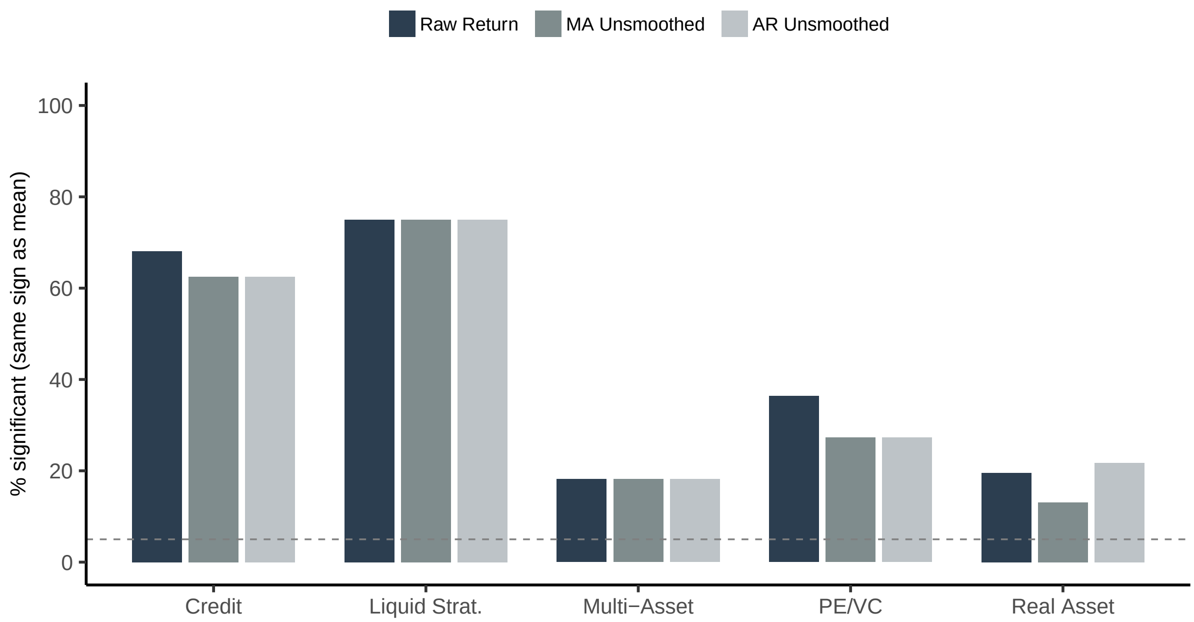
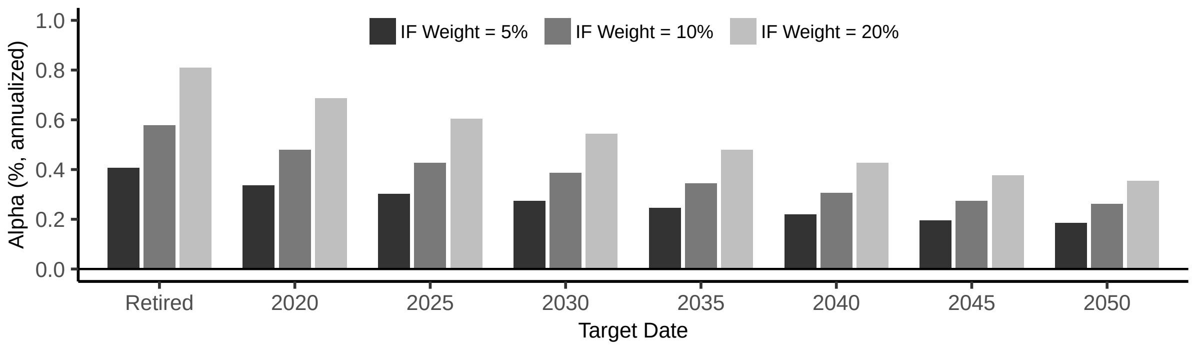
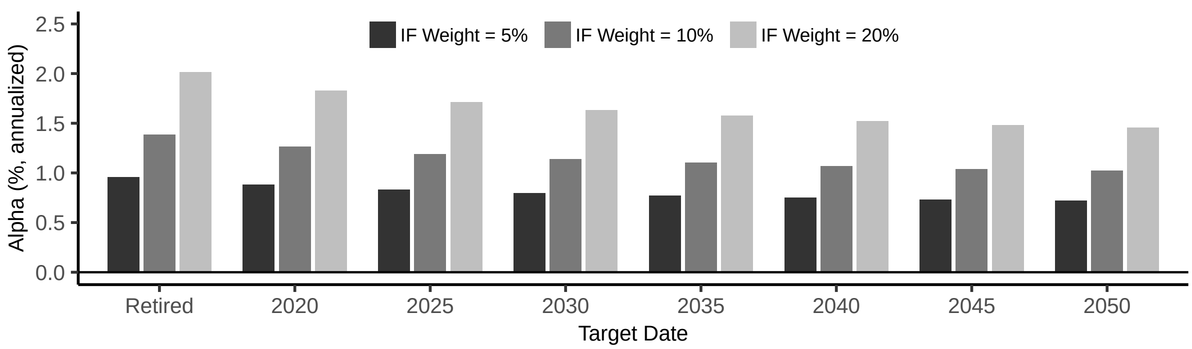


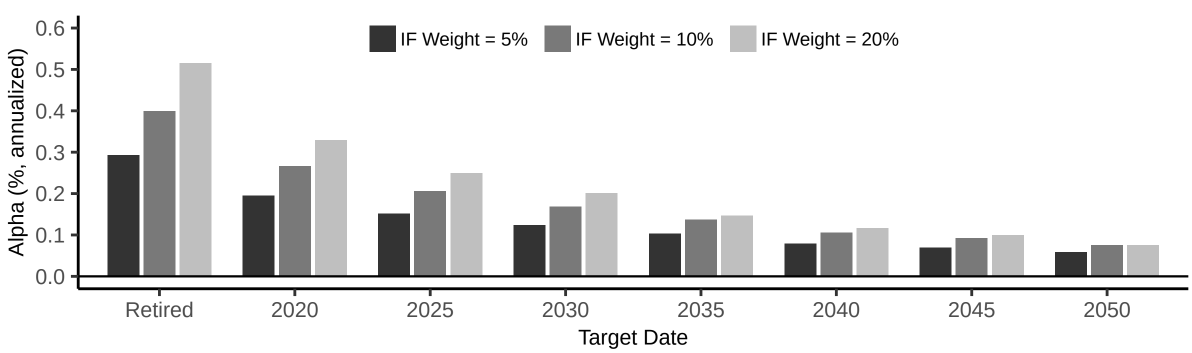
Figure IA.5. The figure plots the percentage of interval funds with statistically significant alphas across investment categories using multi-factor regressions with index-based factors. For each interval-fund share class, we estimate alpha from a time-series regression of excess returns on equity factors derived from equity indexes (S&P 500, SP500 Value-minus-Growth, Russell 2000, Russell 2000 Value-minus-Growth, FTSE NAREIT, FTSE Global, and FTSE Emerging) and fixed income factors derived from monthly changes in rates (Risk-Free Rate, Term Spread, Credit Spread, and High Yield Spread), with standard errors clustered at the month level. The figure shows the percentage of funds with statistically significant alphas ($|t| > 1.96$) whose sign agrees with the category mean. Results are displayed for three return specifications: raw returns (dark grey), MA-unsmoothed returns (medium grey), and AR-unsmoothed returns (light grey). Dashed lines indicate the 5% upper bound on the threshold expected under the null hypothesis of no abnormal performance.



(a) Fixed-Weight Alpha of Credit Interval Funds

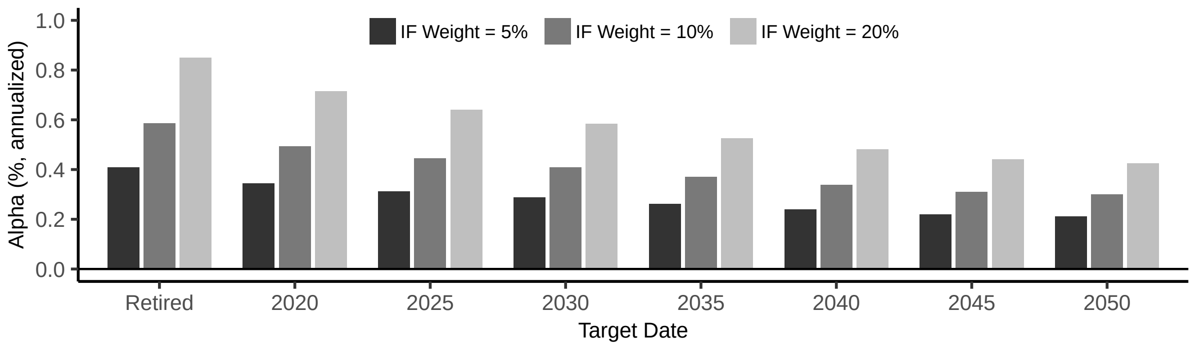


(b) Fixed-Weight Alpha of PE/VC Interval Funds

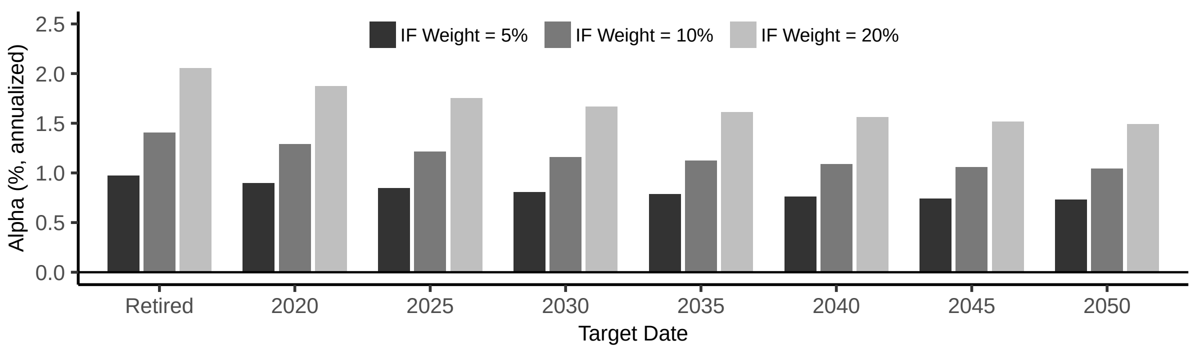


(c) Fixed-Weight Alpha of Real Asset Interval Funds

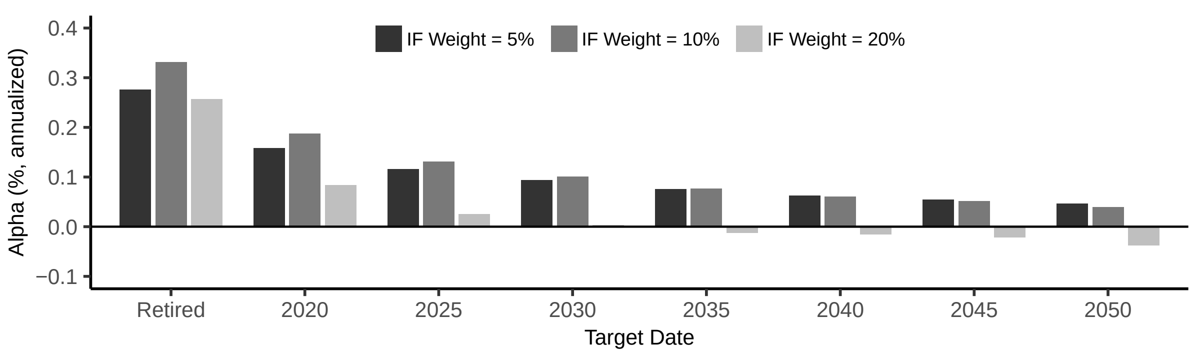
Figure IA.6. Risk-adjusted performance of interval-fund allocations in retirement portfolios (MA-unsmoothed returns). Each panel reports average fixed-weight alpha for interval funds in a given asset class (Credit, PE/VC, and Real Asset) against target-date funds from the Vanguard Target Retirement series, where the x-axis reports target retirement dates (with "Retired" for the income fund). For each fund share class and target-date pair, we first compute the Sharpe ratio of the benchmark target-date fund and the Sharpe ratio of a blended portfolio that combines the interval fund and the target-date fund. We consider three blended portfolios in which the interval fund represents 5% (dark grey), 10% (medium grey), and 20% (light grey) of the total asset allocation, with the remaining share invested in the target-date fund. We then convert the Sharpe-ratio gap into fixed-weight alpha using equation (5). Bars show averages across individual funds within each category.



(a) Fixed-Weight Alpha of Credit Interval Funds



(b) Fixed-Weight Alpha of PE/VC Interval Funds



(c) Fixed-Weight Alpha of Real Asset Interval Funds

Figure IA.7. Risk-adjusted performance of interval-fund allocations in retirement portfolios (AR-unsmoothed returns). Each panel reports average fixed-weight alpha for interval funds in a given asset class (Credit, PE/VC, and Real Asset) against target-date funds from the Vanguard Target Retirement series, where the x-axis reports target retirement dates (with "Retired" for the income fund). For each fund share class and target-date pair, we first compute the Sharpe ratio of the benchmark target-date fund and the Sharpe ratio of a blended portfolio that combines the interval fund and the target-date fund. We consider three blended portfolios in which the interval fund represents 5% (dark grey), 10% (medium grey), and 20% (light grey) of the total asset allocation, with the remaining share invested in the target-date fund. We then convert the Sharpe-ratio gap into fixed-weight alpha using equation (5). Bars show averages across individual funds within each category.

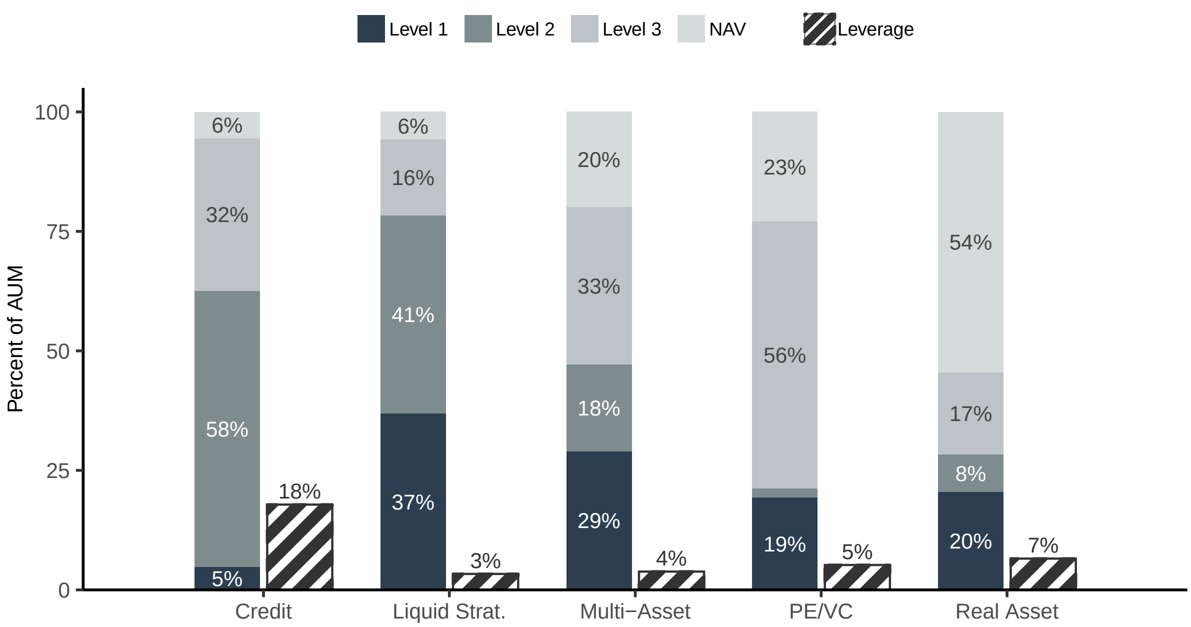
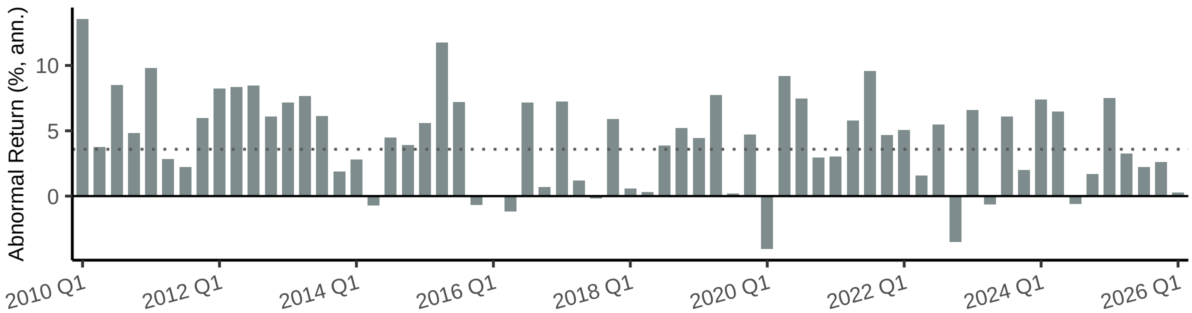
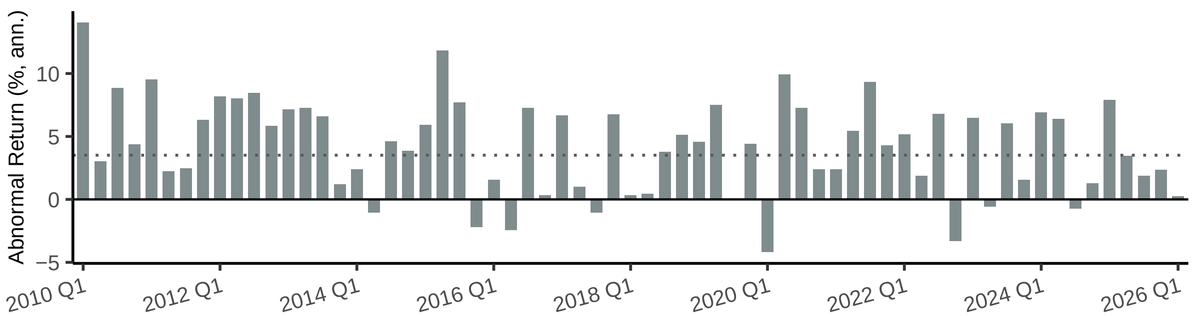


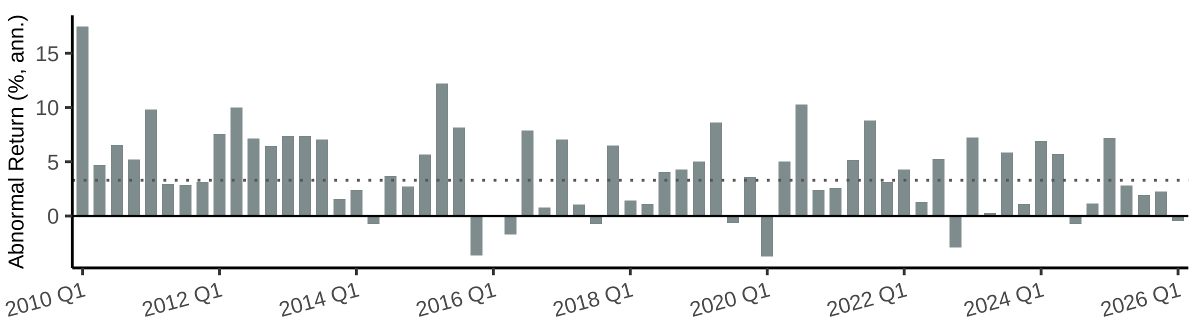
Figure IA.8. The figure shows statistics on the average portfolio liquidity and leverage of interval funds in the Morningstar sample across asset classes. Level 1 assets are valued using quoted prices in active markets and are typically the most liquid; Level 2 assets are valued using observable inputs other than quoted prices, such as dealer quotes or matrix pricing, and are generally less liquid; Level 3 assets are valued using unobservable inputs, such as internal models or management estimates, and are typically the least liquid. The remaining assets are valued at net asset value (NAV) and typically represent shares in other funds. Leverage is total liabilities as a share of total assets under management (AUM). All variables are expressed as percentages of total AUM. Data are from N-PORT filings available on EDGAR from October 2019 to March 2026.



(a) Raw Returns

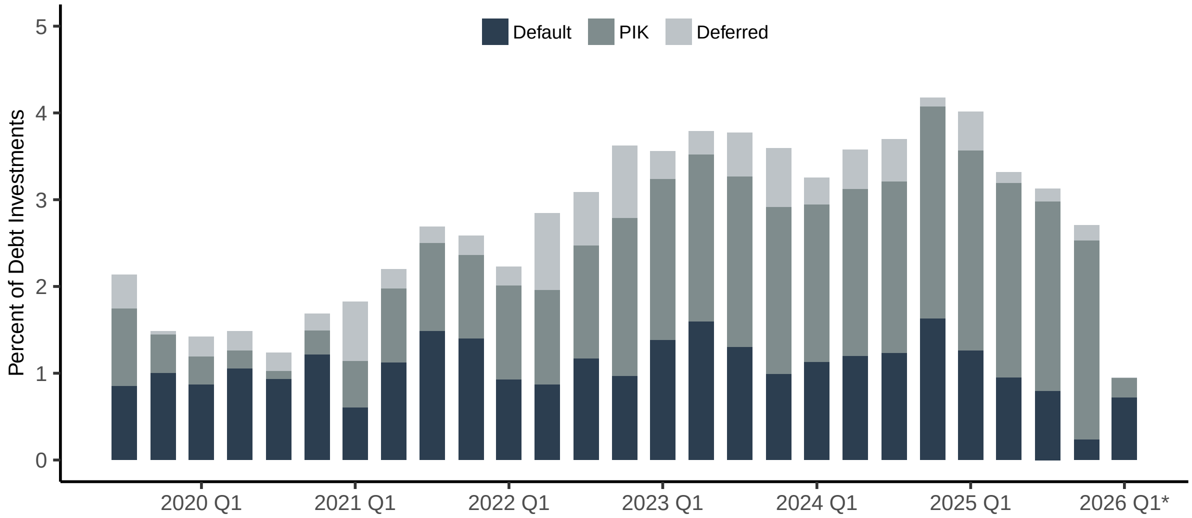


(b) MA-Unsmoothed Returns

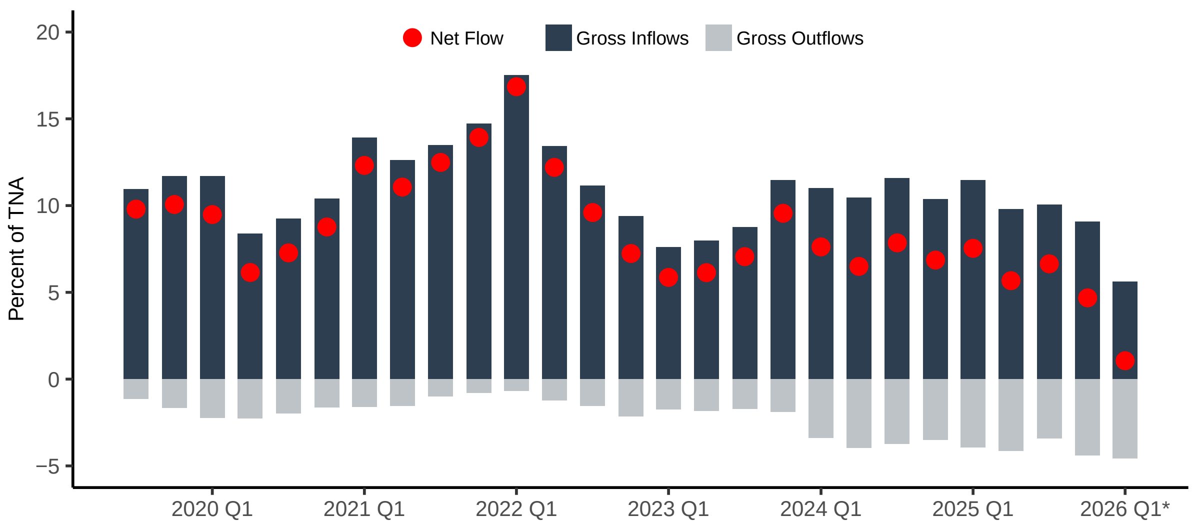


(c) AR-Unsmoothed Returns

Figure IA.9. Quarterly abnormal returns of Credit interval funds. Each panel plots the quarterly mean of abnormal returns (alpha plus regression residual) at the share-class level, averaged across Credit interval funds within each quarter. The dotted line marks the mean alpha estimated from time-series regressions of individual fund excess returns on equity factors derived from equity indexes (S&P 500, SP500 Value-minus-Growth, Russell 2000, Russell 2000 Value-minus-Growth, FTSE NAREIT, FTSE Global, and FTSE Emerging) and fixed income factors derived from monthly changes in rates (Risk-Free Rate, Term Spread, Credit Spread, and High Yield Spread), with standard errors clustered at the month level. Panels (a), (b), and (c) display results for raw returns, MA-unsmoothed returns, and AR-unsmoothed returns, respectively.



(a) Distress Indicators in Debt Portfolio



(b) Quarterly Fund Flows

Figure IA.10. Distress indicators and flows in interval funds. Panel (a) plots quarterly TNA-weighted averages of the portfolio share of distressed debt investments for interval funds. For each quarter, we compute the weighted-average shares of debt investments (by par value) in default, paying interest-in-kind (PIK), or deferring interest, using each fund's share of total debt investments as weight. Panel (b) plots quarterly gross inflows, gross outflows, and net flows for interval funds as a percentage of beginning-of-quarter total net assets (TNA). Dark bars show gross inflows (sales and reinvestments); light bars show gross outflows (redemptions, shown as negative values); the red dot marks net flows. *Data from 2026 Q1 are partial and contain only 14 interval funds, as N-PORT filings are reported with a delay. Data are from N-PORT filings available on EDGAR from October 2019 to March 2026.

Table IA.1. Comparison of pooled investment vehicle structures

Fund type	(1) Clientele	(2) Illiquidity	(3) Leverage	(4) Fees	(5) Redemption
Open-End Mutual Funds & ETFs	Any investor.	Rule 22e–4 caps illiquid investments at 15% of net assets. "In-kind ETFs" largely exempt via creation/redemption.	§18(f) effectively prohibits senior securities; 300% asset coverage applies to bank borrowings.	ICI 2024 asset-weighted avg. ER: 0.40% equity MFs / 0.15% equity ETFs; 0.36% bond MFs / 0.10% bond ETFs.	MFs: daily NAV redemption required (§22(e)). ETFs: in-kind create/redeem via APs (Rule 6c–11); intraday exchange trading.
Interval Funds	Typically any investor.	No statutory illiquid cap.	§18(a): 300% coverage for debt ($\leq 33\frac{1}{3}\%$ of assets), 200% for preferred. Derivatives: Rule 18f–4.	Class A loads up to 5.75%; Class I/C/U typically no front-end load; incentive fees common.	Mandatory. Rule 23c–3 repurchase offers, generally quarterly, for 5%–25% at NAV; pro-rata if oversubscribed.
Tender Offer Funds	Typically eligibility requirements.	No statutory illiquid cap.	Same as interval funds: §18(a) 300%/200% coverage; Rule 18f–4 for derivatives.	Similar to interval funds. Often include performance/incentive fees ("qualified clients"). Retail share-class loads common.	Discretionary board-led tenders under Rule 13e–4 / §23(c)(2); size and timing at board discretion (often quarterly).
Business Development Companies (BDCs)	Eligibility requirements.	No statutory illiquid cap.	§18(a) / §61(a): 200% coverage (1:1 D/E) historically. Under the 2018 SBCAA, may elect 150% (2:1 D/E) with approval.	Externally managed: 1.5%–2.0% mgmt fee + 20% incentive fee. Distribution costs added for non-traded BDCs	Traded BDCs: no redemption; secondary-market trading. Non-traded: discretionary repurchases, quarterly at NAV up to 5%.
REITs	Eligibility requirements.	Listed: exchange-traded liquidity. NTRs: illiquid real estate; liquidity via periodic repurchase programs. Private: none.	No federal statutory limit. Typical 30%–50% debt-to-assets. §857(a) requires $\geq$ 90% distribution of taxable income.	Listed: internal mgmt; no fund-level fee. Legacy NTRs: 10%–15% upfront loads. NAV/perpetual-life NTRs (BREIT, SREIT): 0.85%–1.25% fee + performance fee.	Listed: no redemption; secondary trading. Non-traded / perpetual: monthly repurchases at NAV, capped 5%/qtr, 20%/yr. Private: none.
Private Drawdown Funds	Eligibility requirements.	No statutory illiquid cap.	No fund-level statutory limit; set contractually in the LPA.	"2 and 20": $\sim$ 2% mgmt fee on committed capital + 20% carried interest.	No investor redemption rights. Capital called via drawdowns; returned at GP discretion.

Table IA.2. Summary Statistics: Other Funds

This table presents share-class-level summary statistics for funds other than interval funds, including the mean, standard deviation, and the 25th, 50th, and 75th percentiles. The sample covers monthly data from January 2010 through March 2026. Panel A for passive ETFs, Panel B for active ETFs, Panel C for tender offer funds, Panel D for BDCs, and Panel E for REITs. Variables are defined in Appendix Table A.1. Variables not applicable to a given fund structure are omitted. The final column reports the total number of monthly share-class observations.

Panel A: Passive ETF

	Mean	SD	P25	P50	P75	N.Obs
TNA (millions)	4855.20	29771.84	31.06	179.56	1054.49	79083
Family TNA (billions)	576.43	920.99	9.81	84.89	717.33	79083
Age (years)	6.76	5.39	2.46	5.30	9.80	79083
Fund of Funds (indicator)	0.05	0.23	0.00	0.00	0.00	79083
Flows (monthly %)	2.35	12.92	-0.75	0.00	3.01	79083
Net Return (annualized %)	7.77	55.80	-15.30	7.62	33.69	79083
NAV Return (annualized %)	5.09	56.91	-18.60	4.66	31.23	79083
Distribution Return (annualized %)	2.68	7.16	-0.00	0.11	4.15	79083

Panel B: Active ETF

	Mean	SD	P25	P50	P75	N.Obs
TNA (millions)	493.09	2029.94	19.81	65.02	238.41	45141
Family TNA (billions)	10.88	28.51	0.24	1.32	6.69	45141
Age (years)	4.47	5.07	1.25	2.66	5.49	45141
Fund of Funds (indicator)	0.16	0.37	0.00	0.00	0.00	45141
Flows (monthly %)	3.36	15.11	-0.66	0.00	3.84	45141
Net Return (annualized %)	6.54	52.55	-13.63	6.97	28.16	45141
NAV Return (annualized %)	1.71	59.17	-18.51	2.84	24.99	45141
Distribution Return (annualized %)	4.83	21.61	-0.00	0.42	5.00	45141

Table 3: Summary Statistics (continued).

Panel C: Tender Offer Fund

	Mean	SD	P25	P50	P75	N.Obs
TNA (millions)	634.92	1425.37	18.43	115.27	525.33	3045
Family TNA (billions)	2.22	3.75	0.19	0.70	2.52	3045
Age (years)	3.88	3.11	1.42	2.99	5.51	3045
Fund of Funds (indicator)	0.20	0.40	0.00	0.00	0.00	3045
Flows (monthly %)	4.19	16.25	-0.56	0.06	2.91	3045
Net Return (annualized %)	8.70	24.28	0.00	7.59	18.06	3045
NAV Return (annualized %)	4.88	26.52	-4.43	4.67	16.21	3045
Distribution Return (annualized %)	3.82	10.95	-0.00	0.00	4.27	3045

Panel D: BDC

	Mean	SD	P25	P50	P75	N.Obs
TNA (millions)	2191.27	4881.20	108.07	557.65	1721.87	1628
Family TNA (billions)	7.77	10.76	0.79	3.16	10.56	1628
Age (years)	2.06	1.37	1.08	1.84	2.67	1628
Fund of Funds (indicator)	0.00	0.00	0.00	0.00	0.00	1628
Flows (monthly %)	7.37	20.82	-0.82	-0.29	6.68	1628
Net Return (annualized %)	8.74	7.79	6.56	9.38	12.01	1628
NAV Return (annualized %)	-0.58	8.09	-2.64	0.00	2.38	1628
Distribution Return (annualized %)	9.32	4.63	8.58	9.42	10.19	1628

Panel E: REIT

	Mean	SD	P25	P50	P75	N.Obs
TNA (millions)	1083.75	4178.13	13.33	111.84	461.27	4787
Family TNA (billions)	4.67	12.70	0.25	0.95	1.97	4787
Age (years)	5.11	3.98	2.16	4.07	6.92	4787
Fund of Funds (indicator)	0.00	0.00	0.00	0.00	0.00	4787
Flows (monthly %)	4.24	18.81	-0.78	-0.37	1.08	4787
Net Return (annualized %)	10.60	85.00	0.85	5.60	9.08	4787
NAV Return (annualized %)	0.45	15.81	-3.99	0.48	4.38	4787
Distribution Return (annualized %)	10.15	80.27	4.23	5.16	6.02	4787

Table IA.3. Unsmoothed Returns: Summary Statistics

This table presents share-class-level summary statistics for returns of interval funds used in the factor analysis, including the mean, standard deviation, and the 25th, 50th, and 75th percentiles, based on monthly data from January 2010 through March 2026. Variables include monthly net return, MA(3)-unsmoothed monthly net return, and AR(2) unsmoothed monthly net return. All monthly returns are annualized. This is a restricted sample of interval funds with at least 24 months of data to allow for estimation of the factor regression.

	Mean	SD	P25	P50	P75	Observations
Raw Returns (%, Annualized)	5.50	24.71	-1.12	6.72	14.04	15985
MA-Unsmoothed Returns (%, Annualized)	5.50	26.50	-2.02	6.83	14.98	15985
AR-Unsmoothed Returns (%, Annualized)	5.50	26.84	-1.61	6.78	14.86	15985

Table IA.4. Performance across Investment Categories: Gross Returns at the Portfolio Level

This table presents results on the performance of interval funds relative to index ETFs, estimated at the portfolio level using annualized gross returns as the dependent variable. Gross returns are defined as net returns plus the expense ratio. The sample includes interval fund portfolios and index ETFs over January 2010 to March 2026, with regressions estimated separately by investment category. When multiple share classes are associated with the same portfolio, observations are aggregated: returns are TNA-weighted averages across share classes, portfolio TNA is the sum of share-class TNAs, portfolio age corresponds to the inception date of the oldest share class, and fund flows are aggregated across share classes. Control variables include the one-month lag of the logarithm of portfolio TNA (*TNA*), the logarithm of aggregate fund family TNA (*Family TNA*), the logarithm of one plus portfolio age in years (*Age*), an indicator for fund-of-funds structures (*FoF*), and one-month lagged net percentage flows (*Flows*). All regressions include benchmark-by-time fixed effects. *t*-statistics, based on standard errors clustered at the month-year level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Interval	3.612***	-1.681	1.881	3.378	-0.717
	(4.96)	(-0.73)	(0.49)	(0.45)	(-0.16)
TNA	0.013	-0.030	-0.031	-0.038	-0.046
	(0.16)	(-0.34)	(-0.35)	(-0.43)	(-0.51)
Family TNA	0.172***	0.154***	0.155***	0.158***	0.157***
	(3.21)	(2.76)	(2.80)	(2.79)	(2.86)
Age	-0.451	-0.251	-0.230	-0.227	-0.204
	(-0.92)	(-0.46)	(-0.41)	(-0.41)	(-0.36)
FoF	0.336	0.010	0.104	0.255	-0.089
	(0.79)	(0.02)	(0.23)	(0.56)	(-0.22)
Flows	-0.065***	-0.065***	-0.062**	-0.059**	-0.056**
	(-2.70)	(-2.66)	(-2.53)	(-2.22)	(-2.18)
Benchmark × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	80444	76026	76033	76165	77063
Adjusted R^2	0.764	0.766	0.764	0.756	0.754

Table IA.5. Performance across Investment Categories: Interval Funds vs. Active ETFs

This table presents results on the performance of interval funds relative to active ETFs, estimated using OLS regressions at the share-class level. The dependent variable is annualized net returns, and regressions are estimated separately by investment category. The sample includes interval funds and active ETFs from January 2010 to March 2026. The key independent variable is *Interval*, an indicator equal to one for interval fund share classes, with active ETFs as the omitted comparison group. Controls and fixed effects are the same as in Table 6. *t*-statistics, based on standard errors clustered at the month-year level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Interval	1.916***	-7.255**	-2.182	4.462	-3.184
	(2.64)	(-2.60)	(-0.63)	(0.96)	(-0.68)
TNA	-0.021	-0.096	-0.086	-0.113	-0.070
	(-0.32)	(-0.84)	(-0.82)	(-1.00)	(-0.73)
Family TNA	0.089	0.082	0.077	0.102	0.078
	(1.11)	(0.87)	(0.85)	(1.09)	(0.82)
Age	0.066	0.449	0.448	0.408	0.362
	(0.23)	(1.26)	(1.25)	(1.08)	(1.03)
FoF	-0.019	-0.308	-0.289	-0.250	-0.266
	(-0.04)	(-0.57)	(-0.53)	(-0.45)	(-0.57)
Flows	-0.016	-0.019	-0.018	-0.021	-0.015
	(-0.88)	(-0.84)	(-0.80)	(-0.89)	(-0.70)
Benchmark × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	56020	45221	45268	45441	48637
Adjusted R^2	0.562	0.554	0.549	0.540	0.544

Table IA.6. Performance by Investor Clientele: Single- vs. Multi-Clientele Funds with Accredited Investors

This table presents results on the performance of interval funds relative to index ETFs, estimated separately by clientele structure using an alternative classification that incorporates accreditation status. The dependent variable is annualized net returns at the share-class level, and regressions are estimated separately by investment category. The sample includes interval funds and index ETFs from January 2010 to March 2026. Panel A restricts to single-clientele funds; Panel B restricts to multi-clientele funds. Clientele are classified using Morningstar share class type labels and accreditation status obtained from EDGAR prospectus filings. *Accredited* share classes require investors to meet accredited investor standards. *Inst/HNWI* share classes carry an institutional share class label. *Retail* share classes are noninstitutional classes without accreditation requirements. Index ETFs serve as the omitted comparison group. Controls and fixed effects are the same as in Table 6. *t*-statistics, based on standard errors clustered at the month-year level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Single-Clientele Funds

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Accredited × Interval	2.969***	-0.407	0.996	6.938	-2.700
	(2.74)	(-0.06)	(0.28)	(0.83)	(-0.45)
Inst/HNWI × Interval	2.342***	-8.576*	-0.204	-11.246	-3.111
	(2.66)	(-1.90)	(-0.05)	(-0.89)	(-0.65)
Retail × Interval	1.147	-1.848	0.984	-2.055	0.082
	(1.33)	(-0.69)	(0.24)	(-0.40)	(0.02)
Controls	Yes	Yes	Yes	Yes	Yes
Benchmark × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	82398	79134	79207	79345	79843
Adjusted R^2	0.735	0.737	0.736	0.730	0.729

Panel B: Multi-Clientele Funds

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Accredited × Interval	5.103***				-0.758
	(2.63)				(-0.19)
Inst/HNWI × Interval	5.441***	-6.220	-2.749	5.061	-2.148
	(4.09)	(-1.47)	(-0.56)	(0.72)	(-0.47)
Retail × Interval	4.817***	-4.194	-3.532	4.728	-2.389
	(3.87)	(-1.11)	(-0.73)	(0.67)	(-0.58)
Controls	Yes	Yes	Yes	Yes	Yes
Benchmark × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	86826	79127	79086	79136	81713
Adjusted R^2	0.739	0.738	0.737	0.732	0.721

Table IA.7. Performance across Investment Categories: Interval Funds vs. Alternative Semiliquid Funds

This table presents results on the performance of interval funds relative to alternative nontraded semiliquid vehicles, including tender offer funds, BDCs, and nontraded REITs. The dependent variable is annualized net returns at the share-class level, and regressions are estimated separately by investment category. The sample covers January 2010 to March 2026. The key independent variable is *Interval*, an indicator equal to one for interval fund share classes, with alternative semiliquid funds as the omitted comparison group. Variables are defined in Table A.1. Controls and fixed effects are the same as in Table 6. *t*-statistics, based on standard errors clustered at the month-year level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Interval	0.186	6.216	-5.828*	-4.760***	-8.042***
	(0.21)	(0.60)	(-1.89)	(-2.88)	(-3.71)
TNA	0.021	0.067	0.087	0.038	0.133
	(0.25)	(0.57)	(0.78)	(0.31)	(1.43)
Family TNA	-0.206	-0.358	-0.355	-0.244	-0.554
	(-0.46)	(-0.51)	(-0.51)	(-0.36)	(-0.92)
Age	-3.306***	-7.057***	-7.081***	-7.179***	-5.787***
	(-4.96)	(-4.49)	(-4.49)	(-4.67)	(-5.08)
FoF	0.764	-1.298	-1.125	-0.929	-0.942
	(0.73)	(-0.72)	(-0.64)	(-0.54)	(-0.84)
Flows	-0.124**	-0.236**	-0.232**	-0.227**	-0.194***
	(-2.43)	(-2.48)	(-2.49)	(-2.50)	(-2.71)
Benchmark × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	20471	9784	9880	10156	13196
Adjusted R^2	0.130	0.059	0.035	0.042	0.048

Table IA.8. Performance and Redemption Frequency

This table presents results on the relationship between redemption frequency and interval fund performance, estimated at the portfolio level using annualized gross returns as the dependent variable. Regressions are estimated separately by investment category. The sample includes interval fund portfolios and index ETFs from January 2010 to March 2026. The key independent variables are the interval fund indicator and its interaction with *Long Interval*, an indicator equal to one for funds offering annual or semiannual redemption windows, and zero for funds offering quarterly or monthly redemptions. Control variables are included but omitted from the table for brevity. Control variables include the one-month lag of the logarithm of portfolio TNA (*TNA*), the logarithm of aggregate fund family TNA (*Family TNA*), the logarithm of one plus portfolio age in years (*Age*), an indicator for fund-of-funds structures (*FoF*), and one-month lagged net percentage flows (*Flows*). All regressions include benchmark-by-time fixed effects. *t*-statistics, based on standard errors clustered at the month-year level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Credit (1)	Liq. Strategies (2)	Multi-Asset (3)	PE/VC (4)	Real Asset (5)
Interval	3.992***	-1.681	1.881	2.950	-0.717
	(4.55)	(-0.73)	(0.49)	(0.46)	(-0.16)
Interval $\times$ Long Interval	-2.627*			1.738	
	(-1.78)			(0.24)	
Controls	Yes	Yes	Yes	Yes	Yes
Benchmark $\times$ Time FE	Yes	Yes	Yes	Yes	Yes
Observations	80444	76026	76033	76165	77063
Adjusted R^2	0.764	0.766	0.764	0.756	0.754

Table IA.9. Portfolio Holdings Characteristics of Interval Funds

This table presents portfolio-level summary statistics for the sample of interval funds, including the mean, standard deviation, and the 25th, 50th, and 75th percentiles. Statistics are computed using monthly data from October 2019 through March 2026, corresponding to the period for which N-PORT holdings data are available. Panel A reports the full sample; Panels B through D split by investor clientele. Variables are defined in Appendix Table A.1. The final column reports the total number of monthly portfolio-level observations.

Panel A: Full Sample

	Mean	SD	P25	P50	P75	N.Obs
Level 1 (% of AUM)	11.92	17.82	1.08	5.17	15.47	5376
Level 2 (% of AUM)	33.57	35.26	0.00	19.31	67.59	5376
Level 3 (% of AUM)	31.00	33.24	0.00	17.60	58.72	5376
Liquidity Index (L1+0.5*L2)	28.70	20.74	10.86	27.05	45.07	5376
Leverage (% of AUM)	12.34	13.26	1.55	7.05	20.32	5376

Panel B: Retail

	Mean	SD	P25	P50	P75	N.Obs
Level 1 (% of AUM)	18.45	23.20	2.27	8.83	26.99	1087
Level 2 (% of AUM)	18.50	28.31	0.00	0.92	26.43	1087
Level 3 (% of AUM)	45.11	36.37	0.35	47.13	80.60	1087
Liquidity Index (L1+0.5*L2)	27.70	25.70	4.61	20.30	45.68	1087
Leverage (% of AUM)	12.58	16.02	1.04	4.55	21.59	1087

Panel C: Inst/HNWI

	Mean	SD	P25	P50	P75	N.Obs
Level 1 (% of AUM)	11.53	16.16	1.19	6.10	16.16	1962
Level 2 (% of AUM)	31.30	35.56	0.00	12.69	69.41	1962
Level 3 (% of AUM)	25.92	32.31	0.00	10.42	47.02	1962
Liquidity Index (L1+0.5*L2)	27.18	20.74	9.60	22.86	44.39	1962
Leverage (% of AUM)	7.46	9.34	0.71	3.95	10.34	1962

Panel D: Multi-Class

	Mean	SD	P25	P50	P75	N.Obs
Level 1 (% of AUM)	9.19	15.33	0.42	3.73	10.53	2327
Level 2 (% of AUM)	42.52	35.25	4.04	38.23	76.18	2327
Level 3 (% of AUM)	28.70	30.59	1.02	16.43	51.88	2327
Liquidity Index (L1+0.5*L2)	30.45	17.82	16.11	31.05	45.09	2327
Leverage (% of AUM)	16.34	13.30	4.21	13.77	27.13	2327

Table IA.10. Redemption Events

Fund-level summary statistics for the Morningstar sample and for retail and institutional or high net worth share classes separately for funds that have redemption data by share class. Offered Amount is the percentage of shares offered by the fund for redemption. Potential Increase is the additional percentage that may be offered for repurchase before the fund begins to prorate redemptions. Start to Deadline is the number of days from the start of the repurchase offer to the redemption request deadline. Deadline to Pricing is the number of days between the redemption deadline and the pricing date at which the repurchase NAV is determined. Amount Tendered is the percentage of outstanding shares tendered on a given redemption date. Amount Repurchased is the percentage of outstanding shares repurchased on a given redemption date. Oversubscribed is an indicator of whether the number of shares tendered exceeds the number of shares offered for redemption at the fund level. Prorated is an indicator of whether the shares repurchased are fewer than the shares tendered, conditional on the offer being oversubscribed.

	All Share Classes				Retail Share Classes				Inst./HNWI Share Classes			
	mean	p50	sd	N	mean	p50	sd	N	mean	p50	sd	N
Offered amount (%)	6.16	5.00	3.92	1,787	6.17	5.00	5.36	564	6.42	5.00	3.69	572
Potential increase (%)	1.73	2.00	0.68	1,847	1.69	2.00	0.72	564	1.65	2.00	0.76	572
Start to deadline	26.55	30	9.623	1,155	30.57	31	4.749	319	29.42	30	5.831	374
Deadline to pricing	3.473	0	89.74	1,676	9.213	0	160.2	520	1.816	0	17.11	506
Amount tendered (%)	4.04	1.73	9.12	968	6.96	1.67	14.6	311	2.94	1.65	5.19	324
Amount repurchased (%)	3.05	2.25	4.04	1,631	2.90	2.15	3.96	409	2.75	1.67	3.47	428
Oversubscribed (0-1)	0.158	0	0.365	1,592	0.223	0	0.417	524	0.149	0	0.357	482
Prorated (0-1)	0.567	1	0.498	104	0.662	1	0.477	68	0.560	1	0.507	25

B Proofs

Using equation (1), write

$$R_2(\ell,m) = \iota v + 2\delta_S + \ell(\delta_L - 2\delta_S) - \gamma\left[q(\ell,m) + \frac{\max\{0, \ell m - q(\ell,m)\}}{1-m}\right] - 2f.$$

For compact derivations below, define expected liquidation burden directly as

$$C_e(\ell;\eta) := \mathrm{E}_z\left[q\big(\ell, m_e(z)\big) + \frac{\max\{0, \ell m_e(z) - q(\ell, m_e(z))\}}{1-m_e(z)}\right], \qquad C_e^* := C_e(\ell^*;\eta).$$

Proof of Prediction 1. Set $\iota = 0$ in equation (1). Since informed redemptions are inactive, $m_0(z) = m_1(z) = \min\{\lambda, z\}$. For any fixed allocation,

$$R_2\big(\ell, \min\{\lambda,z\}\big) = 2\delta_S + \ell(\delta_L - 2\delta_S) - \gamma\left[q\big(\ell, \min\{\lambda,z\}\big) + \frac{\max\{0, \ell\min\{\lambda,z\} - q(\ell,\min\{\lambda,z\})\}}{1-\min\{\lambda,z\}}\right] - 2f.$$

Evaluating at $\ell = \ell^*$ and taking expectations,

$$\mathcal{V}^* = \mathrm{E}_z\big[R_2(\ell^*, \min\{\lambda,z\})\big] = 2\delta_S + \ell^*(\delta_L - 2\delta_S) - \gamma C^* - 2f,$$

where

$$C^* := \mathrm{E}_z\left[q\big(\ell^*, \min\{\lambda,z\}\big) + \frac{\max\{0, \ell^*\min\{\lambda,z\} - q(\ell^*,\min\{\lambda,z\})\}}{1-\min\{\lambda,z\}}\right].$$

Since $\iota = 0$ implies $m_0(z) = m_1(z) = \min\{\lambda,z\}$, then $C_0^* = C_1^* =: C^*$, and

$$\mathcal{V}^* = 2\delta_S + \ell^*(\delta_L - 2\delta_S) - \gamma C^* - 2f.$$

By the envelope theorem,

$$\frac{\partial \mathcal{V}^*}{\partial \delta_L} = \ell^* > 0, \qquad \frac{\partial \mathcal{V}^*}{\partial \delta_S} = 2(1-\ell^*) > 0,$$

where the strict inequalities use the maintained interiority assumption $\ell^* \in (0,1)$. Therefore, expected excess return is strictly increasing in both illiquidity premia.

Define

$$L(\ell,m) := q(\ell,m) + \frac{\max\{0, \ell m - q(\ell,m)\}}{1-m}.$$

If $m \le 1-\ell$, then $q(\ell,m) = 0$ and $L(\ell,m) = \ell m/(1-m) \le m$ because $\ell \le 1-m$. If $m > 1-\ell$, then $q(\ell,m) = m-(1-\ell)$ and $\ell m - q(\ell,m) = (1-\ell)(1-m) \ge 0$, so

$$L(\ell,m) = q(\ell,m) + \frac{(1-\ell)(1-m)}{1-m} = m,$$

because $m \le \lambda < 1$ implies $1-m > 0$. Hence $L(\ell,m) \le m \le \lambda$, implying $C^* \le \lambda$. Therefore,

$$\mathcal{V}^* \ge 2\delta_S + \ell^*(\delta_L - 2\delta_S) - \gamma\lambda - 2f \ge 2(f+\gamma) - \gamma\lambda - 2f = \gamma(2-\lambda) > 0,$$

where we use $\delta_L > 2\delta_S$, $\delta_S \geq f+\gamma$, and $\lambda < 1$. This proves strict positivity and the stated lower bound. $\square$

Proof of Prediction 2. Consider a limiting case where $\delta_L \to \delta_S = \gamma = f = 0$. Equation (1) implies $R_2(\ell,m) = \iota v$. Hence, with $\iota = 1$, $R_2(\ell,m) = v$ and equation (3) becomes

$$\mathcal{V}^* = \pi(\eta)v_1 + (1-\pi(\eta))v_0.$$

Hence, underperformance occurs iff

$$\pi(\eta)v_1 + (1-\pi(\eta))v_0 < 0,$$

equivalently iff

$$\pi(\eta) < \frac{-v_0}{v_1 - v_0}.$$

Outside that limiting case, illiquidity premia enter as

$$2\delta_S + \ell^*(\delta_L - 2\delta_S),$$

which raises the expected excess return for a fixed allocation when premia increase. In particular, sufficiently large long-illiquidity premium δ_L offsets weak incentive states, yielding the stated requirement for positive expected excess return. $\square$

Proof of Prediction 3. Define

$$A_e(\ell;\eta) := \iota v_e + 2\delta_S + \ell(\delta_L - 2\delta_S) - \gamma C_e(\ell;\eta) - 2f, \qquad A_e^* := A_e(\ell^*;\eta),$$

and

$$\mathcal{V}(\ell;\eta) := \pi(\eta)A_1(\ell;\eta) + (1-\pi(\eta))A_0(\ell;\eta), \qquad \mathcal{V}^* := \mathcal{V}(\ell^*;\eta),$$

so that $\mathcal{V}^* = \pi(\eta)A_1^* + (1-\pi(\eta))A_0^*$. Applying the chain rule and using the envelope condition,

$$\frac{d\mathcal{V}^*}{d\eta} = \underbrace{\pi(\eta)\left.\frac{\partial A_1(\ell;\eta)}{\partial\eta}\right|_{\ell=\ell^*} + (1-\pi(\eta))\left.\frac{\partial A_0(\ell;\eta)}{\partial\eta}\right|_{\ell=\ell^*}}_{\text{liquidity transformation channel}} + \underbrace{(A_1^* - A_0^*)\frac{d\pi(\eta)}{d\eta}}_{\text{incentive channel}}. \qquad \text{(IA.1)}$$

Define

$$\xi^*(u) := \begin{cases} \dfrac{\ell^*}{(1-u)^2}, & u \in [0, 1-\ell^*], \\ 1, & u \in (1-\ell^*, \lambda]. \end{cases}$$

Then, holding $\ell = \ell^*$ fixed,

$$C_e^* = \mathrm{E}_z\left[q\big(\ell^*, m_e(z)\big) + \frac{\max\{0, \ell^* m_e(z) - q(\ell^*, m_e(z))\}}{1 - m_e(z)}\right] = \int_0^\lambda \xi^*(u)\big[1 - H_\eta(u - \iota(1-e)\eta)\big]\,du,$$

and its direct sensitivity is

$$\mathcal{S}_e^* := \left.\frac{\partial C_e(\ell;\eta)}{\partial \eta}\right|_{\ell=\ell^*} = -\int_0^\lambda \xi^*(u)\left[\frac{\partial H_\eta(u-\iota(1-e)\eta)}{\partial \eta} - \iota(1-e)h_\eta(u-\iota(1-e)\eta)\right]du.$$

Hence

$$\left.\frac{\partial A_e(\ell;\eta)}{\partial \eta}\right|_{\ell=\ell^*} = -\gamma\mathcal{S}_e^*.$$

If $\iota=0$, then $\mathcal{S}_0^*=\mathcal{S}_1^*$ and

$$\frac{d\mathcal{V}^*}{d\eta} = -\gamma\mathcal{S}_1^*.$$

Under the maintained left-shift assumption for H_η, $\partial H_\eta(u)/\partial\eta \geq 0$ and $\xi^*(u)\geq 0$, so $\mathcal{S}_1^*\leq 0$ and therefore $d\mathcal{V}^*/d\eta \geq 0$.

If $\iota=1$, define

$$\mathcal{S}_1^* = -\int_0^\lambda \xi^*(u)\frac{\partial H_\eta(u)}{\partial \eta}\,du \leq 0,$$

and

$$\mathcal{S}_0^* = -\int_0^\lambda \xi^*(u)\left[\frac{\partial H_\eta(u-\eta)}{\partial \eta} - h_\eta(u-\eta)\right]du,$$

where $\mathcal{S}_0^*$ is sign-ambiguous because it combines distribution-shift and direct informed-redemption terms.

For the incentive channel,

$$\frac{d\pi(\eta)}{d\eta} = \iota f g\big(\iota f\Psi(\eta,\lambda)\big)\frac{\partial\Psi(\eta,\lambda)}{\partial\eta},$$

with

$$\frac{\partial\Psi(\eta,\lambda)}{\partial\eta} = H_\eta(\lambda-\eta)+\mathcal{P}^*, \qquad \mathcal{P}^* := \left.\frac{\partial}{\partial s}\mathrm{E}_{z\sim H_s}[\min\{\eta,(\lambda-z)_+\}]\right|_{s=\eta}.$$

Under the maintained left-shift assumption, $\mathcal{P}^*\geq 0$, so $\partial\Psi/\partial\eta\geq 0$ and therefore $d\pi(\eta)/d\eta \geq 0$.

Combining with equation (IA.1),

$$\frac{d\mathcal{V}^*}{d\eta} = \underbrace{-\gamma\pi(\eta)\mathcal{S}_1^*}_{\text{high-effort liquidity transformation channel}} + \underbrace{-\gamma(1-\pi(\eta))\mathcal{S}_0^*}_{\text{low-effort liquidity transformation channel}} + \underbrace{(A_1^*-A_0^*)\frac{d\pi(\eta)}{d\eta}}_{\text{incentive channel}}.$$

The high-effort liquidity transformation term is weakly positive; the low-effort liquidity transformation term is ambiguous. For $\iota=1$,

$$A_1^*-A_0^* = (v_1-v_0)-\gamma(C_1^*-C_0^*).$$

Because $m_0(z)=\min\{\lambda,z+\eta\}\geq m_1(z)=\min\{\lambda,z\}$ pointwise and $L(\ell,m)=q(\ell,m)+\max\{0,\ell m-q(\ell,m)\}/(1-m)$ is increasing in m (shown above), we have $C_0^*\geq C_1^*$. Together with the maintained strict ordering $v_1 > v_0$, this implies

$$A_1^* - A_0^* > 0.$$

Therefore, because $d\pi(\eta)/d\eta \geq 0$, if

$$(1-\pi(\eta))\mathcal{S}_0^* \leq -\pi(\eta)\mathcal{S}_1^*. \tag{IA.2}$$

the total effect is weakly positive; otherwise, the sign is ambiguous. $\square$